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(19) **United States**(12) **Patent Application Publication**
SUGIO et al.(10) **Pub. No.: US 2025/0280144 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **CODING METHOD AND APPARATUS WITH
CANDIDATE MOTION VECTORS****Publication Classification**(71) Applicant: **Sun Patent Trust**, New York, NY (US)(72) Inventors: **Toshiyasu SUGIO**, Osaka (JP);
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MATSUNOBU**, Osaka (JP)(73) Assignee: **Sun Patent Trust**, New York, NY (US)(21) Appl. No.: **19/202,748**(22) Filed: **May 8, 2025**(51) **Int. Cl.****H04N 19/52** (2014.01)
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H04N 19/172 (2014.01)
H04N 19/176 (2014.01)
H04N 19/503 (2014.01)
H04N 19/56 (2014.01)(52) **U.S. Cl.****CPC** **H04N 19/52** (2014.11); **H04N 19/105**
(2014.11); **H04N 19/503** (2014.11); **H04N**
19/56 (2014.11); **H04N 19/107** (2014.11);
H04N 19/172 (2014.11); **H04N 19/176**
(2014.11)**Related U.S. Application Data**

(63) Continuation of application No. 18/394,461, filed on Dec. 22, 2023, now Pat. No. 12,323,616, which is a continuation of application No. 18/087,516, filed on Dec. 22, 2022, now Pat. No. 11,895,324, which is a continuation of application No. 17/339,118, filed on Jun. 4, 2021, now Pat. No. 11,575,930, which is a continuation of application No. 16/224,209, filed on Dec. 18, 2018, now Pat. No. 11,076,170, which is a continuation of application No. 15/629,101, filed on Jun. 21, 2017, now Pat. No. 10,212,450, which is a continuation of application No. 15/266,004, filed on Sep. 15, 2016, now Pat. No. 9,723,322, which is a continuation of application No. 13/479,636, filed on May 24, 2012, now Pat. No. 9,485,518.

(60) Provisional application No. 61/490,747, filed on May 27, 2011.

(57)

ABSTRACT

A moving picture coding apparatus for coding a current block of a picture is provided. A motion vector candidate list including motion vector candidates is generated and, for each of the motion vector candidates, a value is also generated. A second candidate is arranged before a first candidate in the motion vector candidate list. The first candidate is sorted before the second candidate based on the value of the first candidate and the value of the second candidate, with the value of the first candidate being higher than the value of the second candidate. A motion vector candidate to use to code the current block is selected, and an index to identify the selected motion vector candidate is coded. For each candidate of the motion vector candidates, the value is based on a number of times the candidate is selected as a motion vector predictor for neighboring blocks.

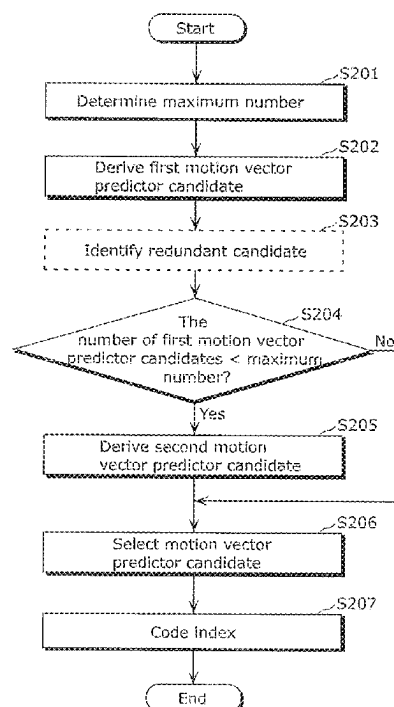


FIG. 1A

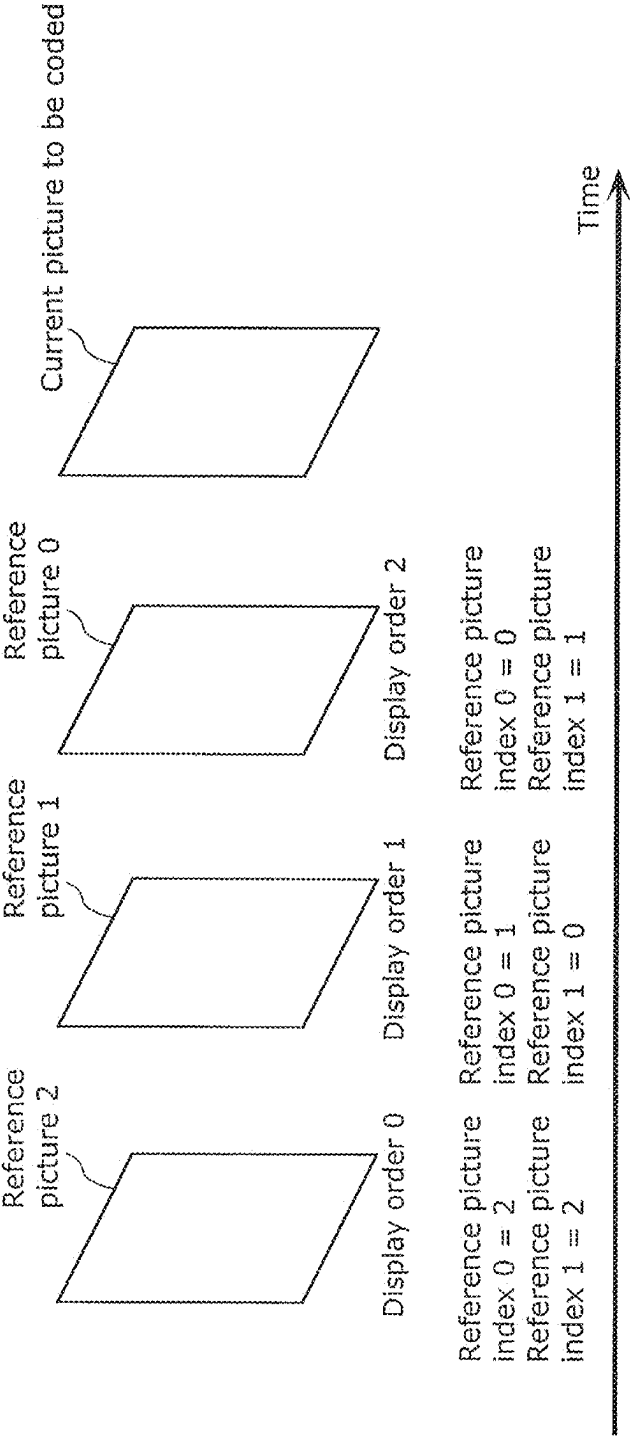


FIG. 1B

Reference picture list 0

Reference picture index 0	Display order
0	2
1	1
2	0

FIG. 1C

Reference picture list 1

Reference picture index 1	Display order
0	1
1	2
2	0

FIG. 2

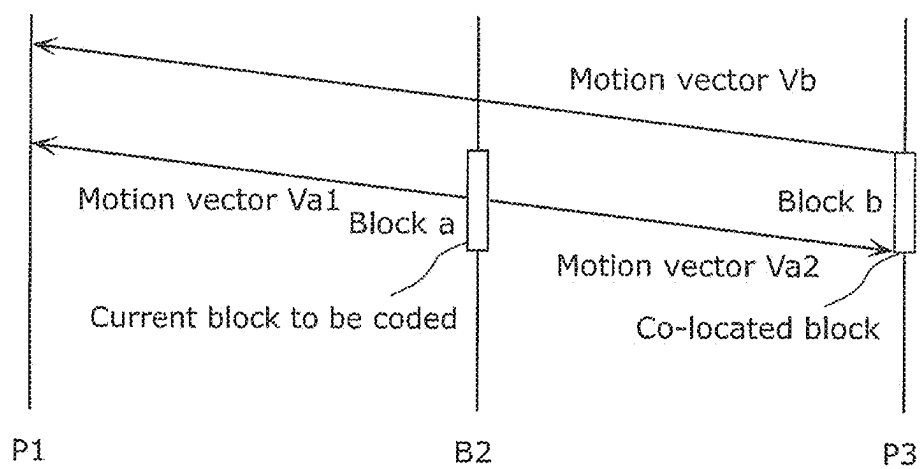


FIG. 3

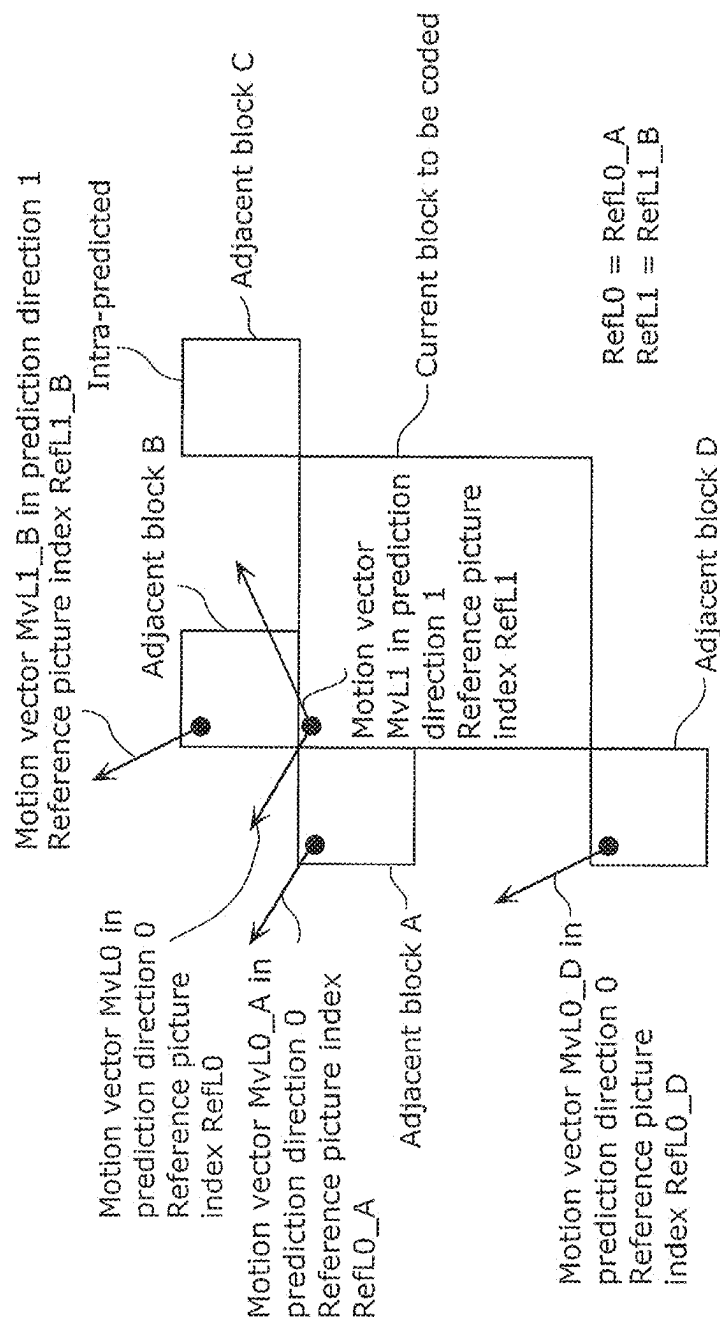


FIG. 4

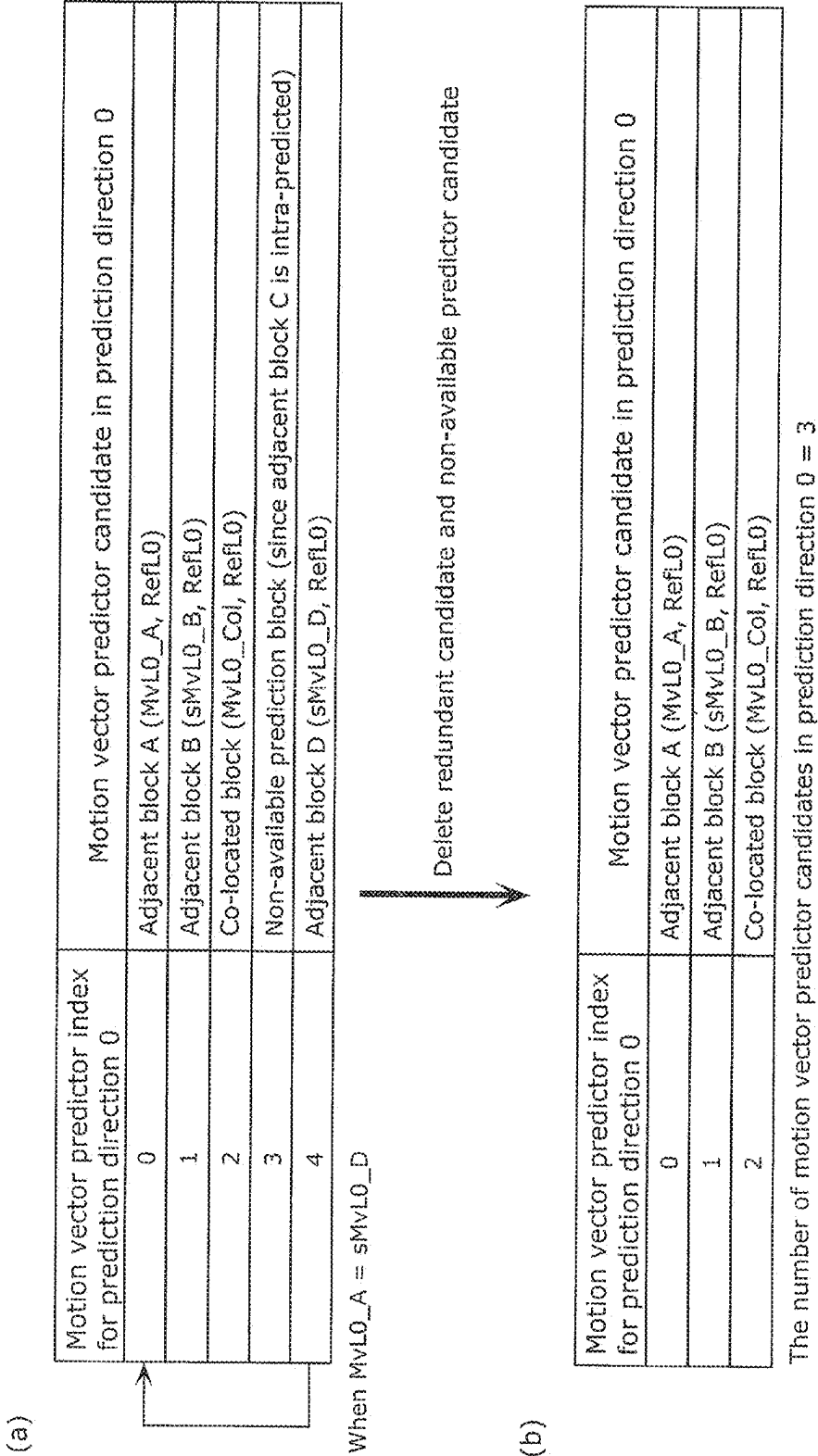


FIG. 5

(a)

Motion vector predictor index for prediction direction 1	Motion vector predictor candidate in prediction direction 1
0	Adjacent block A (sMvL1_A, RefL1)
1	Adjacent block B (MvL1_B, RefL1)
2	Co-located block (MvL1_Col, RefL1)
3	Non-available prediction block (since adjacent block C is intra-predicted)
4	Adjacent block D (sMvL1_D, RefL1)

When $MvL1_B = MvL1_Col = sMvL1_D$

Delete redundant candidate and non-available predictor candidate

(b)

Motion vector predictor index for prediction direction 1	Motion vector predictor candidate in prediction direction 1
0	Adjacent block A (sMvL1_A, RefL1)
1	Adjacent block B (MvL1_B, RefL1)

The number of motion vector predictor candidates in prediction direction 1 = 2

FIG. 6

Motion vector predictor candidate list size = 2

Motion vector predictor index	Assigned bit string
0	0
1	1

Motion vector predictor candidate list size = 3

Motion vector predictor index	Assigned bit string
0	0
1	10
2	11

Motion vector predictor candidate list size = 4

Motion vector predictor index	Assigned bit string
0	0
1	10
2	110
3	111

Motion vector predictor candidate list size = 5

Motion vector predictor index	Assigned bit string
0	0
1	10
2	110
3	1110
4	1111

FIG. 7

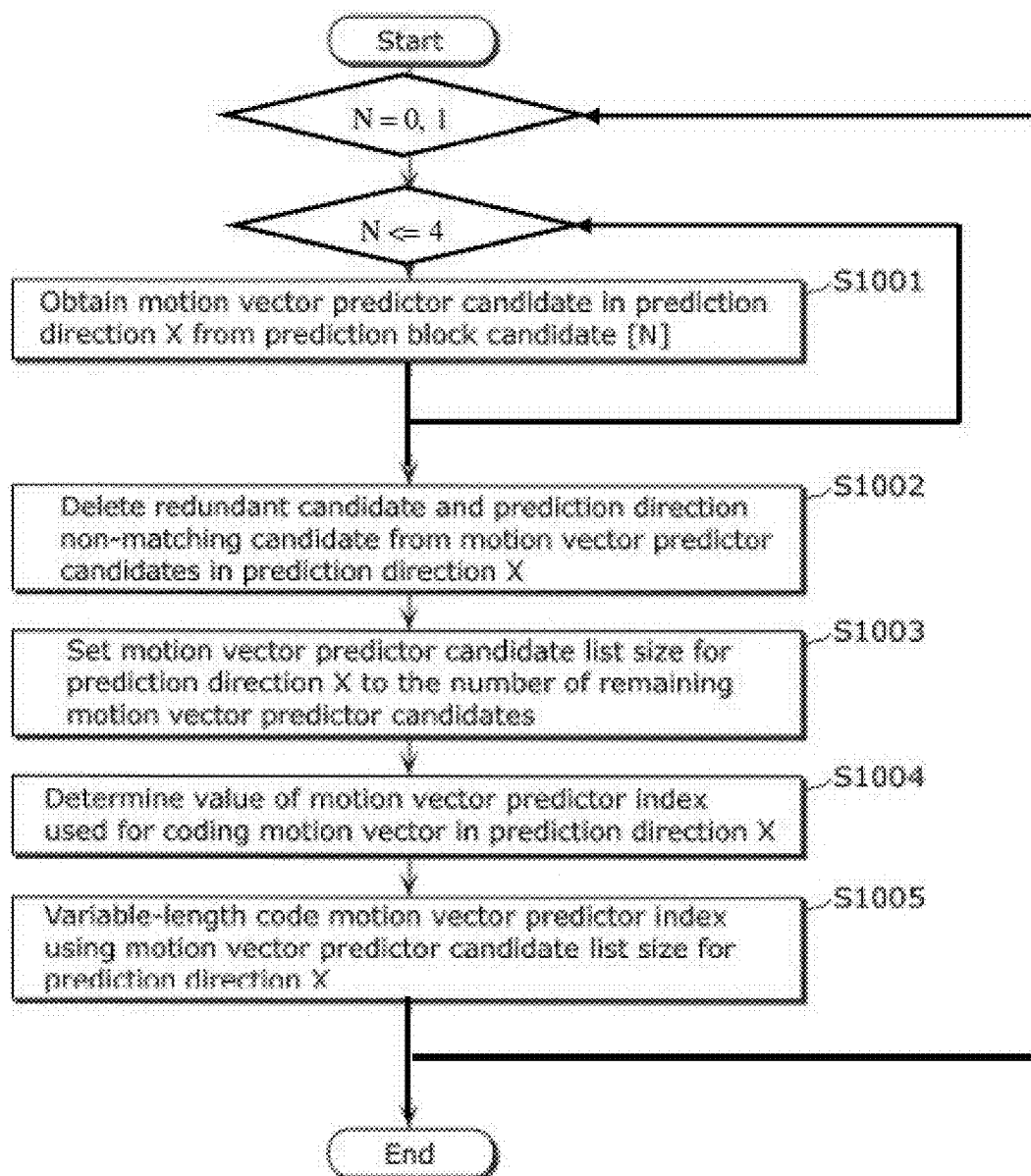


FIG. 8A

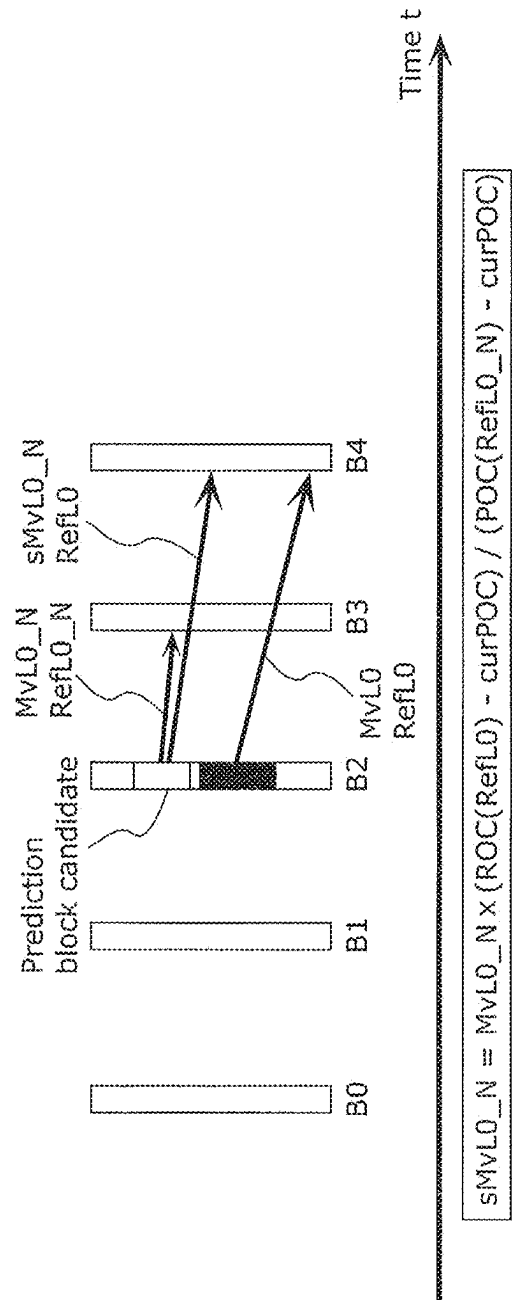


FIG. 8B

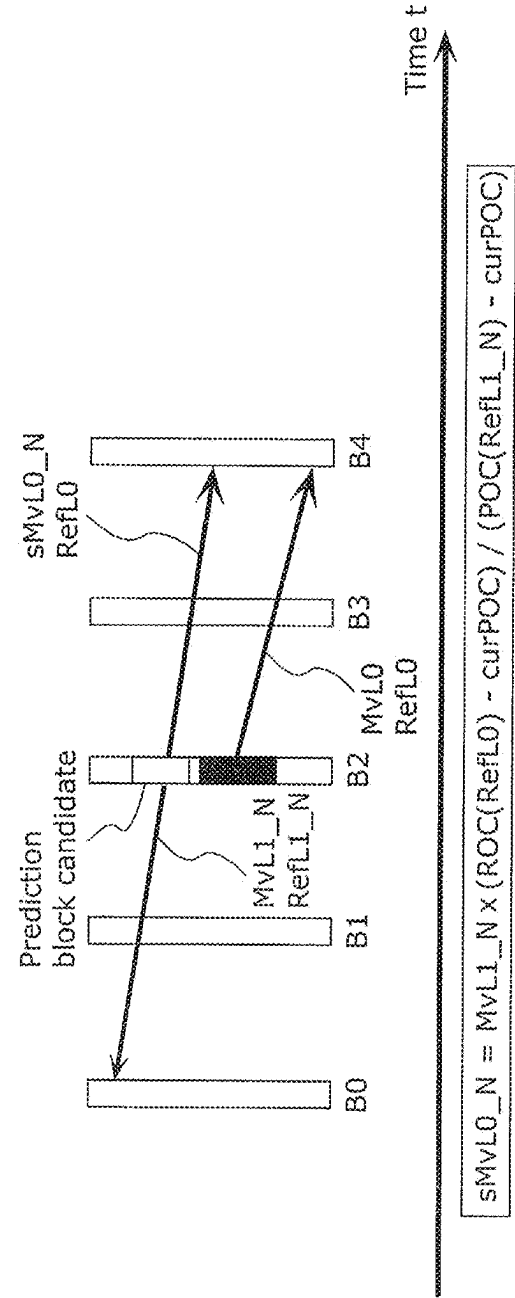


FIG. 10

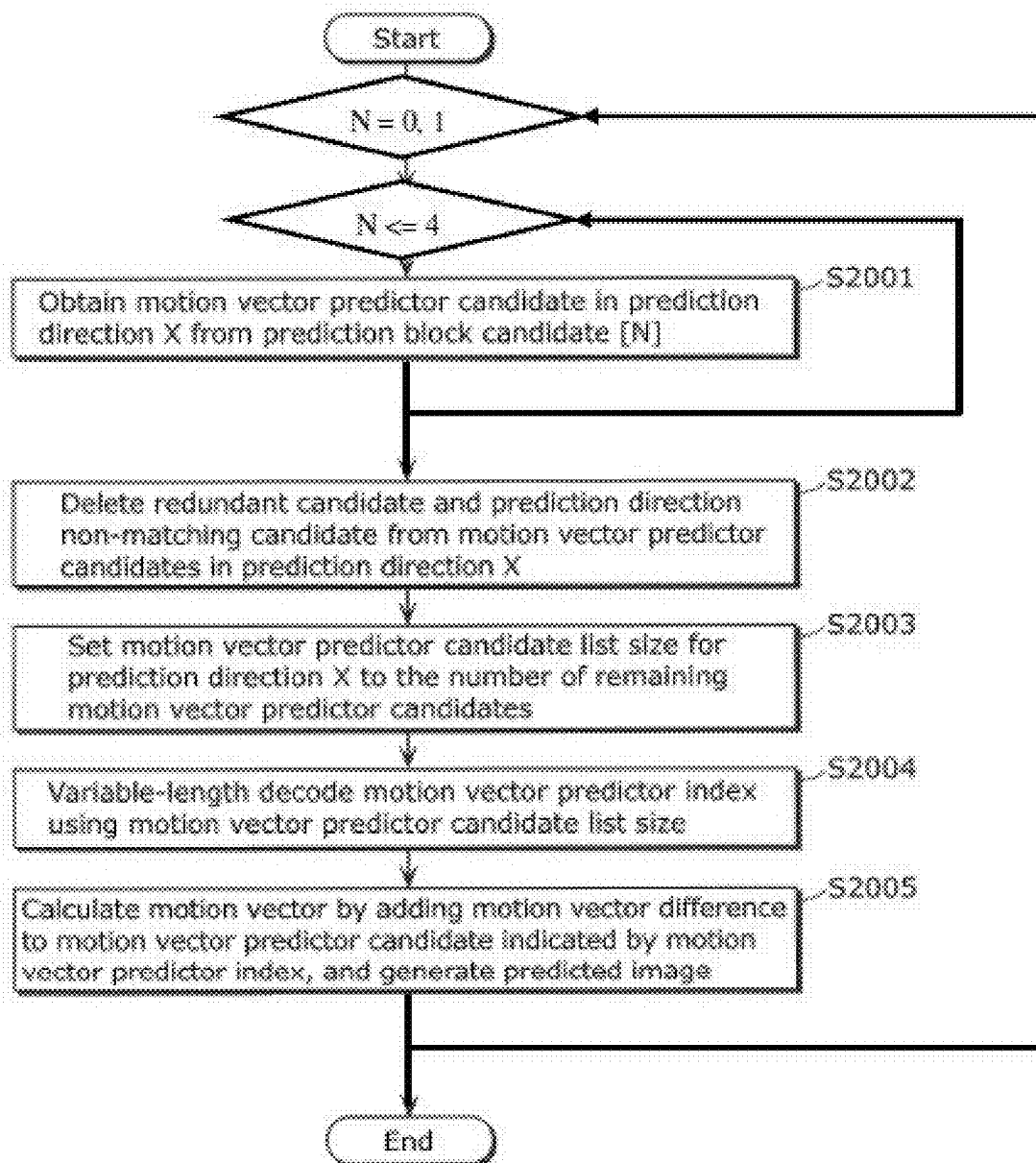


FIG. 11

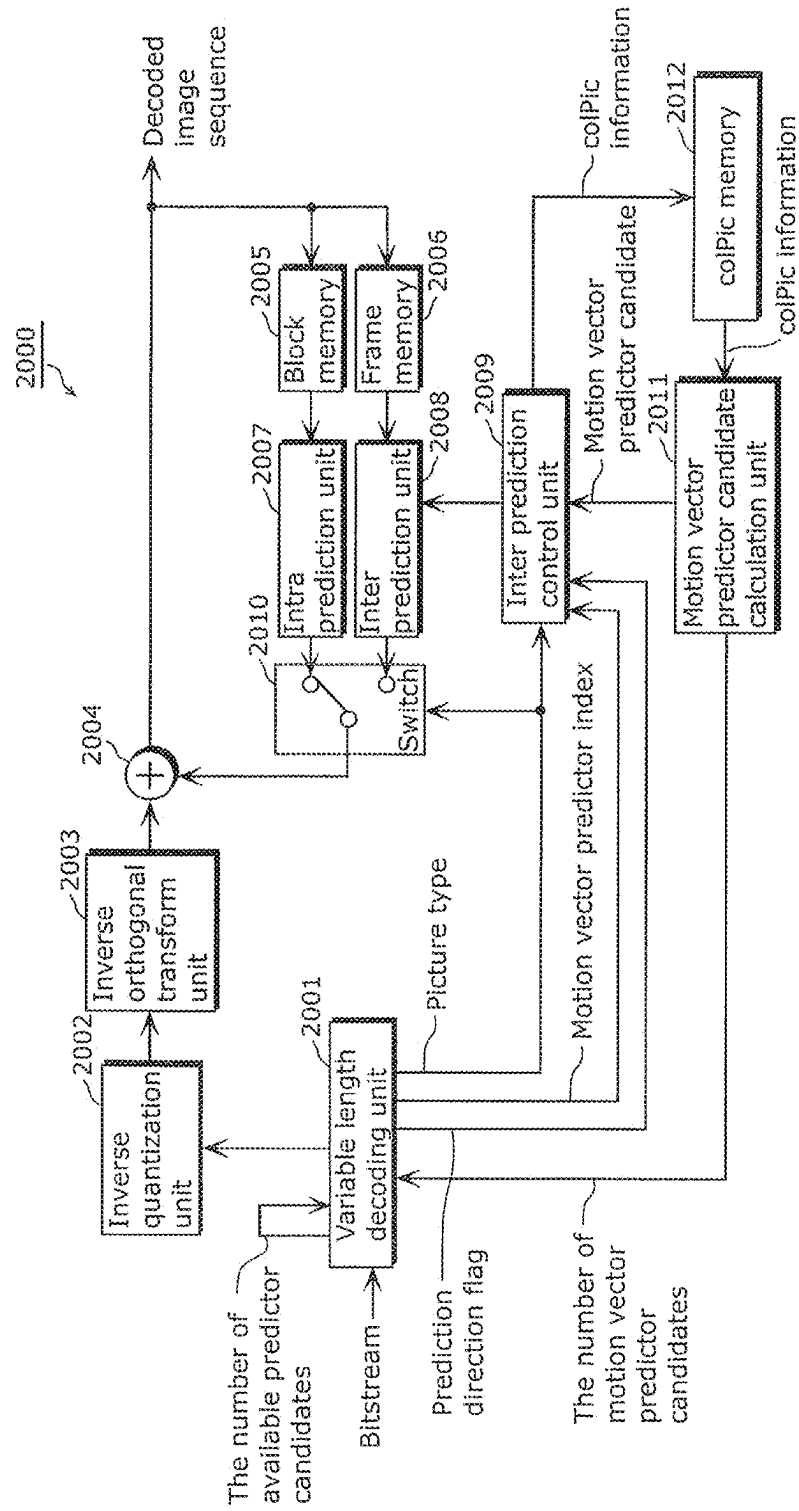


FIG. 13

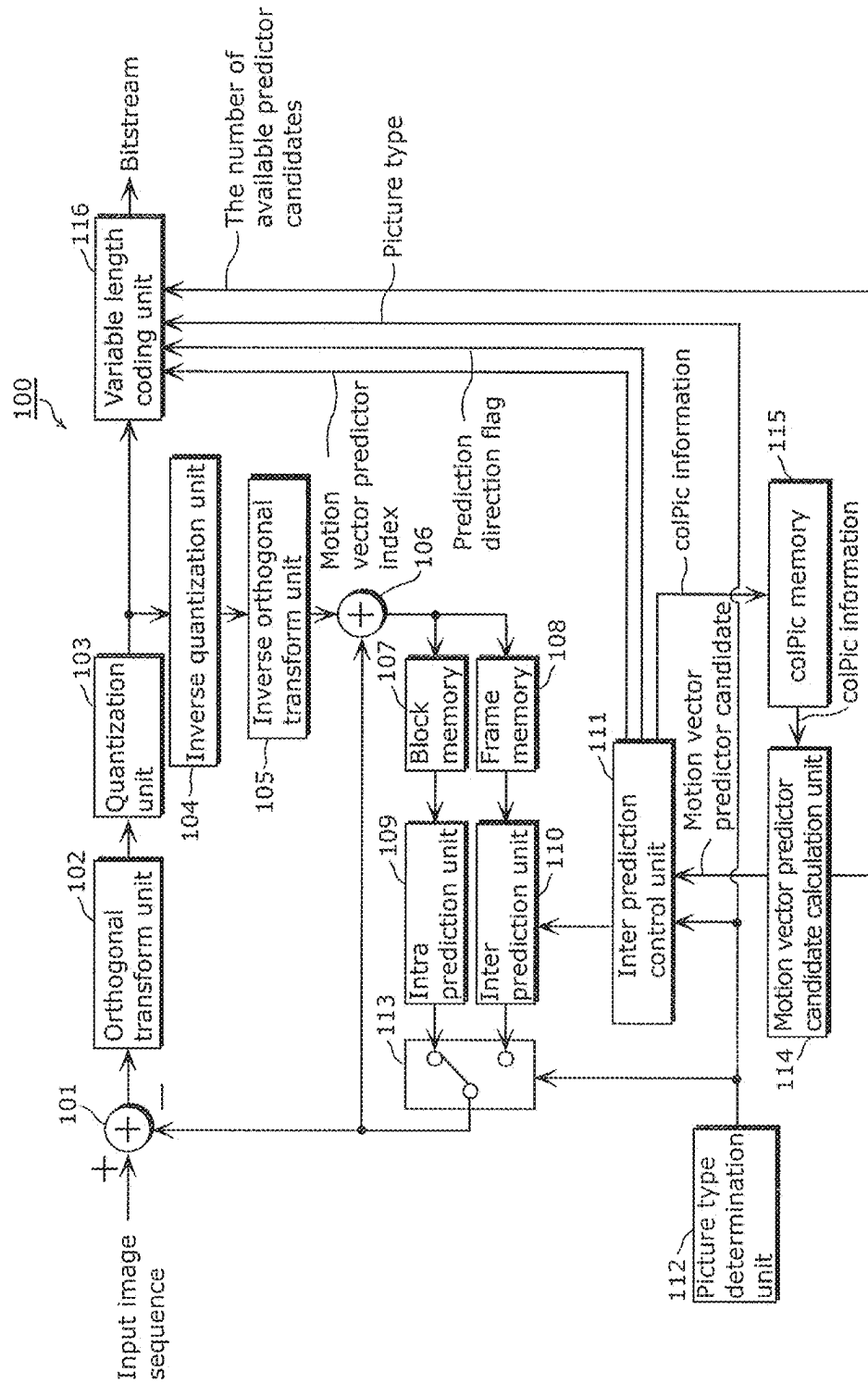


FIG. 14

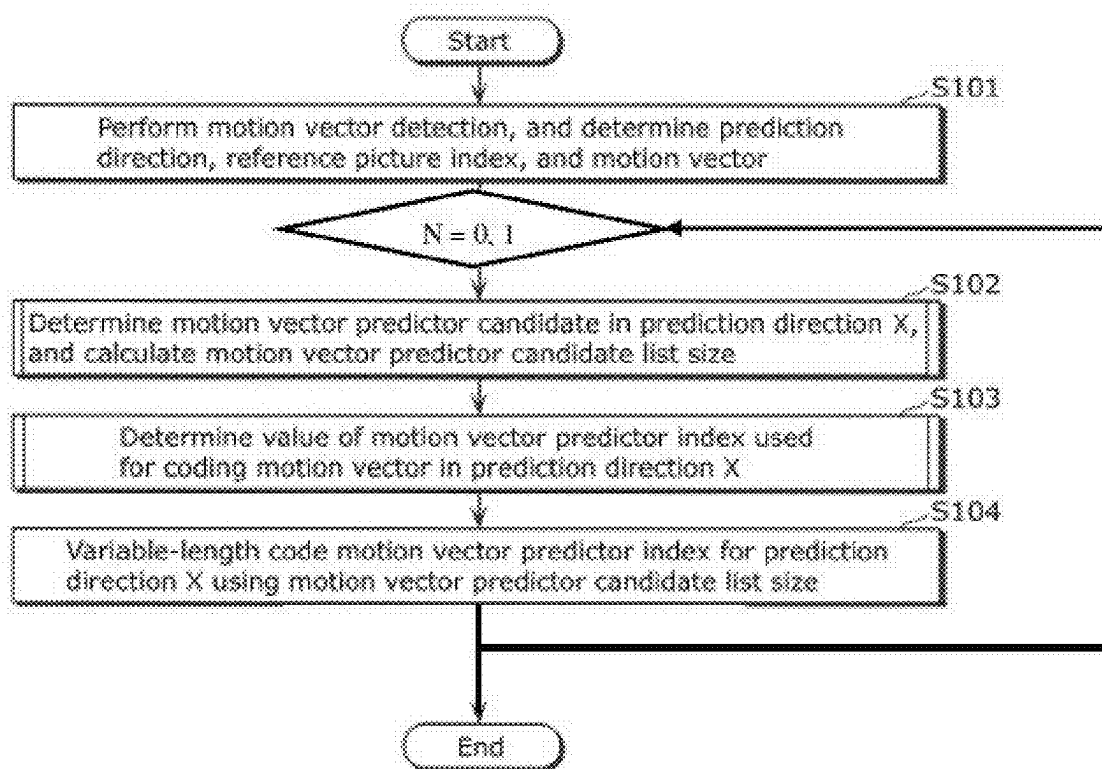


FIG. 15

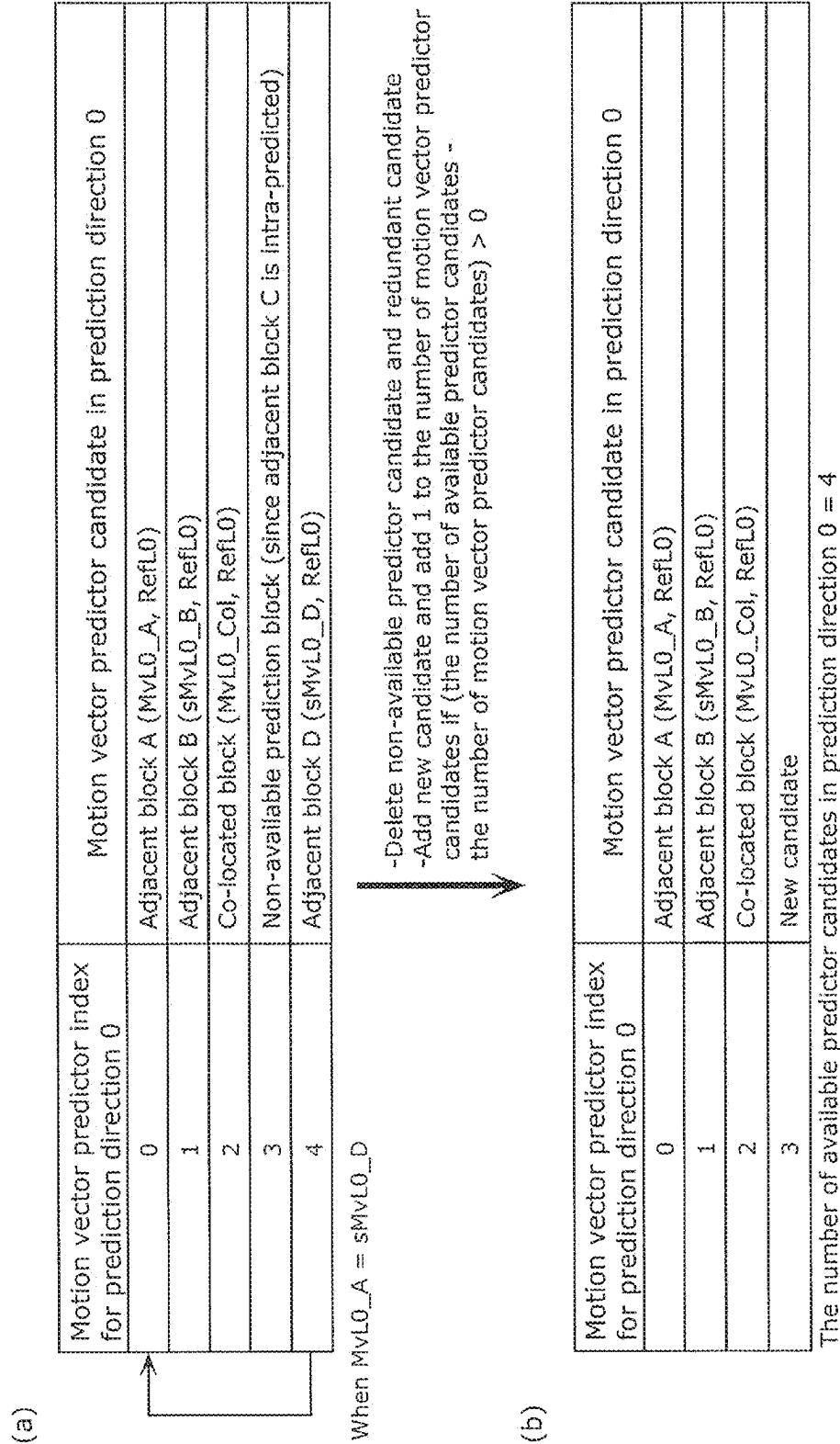


FIG. 17

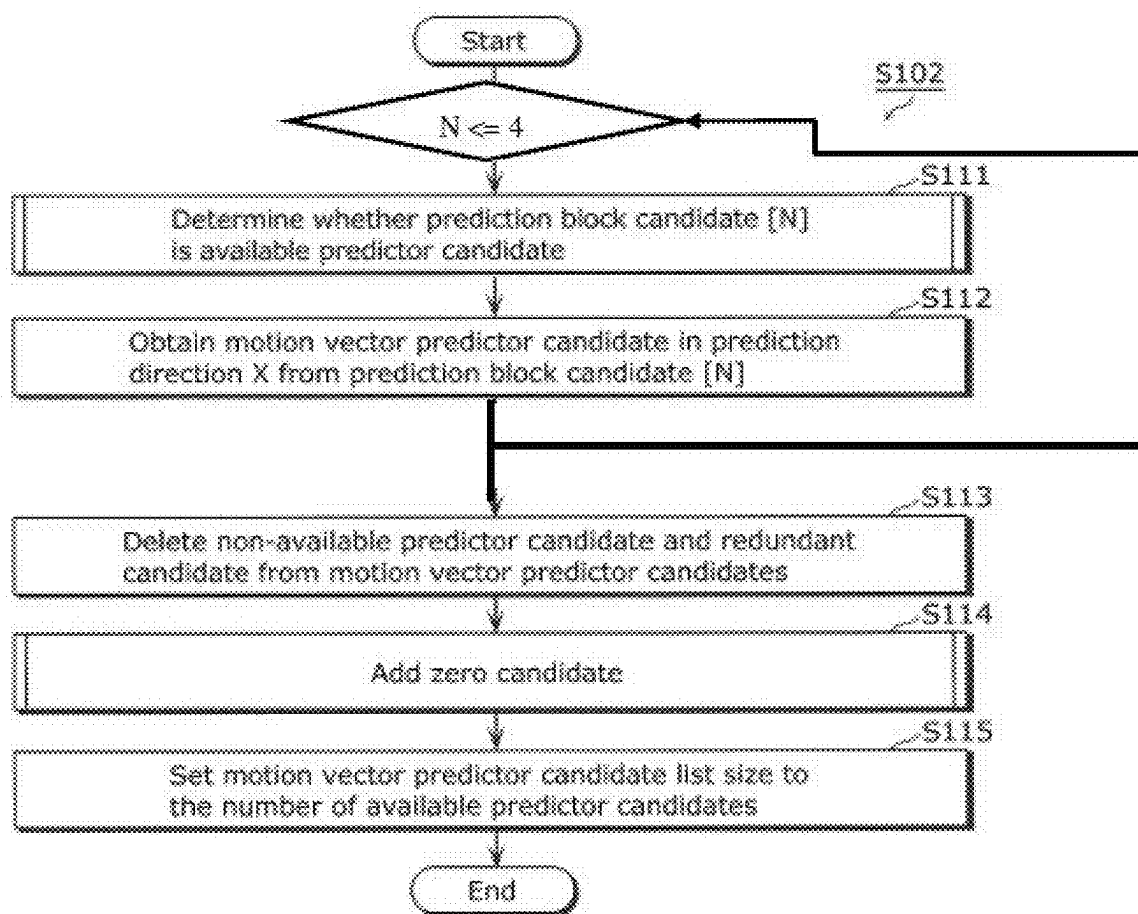


FIG. 18

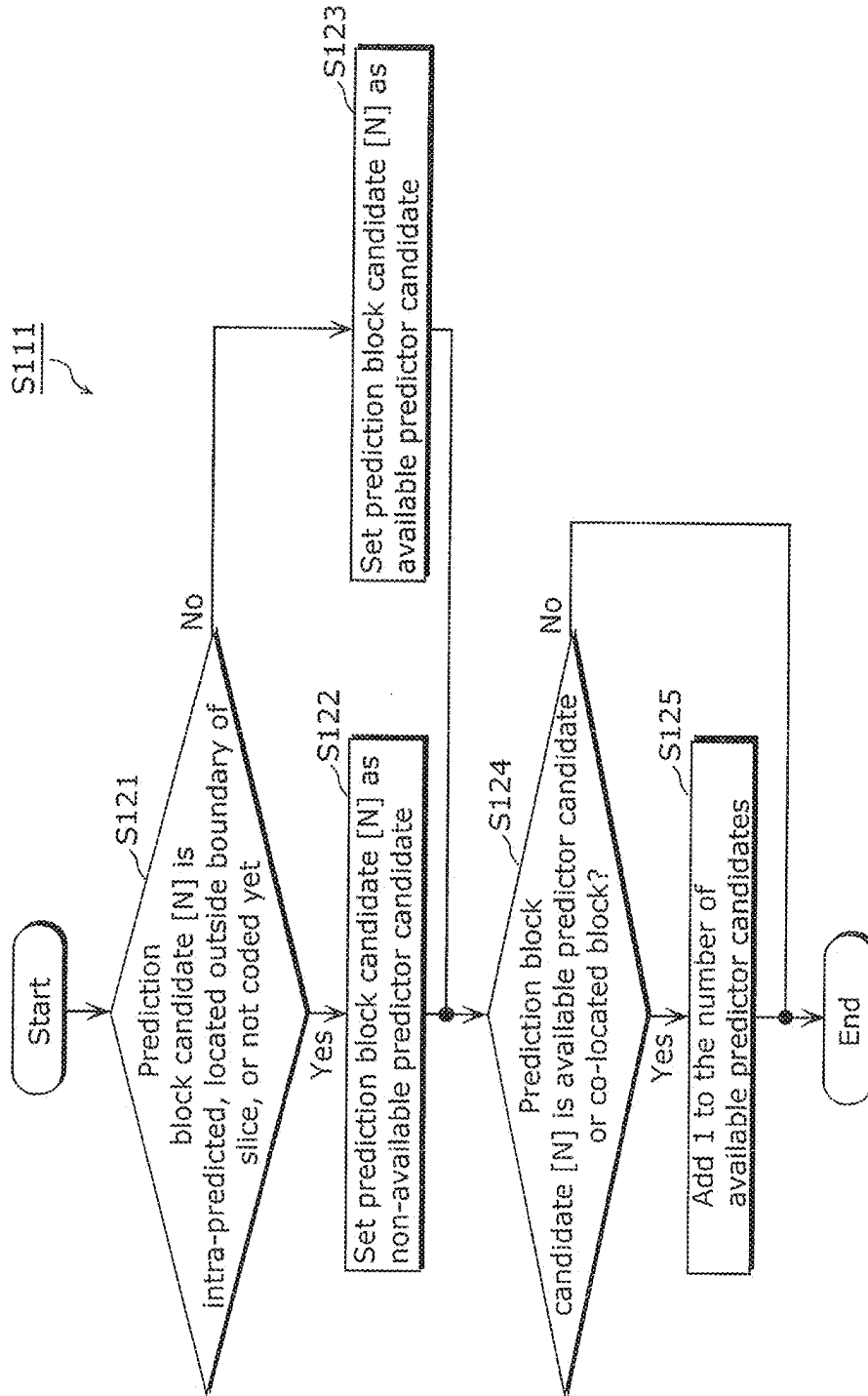


FIG. 19

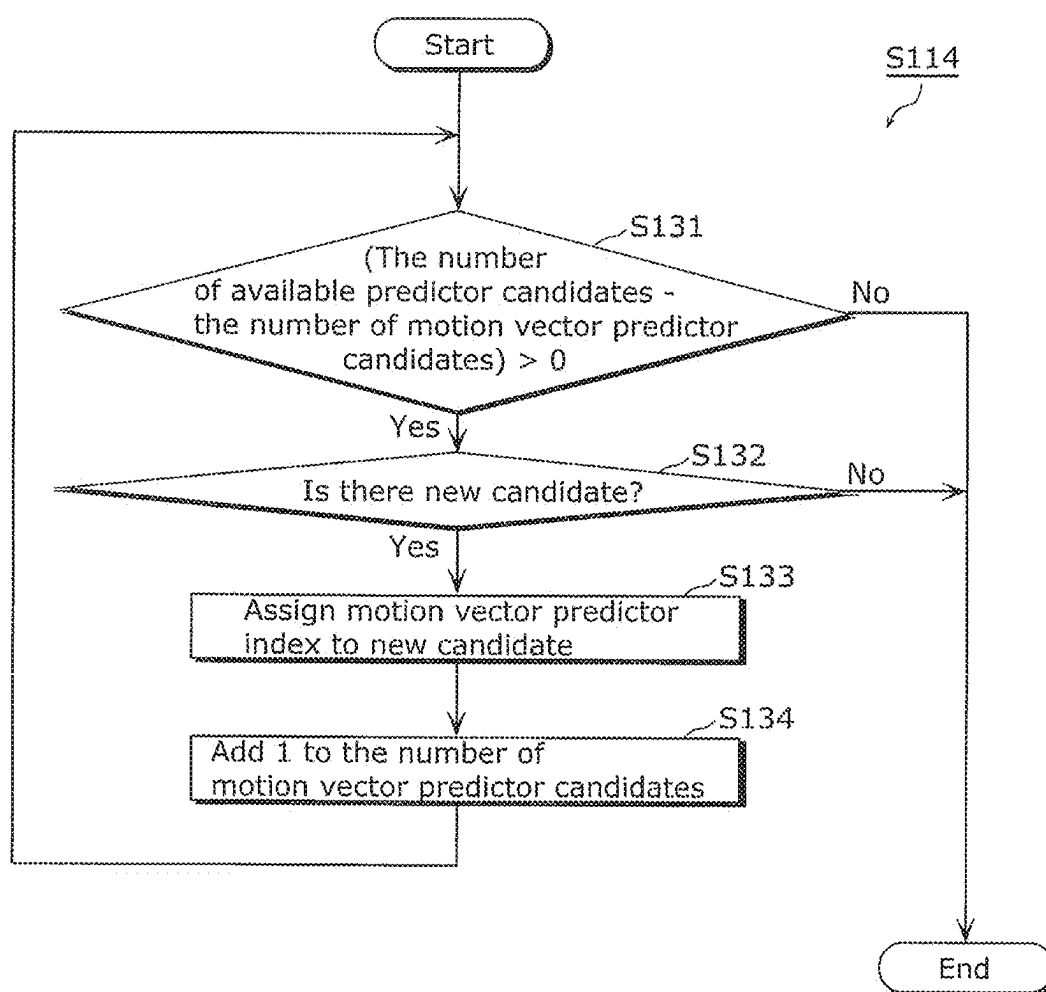


FIG. 20

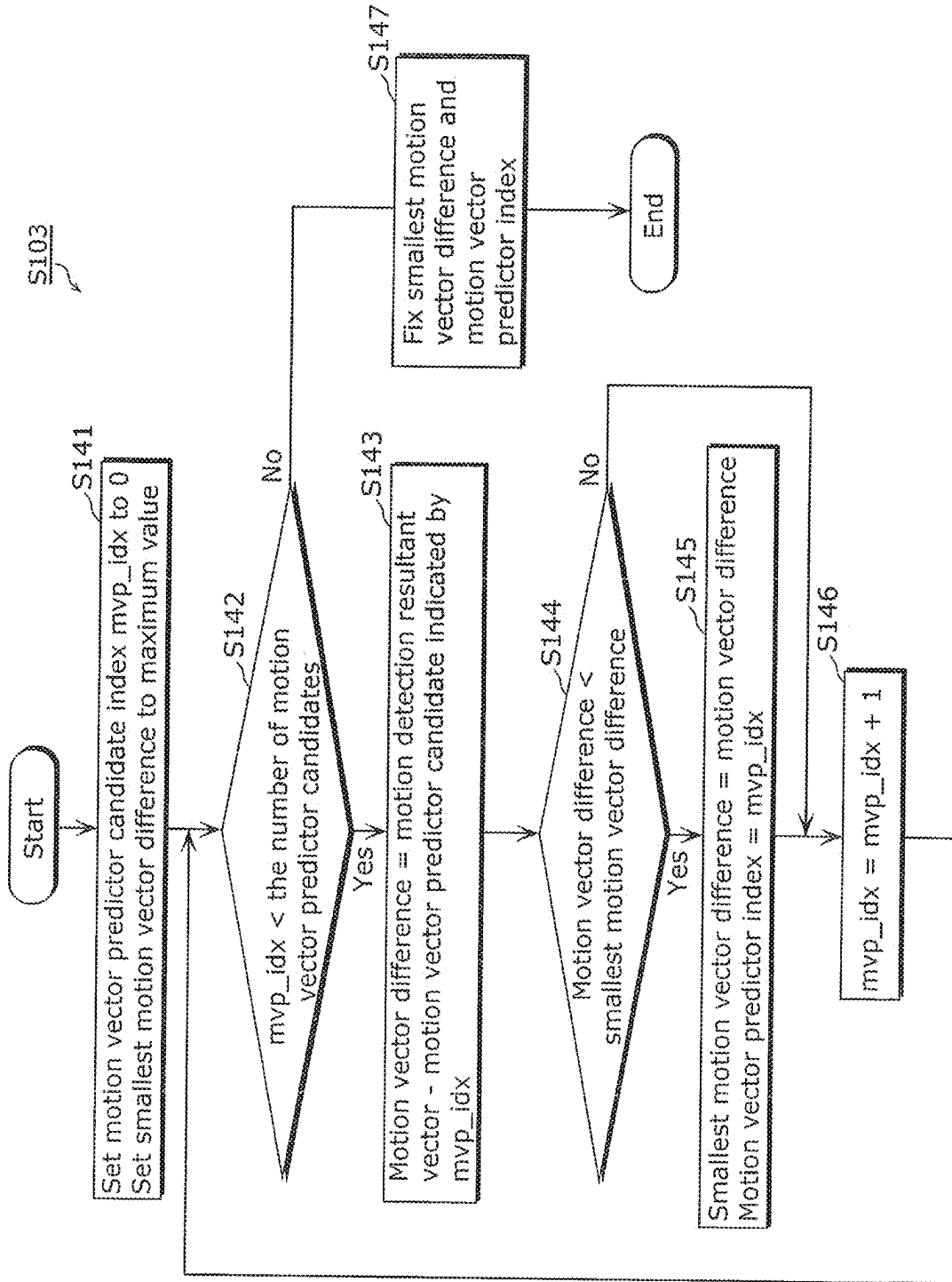


FIG. 21

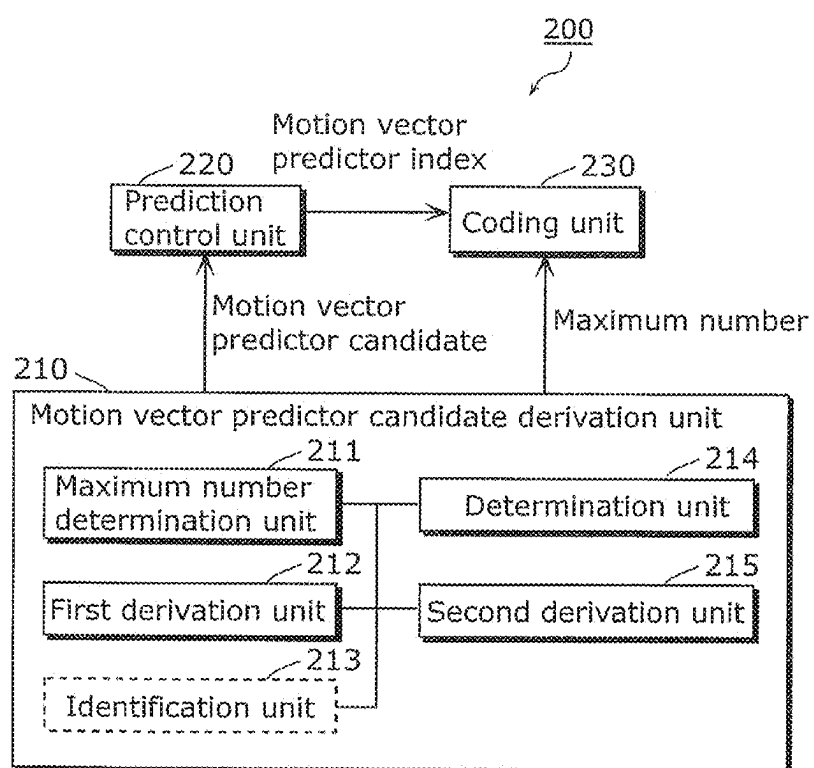


FIG. 22

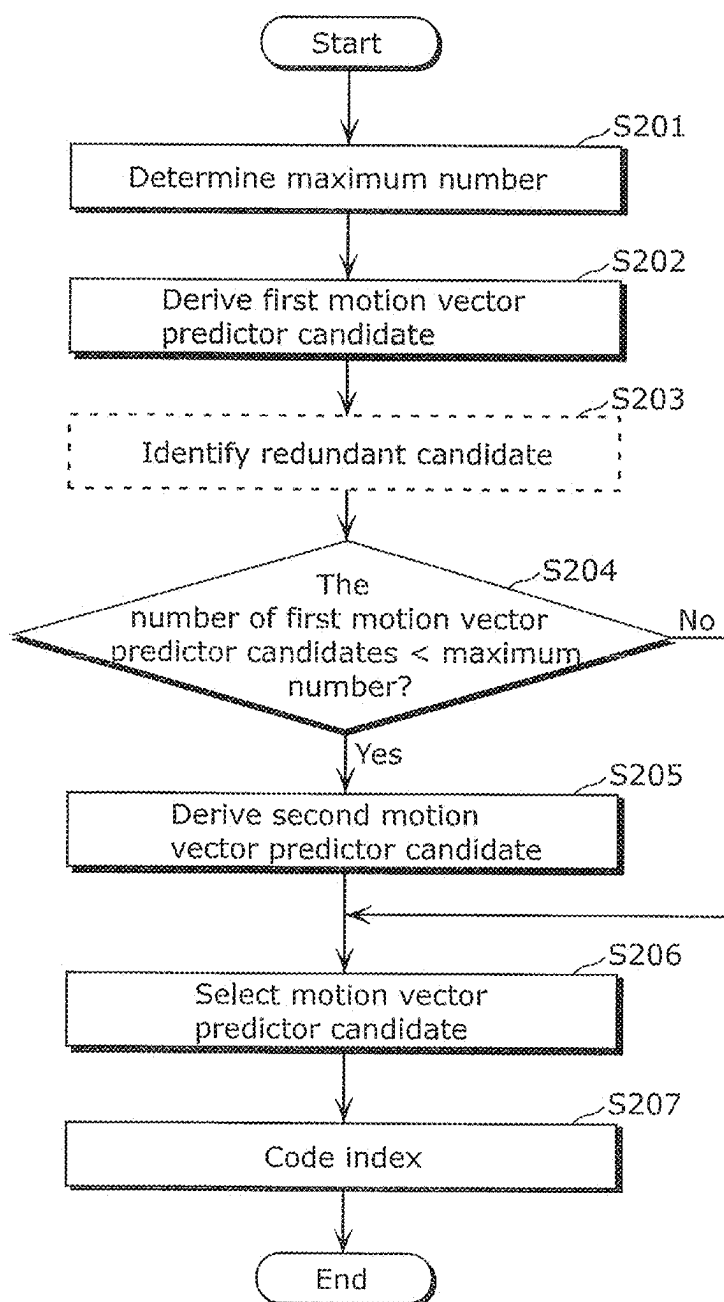


FIG. 23

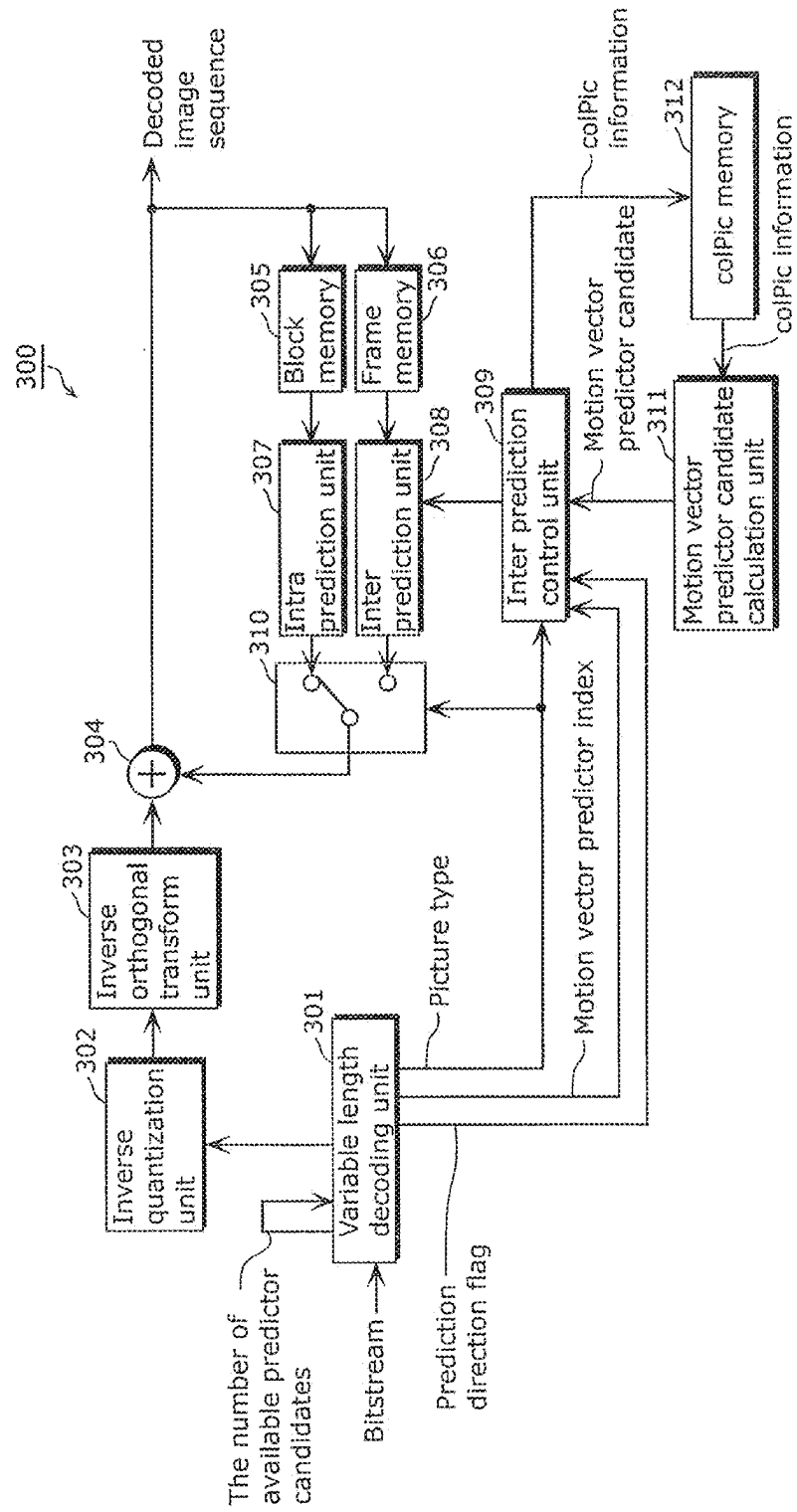


FIG. 24

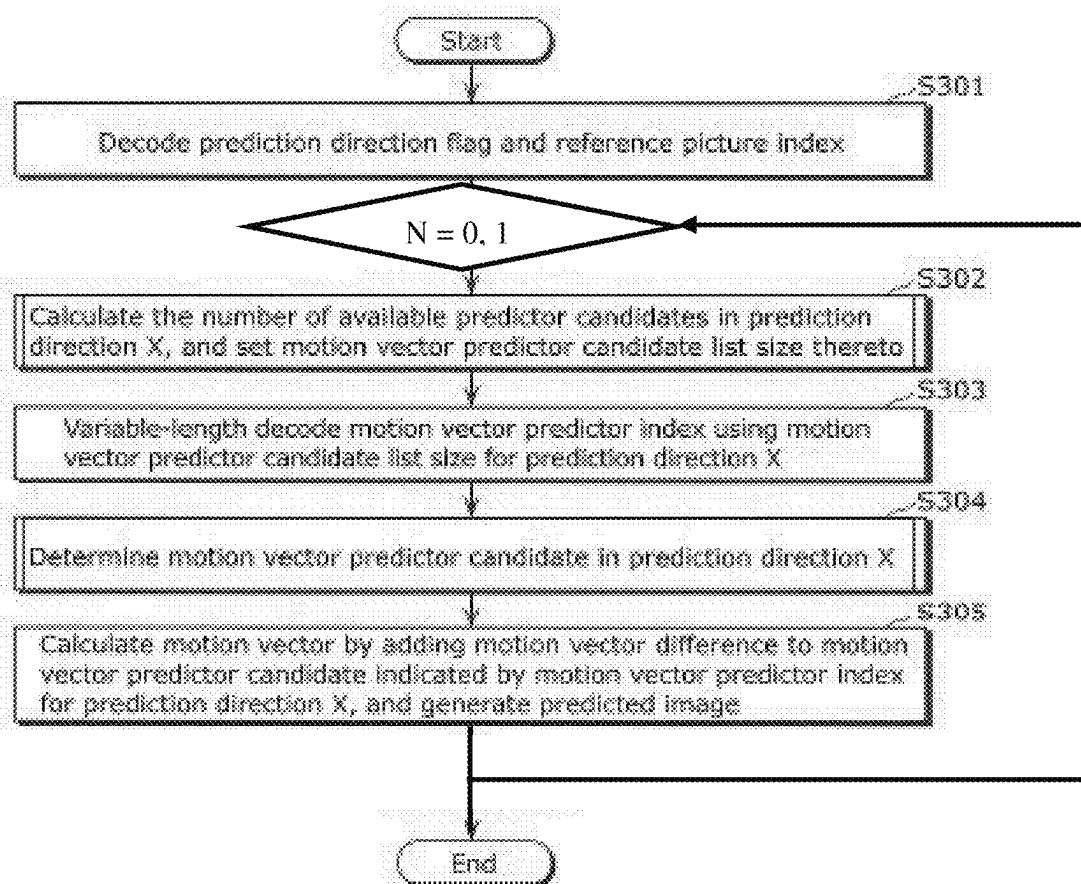


FIG. 25

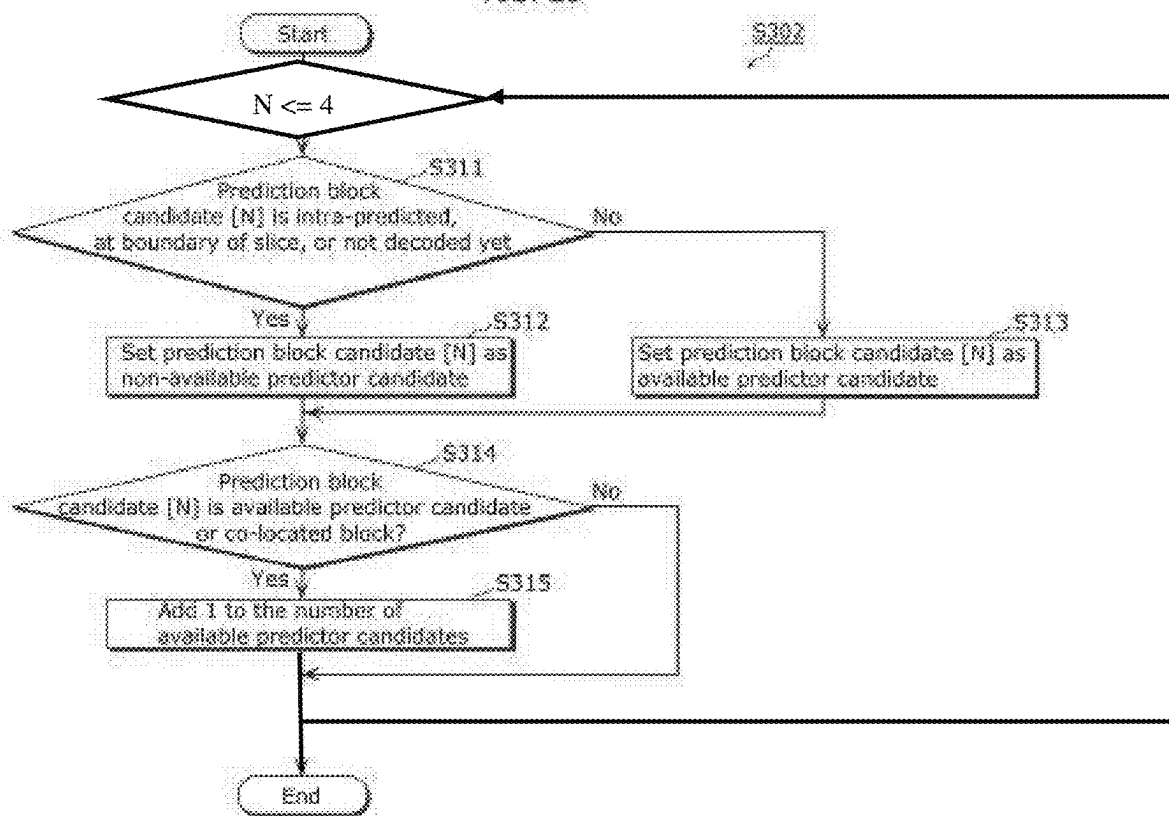


FIG. 26

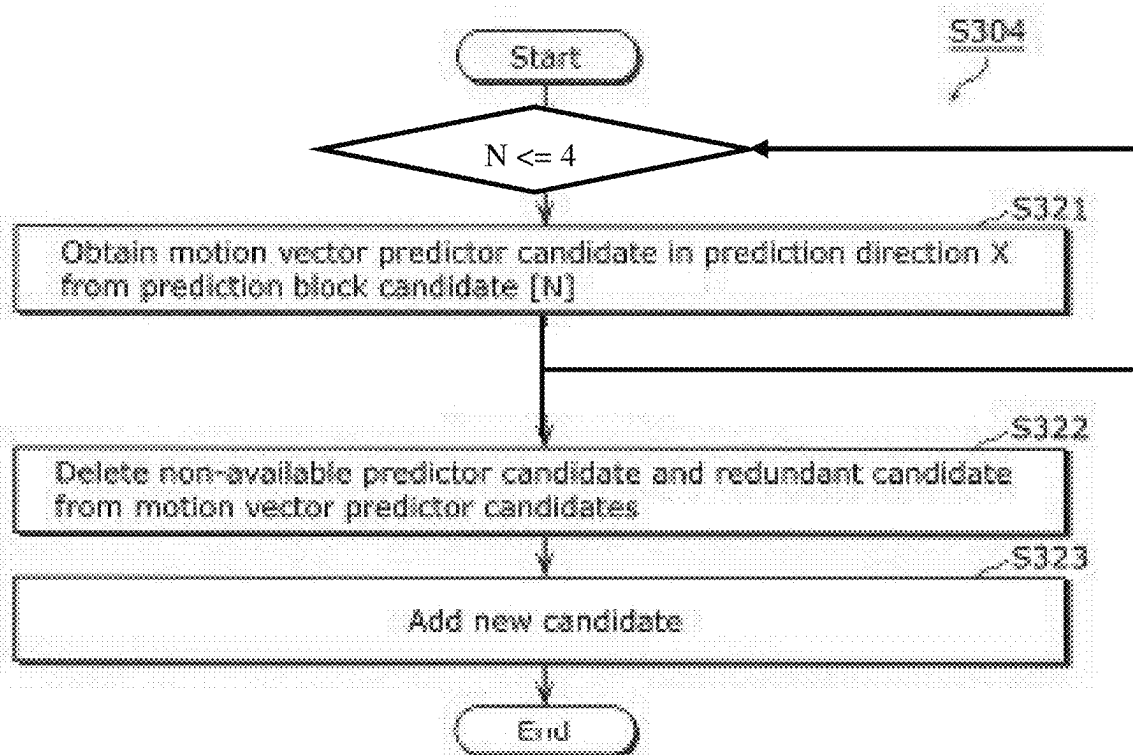


FIG. 27

	prediction_unit(x0, y0, log2PUWidth, log2PUHeight, PartIdx, InferredMergeFlag){	Descriptor
	if(skip_flag[x0][y0]){	
	...	
	} else if(PredMode == MODE_INTRA){	
	...	
Prediction direction flag	} else { /* MODE_INTER */	
	if(!InferredMergeFlag)	
Motion vector predictor candidate list size = the number of available predictor candidates	merge_flag[x0][y0]	u(l) ae(v)
	if(merge_flag[x0][y0] && NumMergeCand > 1){	
	...	
	} else {	
	if(slice_type == B)	
	inter_pred_flag[x0][y0]	ue(v) ae(v)
	if(inter_pred_flag[x0][y0] == Pred_LC){	
	...	
Motion vector predictor index	if(NumMVPCand(LcToLx) > 1)	
	mvp_idx_lc[x0][y0]	ue(v) ae(v)
	}	
Motion vector predictor candidate list size = the number of available predictor candidates	else { /* Pred_L0 or Pred_BI */	
	if(num_ref_idx_l0_active_minus1 > 0)	
	...	
	if(NumMVPCand(L0) > 1)	
	mvp_idx_l0[x0][y0]	ue(v) ae(v)
	}	
	if(inter_pred_flag[x0][y0] == Pred_BI){	
	...	
Motion vector predictor index	if(NumMVPCand(L1) > 1)	
	mvp_idx_l1[x0][y0]	ue(v) ae(v)
	}	
	}	
Motion vector predictor candidate list size = the number of available predictor candidates	}	
	}	
Motion vector predictor index		

FIG. 28

	prediction_unit(x0, y0, log2PUWidth, log2PUHeight, PartIdx, InferredMergeFlag){	Descriptor
	if(skip_flag[x0][y0]){	
	...	
	} else if(PredMode == MODE_INTRA){	
	...	
	} else { /* MODE_INTER */	
	if(!InferredMergeFlag)	
Prediction direction flag	merge_flag[x0][y0]	u(l) ae(v)
	if(merge_flag[x0][y0] && NumMergeCand > 1){	
	...	
	} else {	
	if(slice_type == B)	
	inter_pred_flag[x0][y0]	ue(v) ae(v)
	if(inter_pred_flag[x0][y0] == Pred_LC){	
	...	
	mvp_idx_lc[x0][y0]	ue(v) ae(v)
Motion vector predictor index	}	
	else { /* Pred_L0 or Pred_BI */	
	if(num_ref_idx_l0_active_minus1 > 0)	
	...	
	mvp_idx_l0[x0][y0]	ue(v) ae(v)
Motion vector predictor index	}	
	if(inter_pred_flag[x0][y0] == Pred_BI){	
	...	
	mvp_idx_l1[x0][y0]	ue(v) ae(v)
Motion vector predictor index	}	
	}	
	}	
	}	

FIG. 29

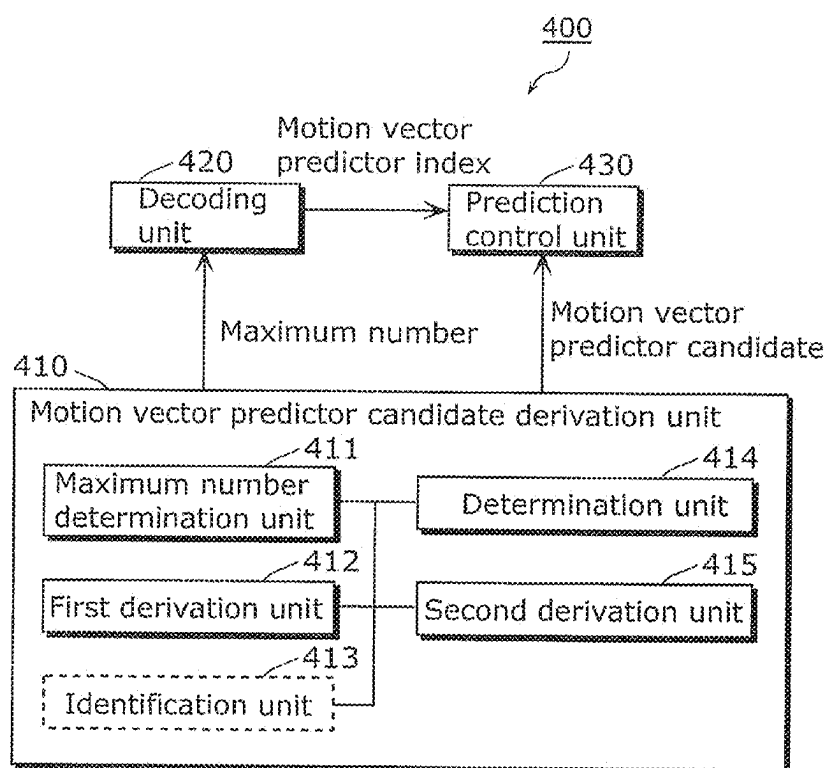
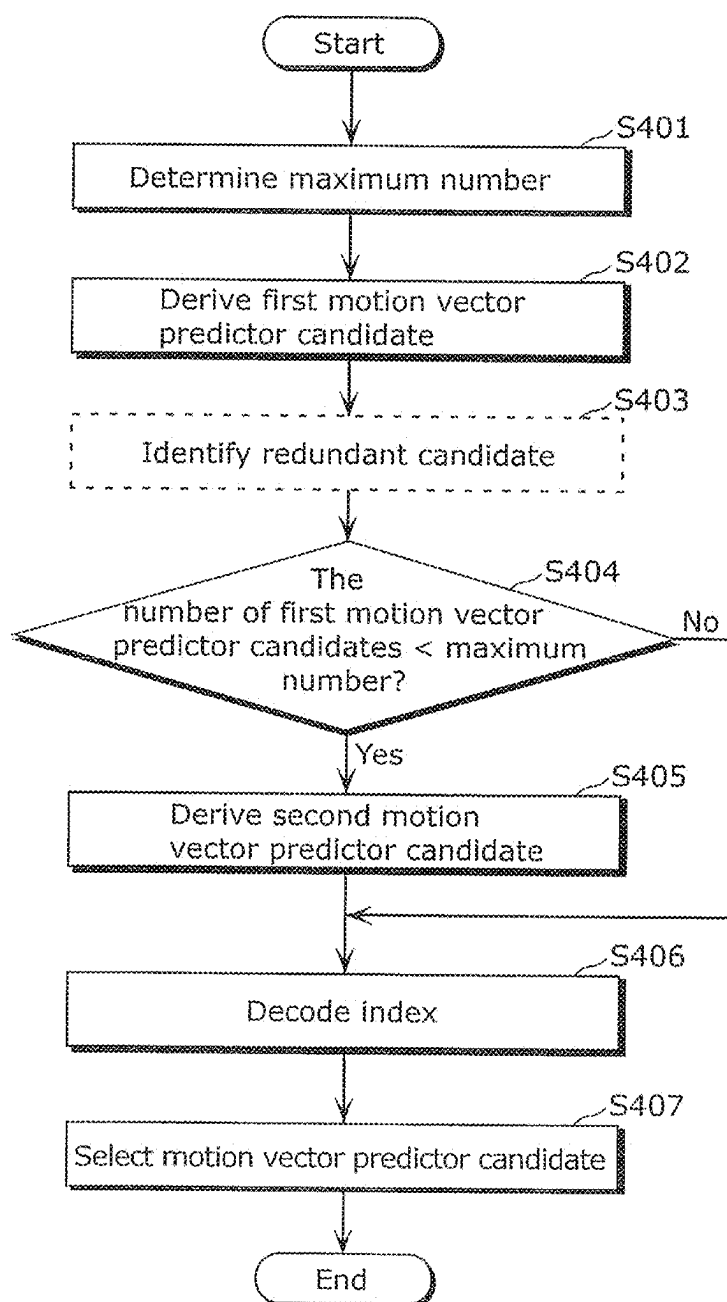


FIG. 30



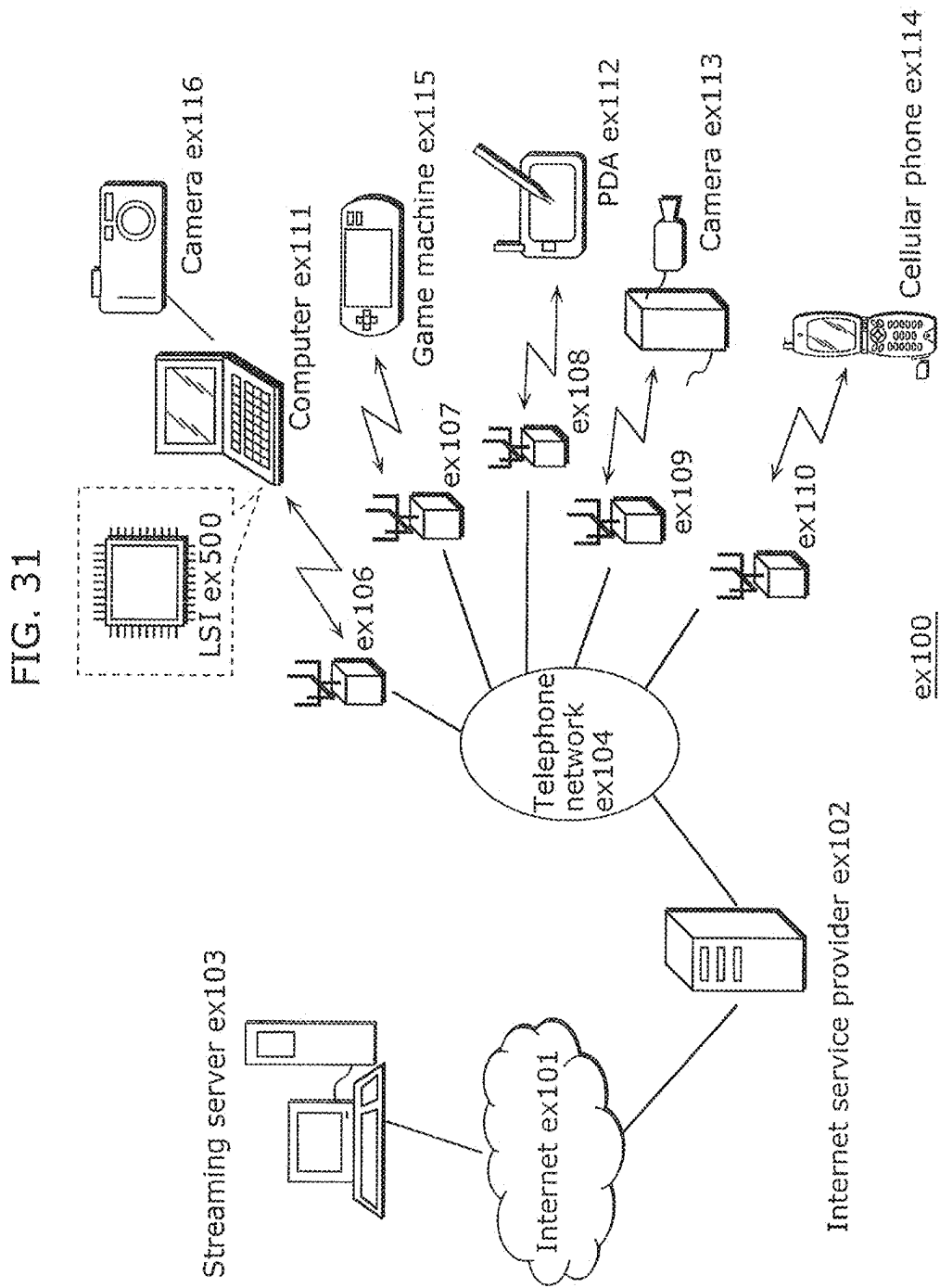


FIG. 32

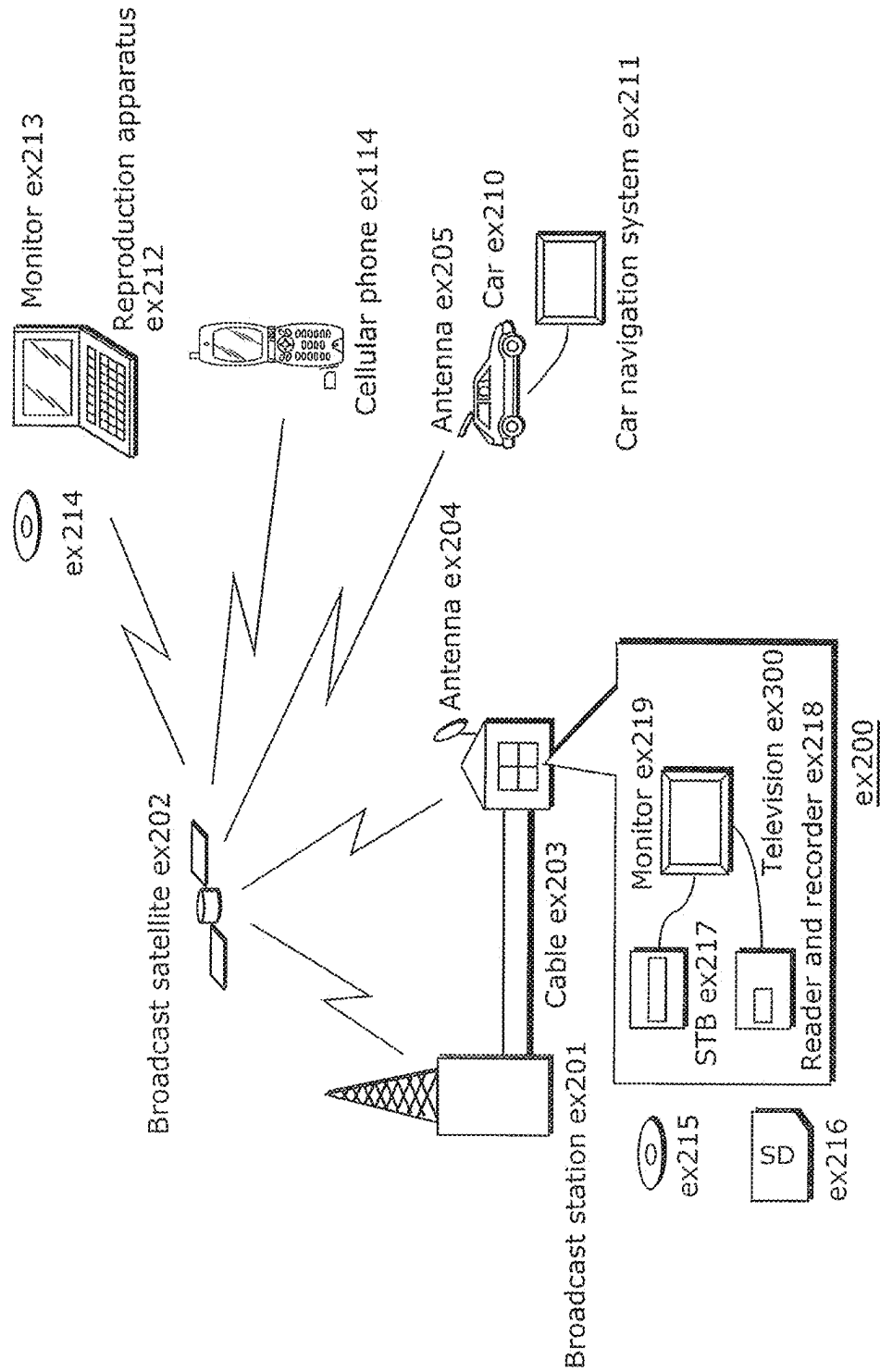


FIG. 33

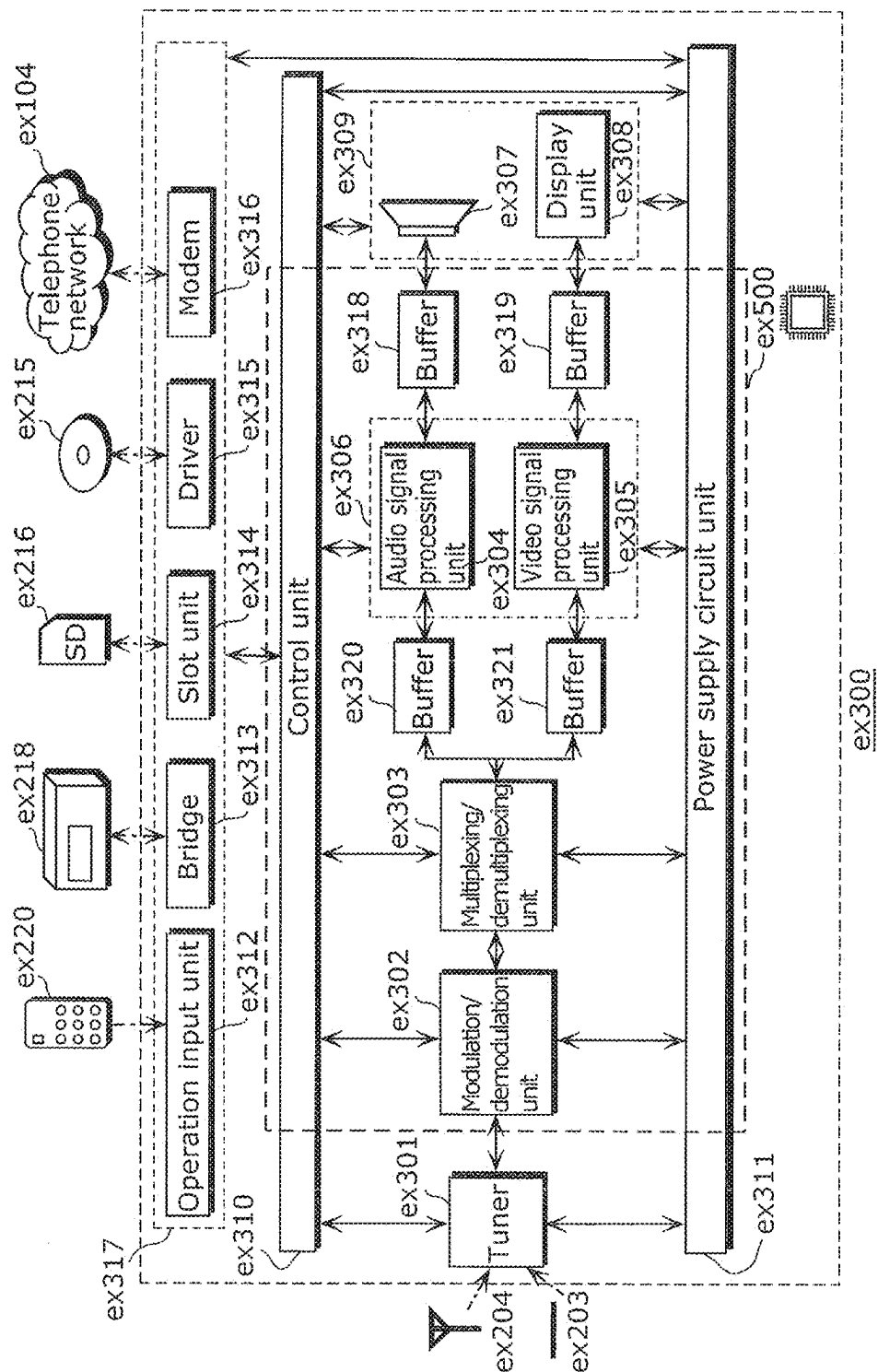


FIG. 34

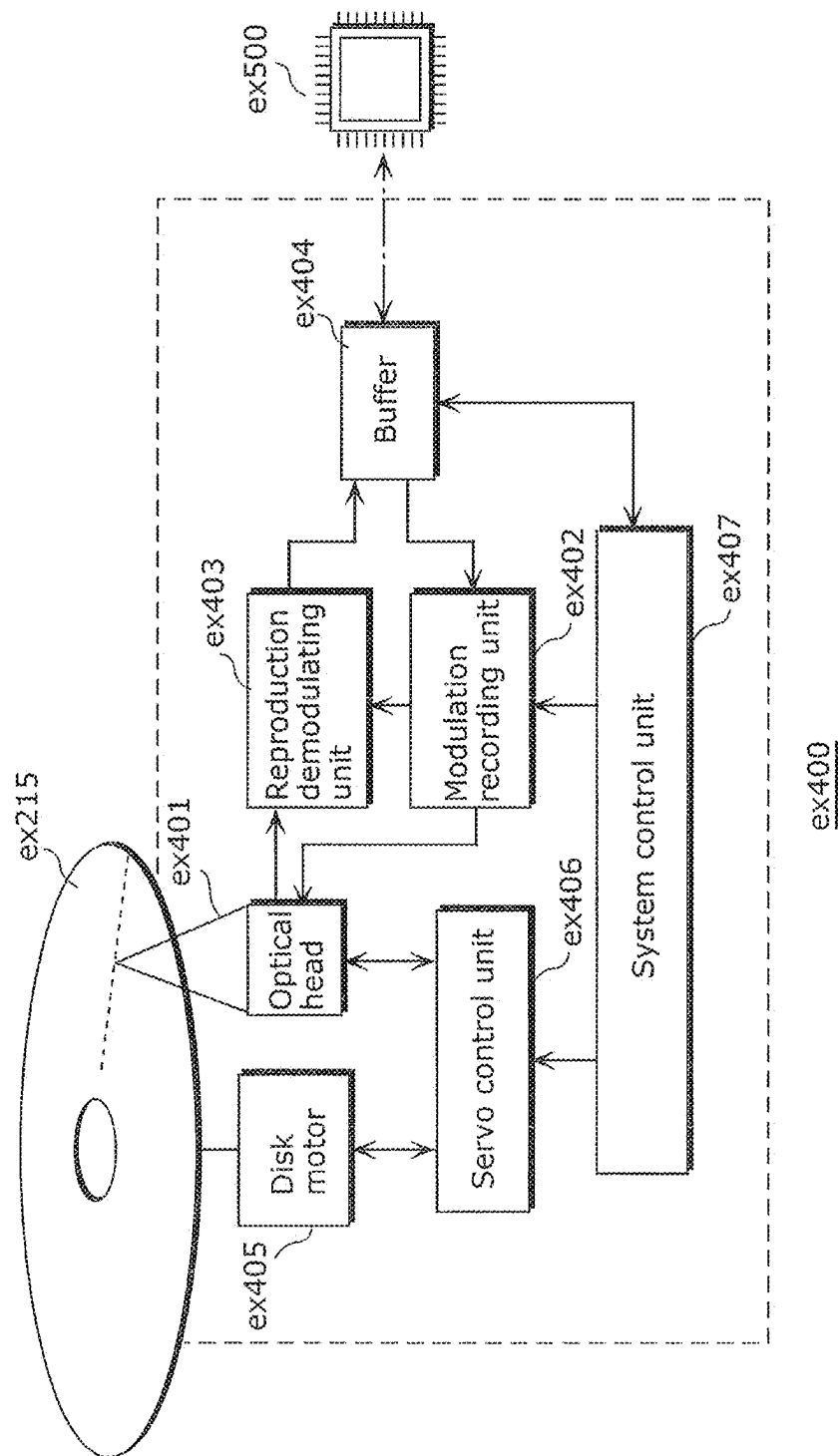


FIG. 35

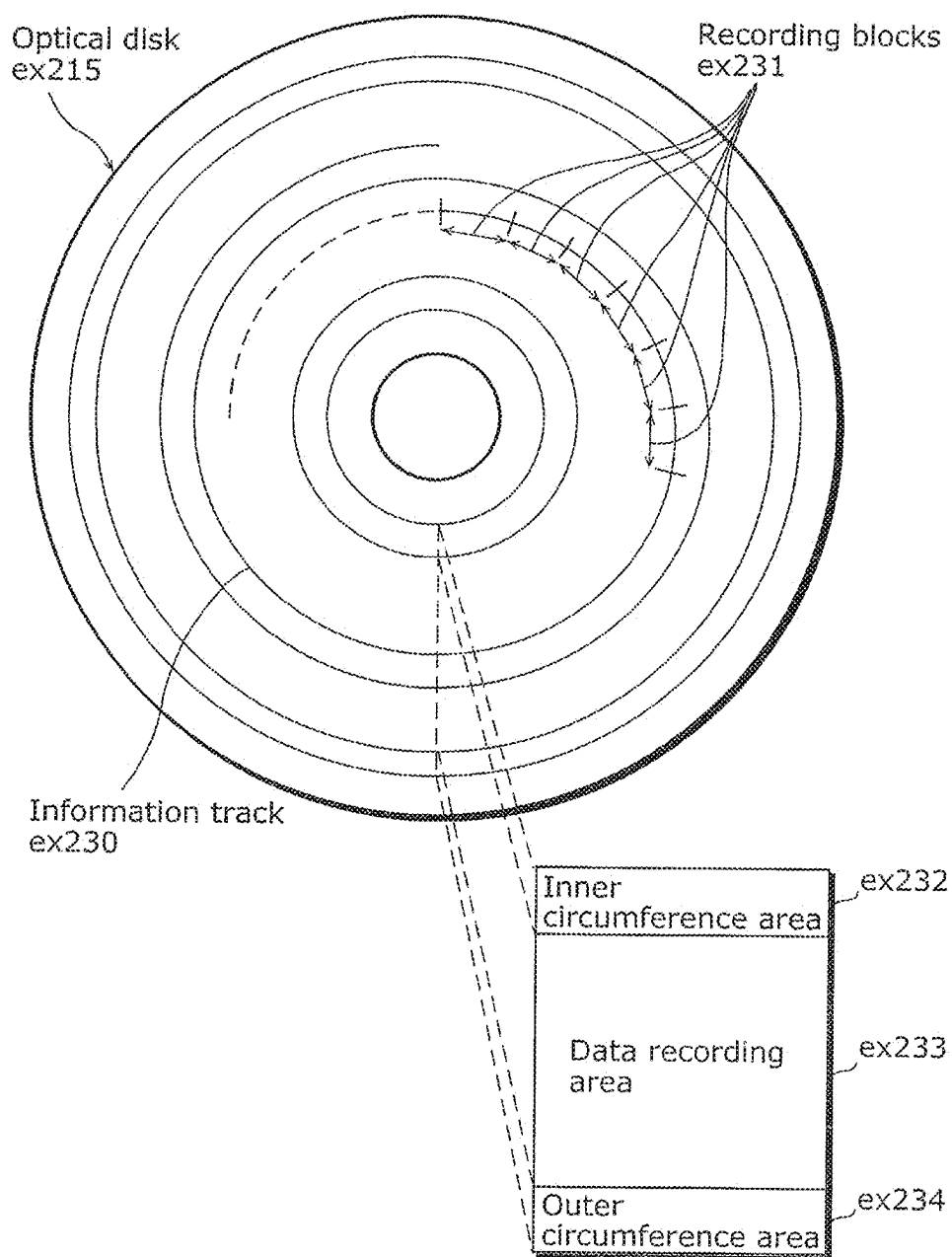


FIG. 36A

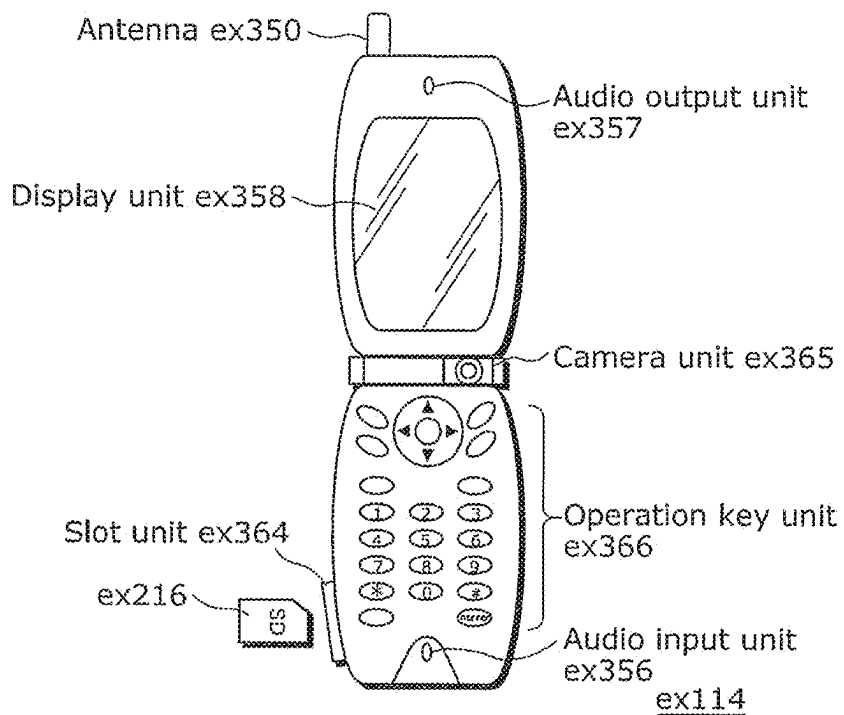


FIG. 36B

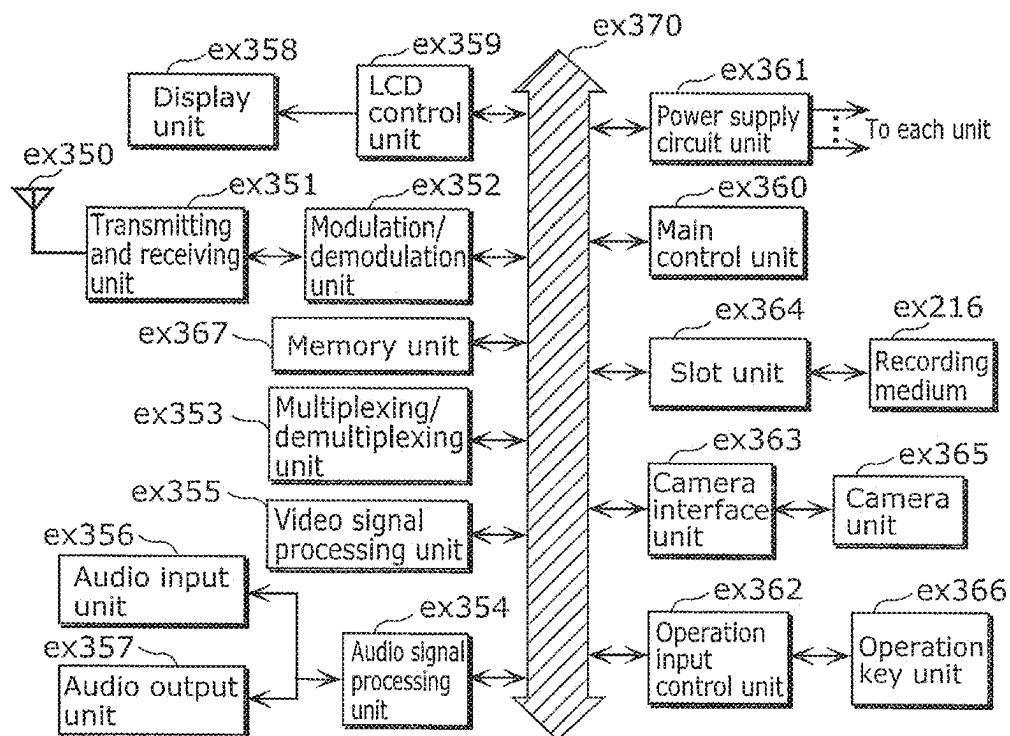


FIG. 37

Video stream (PID=0x1011, Primary video)
Audio stream (PID=0x1100)
Audio stream (PID=0x1101)
Presentation graphics stream (PID=0x1200)
Presentation graphics stream (PID=0x1201)
Interactive graphics stream (PID=0x1400)
Video stream (PID=0x1B00, Secondary video)
Video stream (PID=0x1B01, Secondary video)

FIG. 38

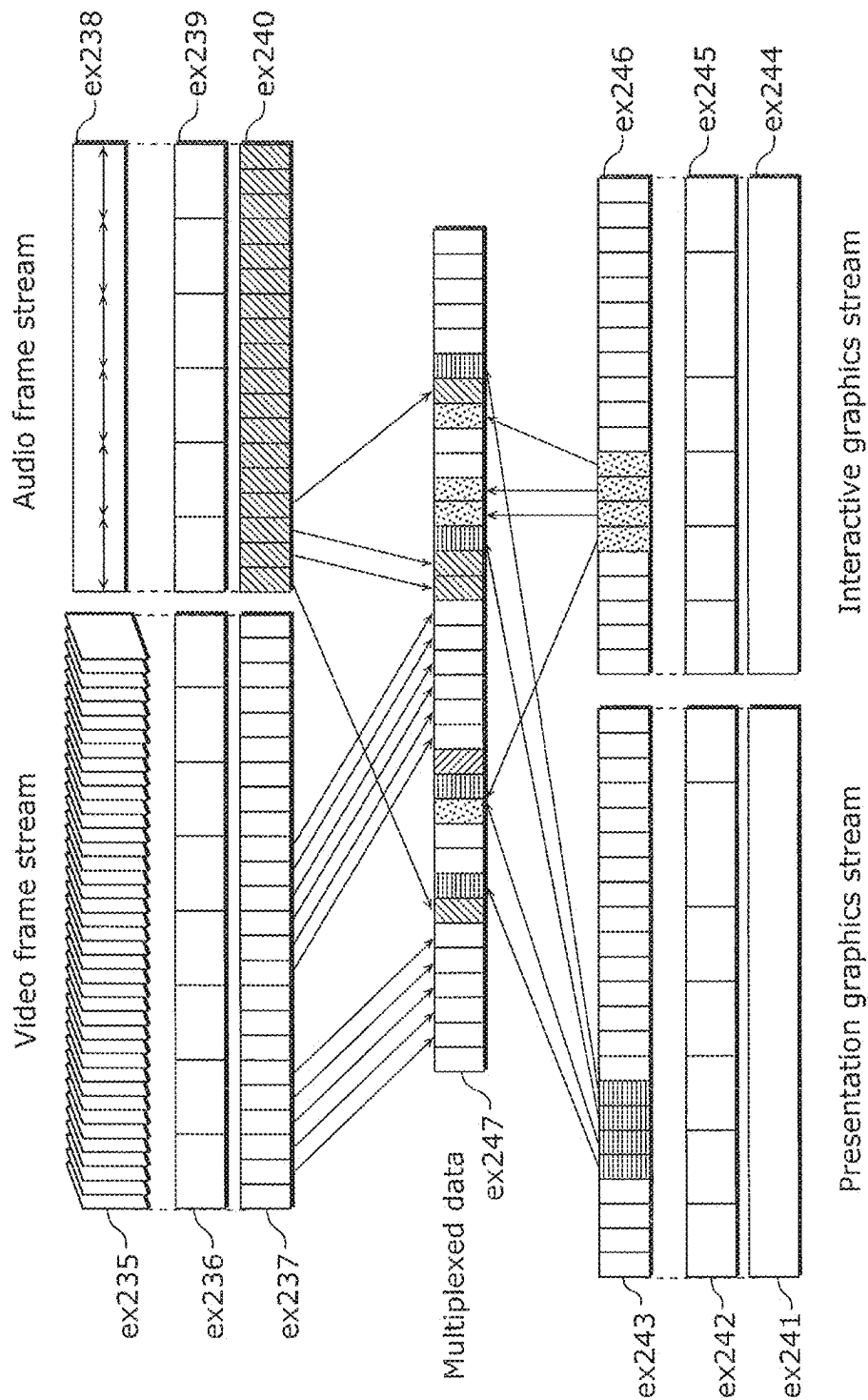


FIG. 39

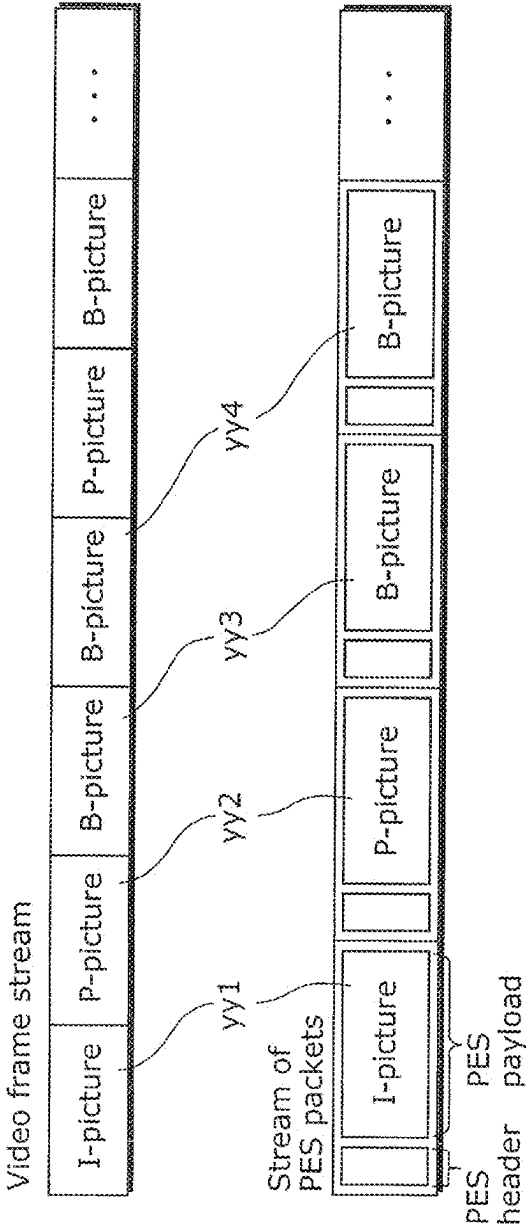


FIG. 40

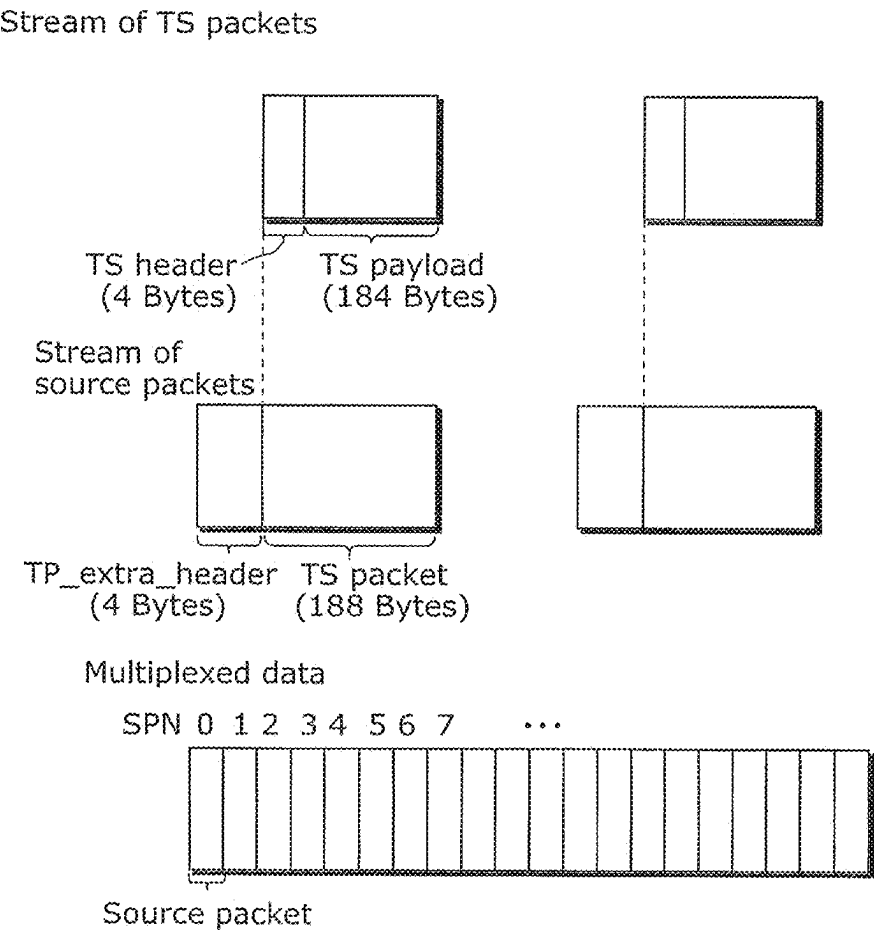


FIG. 41

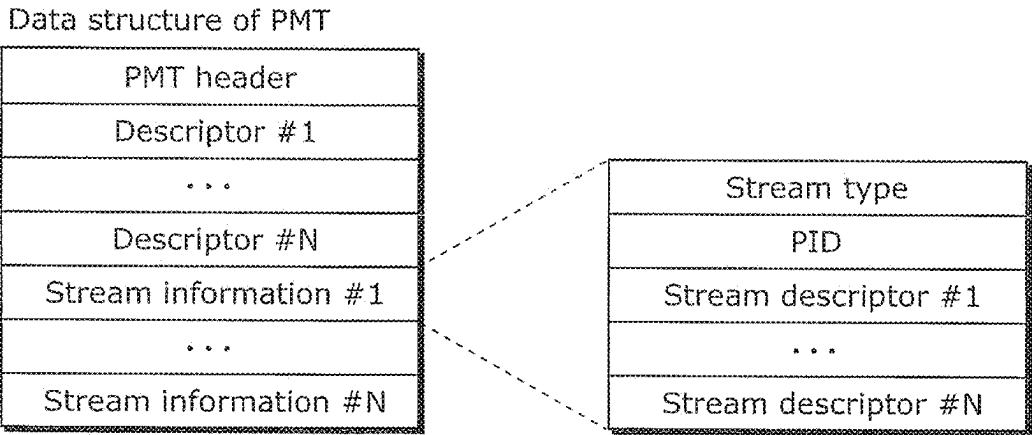


FIG. 42

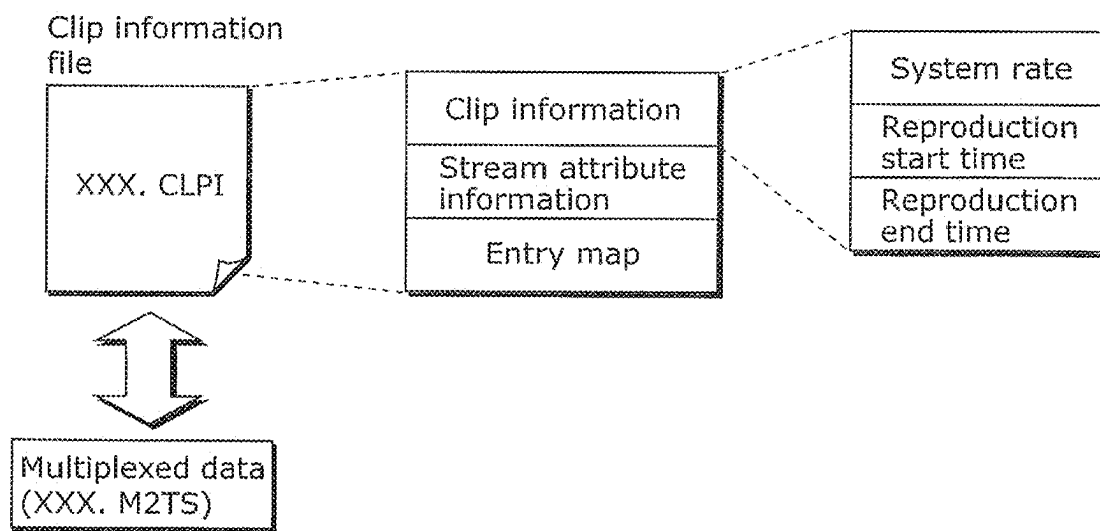


FIG. 43

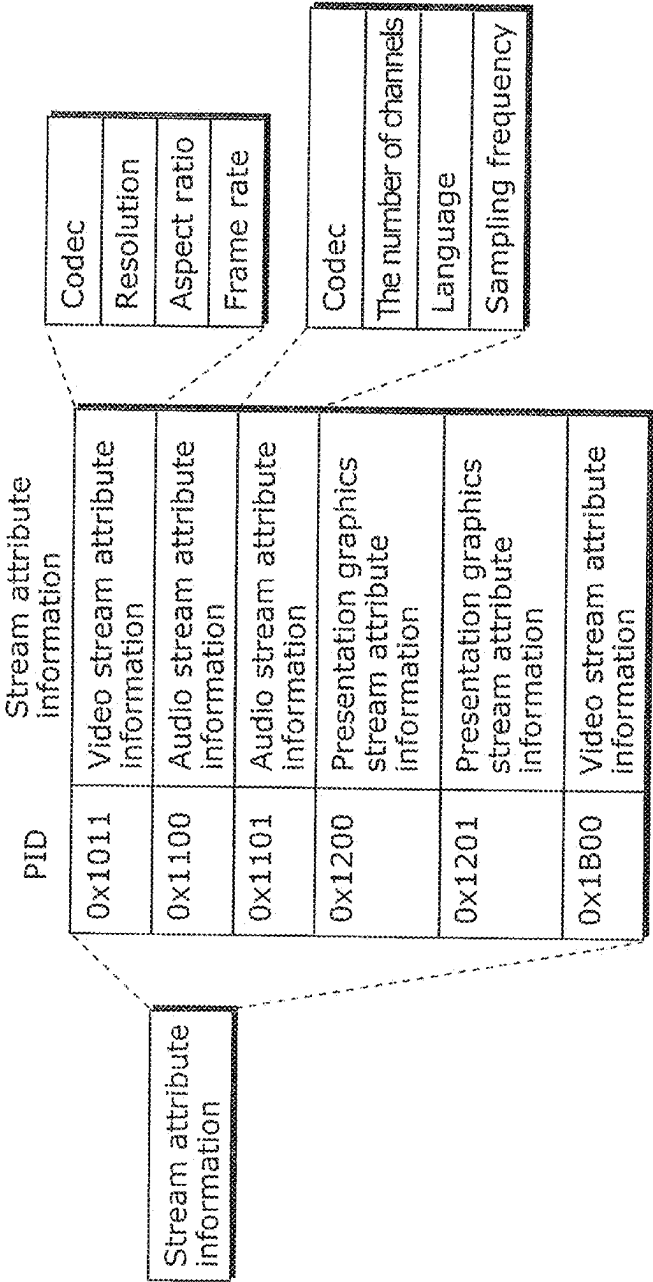


FIG. 44

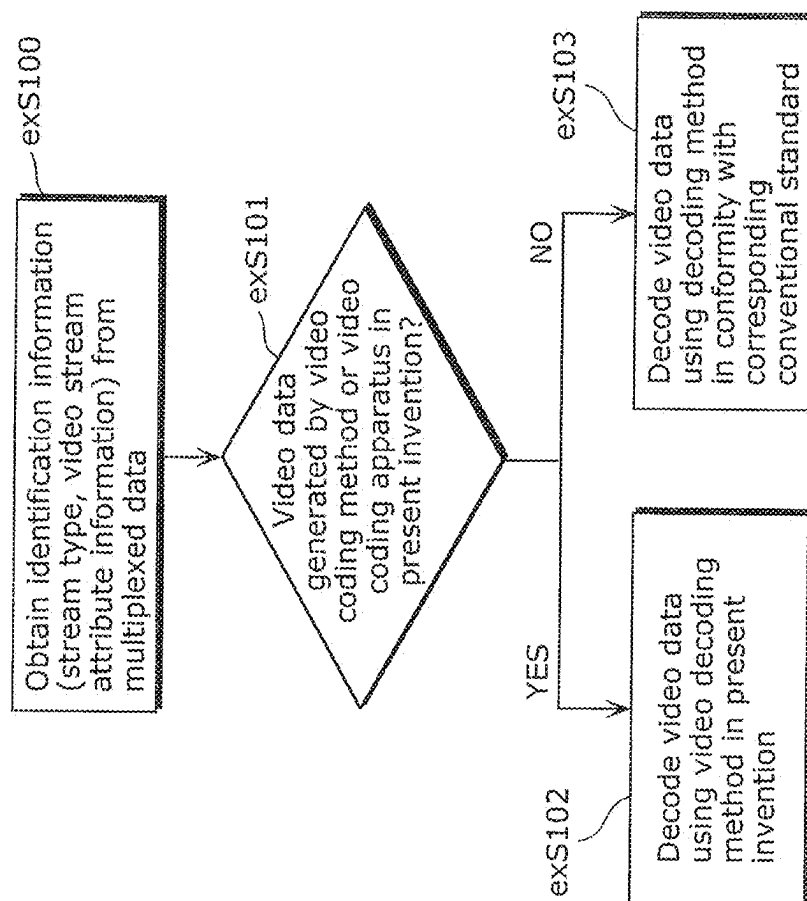


FIG. 45

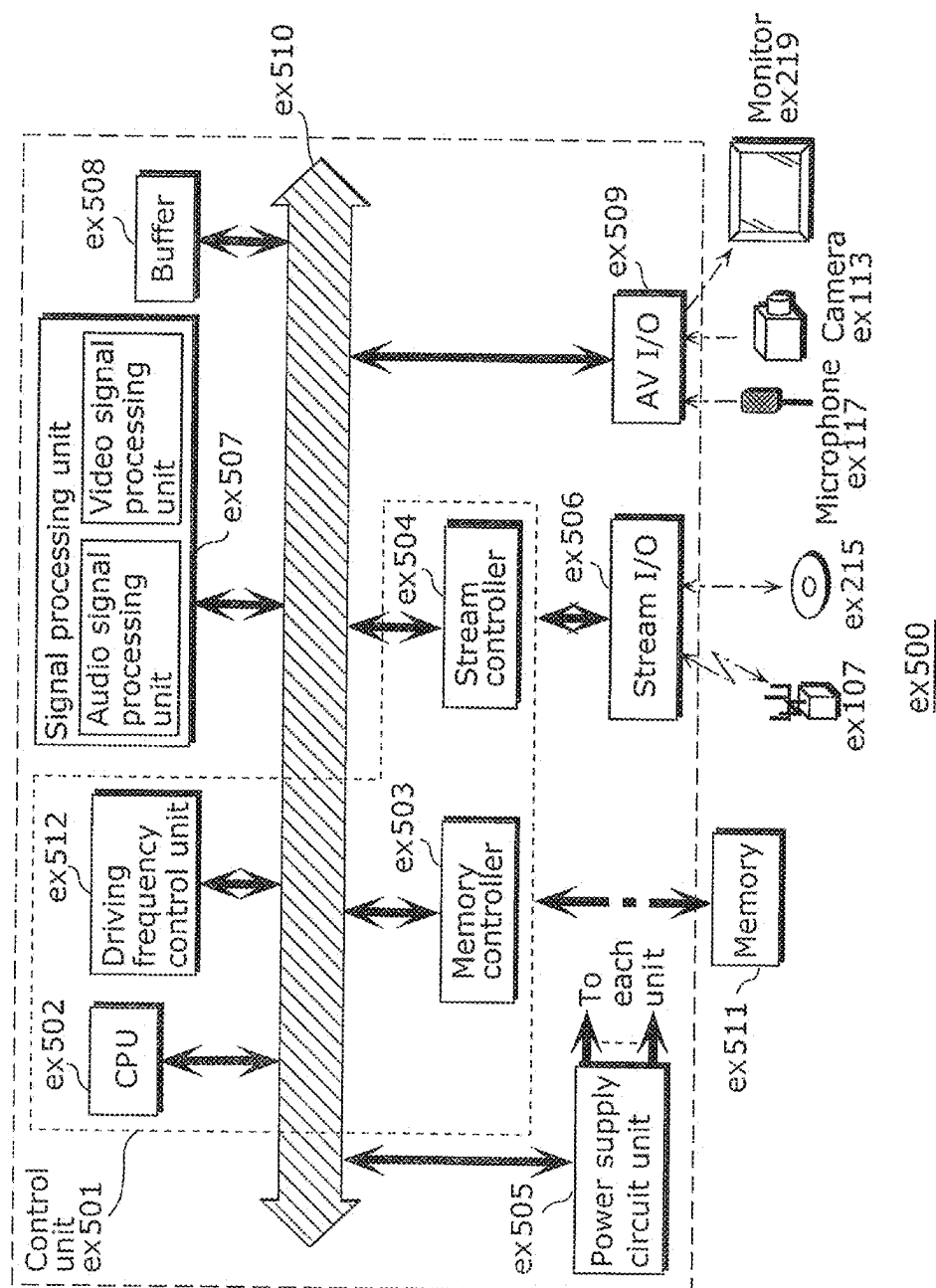


FIG. 46

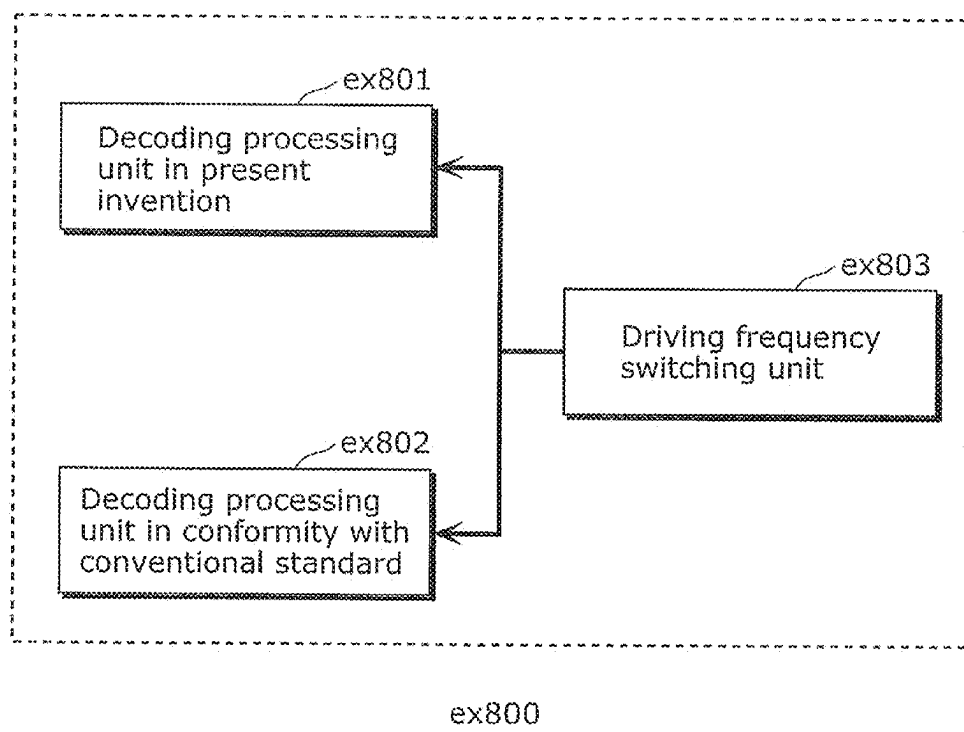


FIG. 47

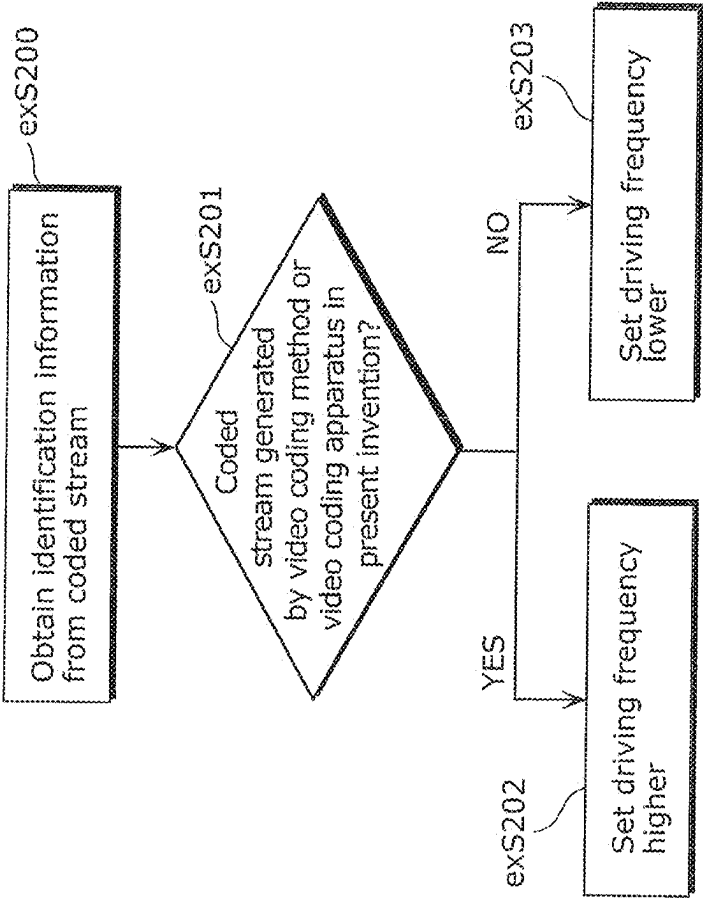
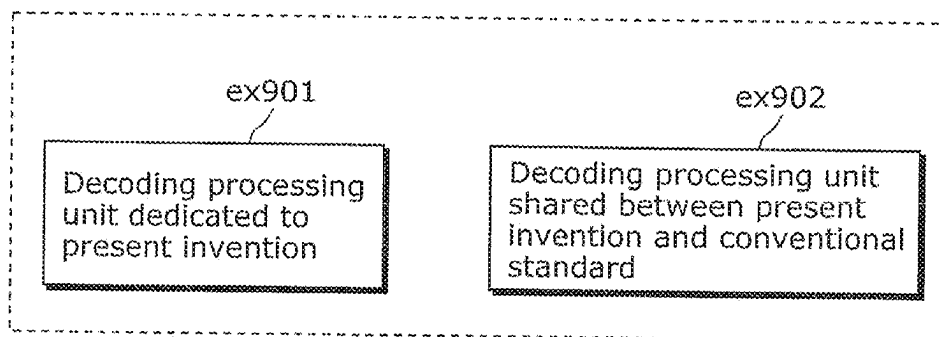


FIG. 48

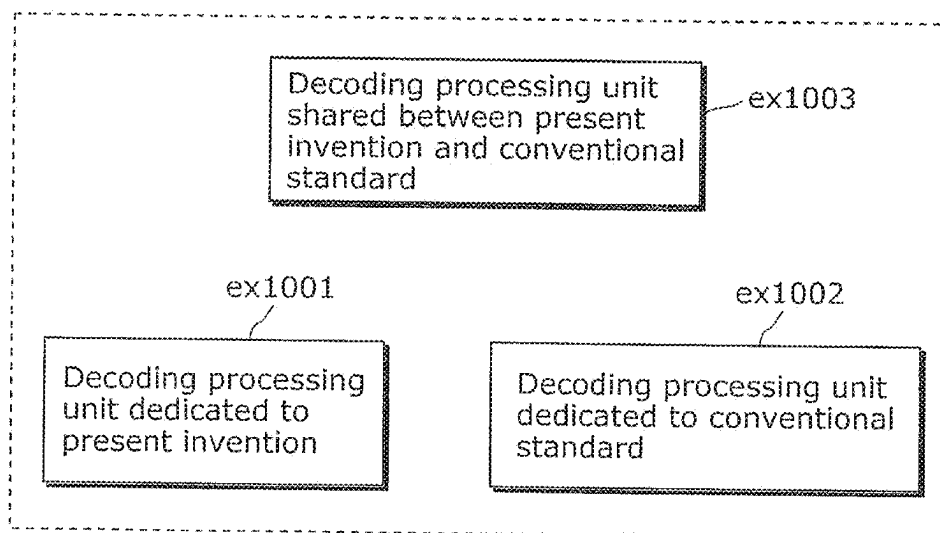
Corresponding standard	Driving frequency
MPEG-4 AVC	500 MHz
MPEG-2	350 MHz
⋮	⋮

FIG. 49A



ex900

FIG. 49B



ex1000

CODING METHOD AND APPARATUS WITH CANDIDATE MOTION VECTORS

CROSS REFERENCE TO RELATED APPLICATION

[0001] This is a continuation of U.S. patent application Ser. No. 18/394,461, filed Dec. 22, 2023, which is a continuation of U.S. patent application Ser. No. 18/087,516, filed Dec. 22, 2022 and now U.S. Pat. No. 11,895,324 issued Feb. 6, 2024, which is a continuation of U.S. patent application Ser. No. 17/339,118, filed Jun. 4, 2021 and now U.S. Pat. No. 11,575,930 issued Feb. 7, 2023, which is a continuation of U.S. patent application Ser. No. 16/224,209, filed Dec. 18, 2018 and now U.S. Pat. No. 11,076,170 issued Jul. 27, 2021, which is a continuation U.S. patent application Ser. No. 15/629,101, filed Jun. 21, 2017 and now U.S. Pat. No. 10,212,450 issued Feb. 19, 2019, which is a continuation of U.S. patent application Ser. No. 15/266,004, filed Sep. 15, 2016 and now U.S. Pat. No. 9,723,322 issued Aug. 1, 2017, which is a continuation of U.S. patent application Ser. No. 13/479,636, filed May 24, 2012 and now U.S. Pat. No. 9,485,518 issued Nov. 1, 2016, which claims the benefit of U.S. Prov. Pat. Appl. No. 61/490,747, filed May 27, 2011. The entire disclosure of each of the above-identified documents, including the specification, drawings, and claims, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to a moving picture coding method and a moving picture decoding method.

BACKGROUND ART

[0003] In moving picture coding processing, in general, the amount of information is reduced by utilizing redundancy in the spatial direction and the temporal direction which moving pictures have. Here, in general, transform to a frequency domain is used as a method utilizing redundancy in the spatial direction. Further, inter-picture prediction (hereinafter, referred to as “inter prediction”) coding processing is used as a method utilizing redundancy in the temporal direction. In inter prediction coding processing, when a picture is coded, a coded picture that appears before or after a current picture to be coded in the display time order is used as a reference picture. A motion vector is derived by performing motion detection on the current picture relative to the reference picture. Then, redundancy in the temporal direction is eliminated by calculating a difference between image data of the current picture and predicted image data obtained by performing motion compensation based on the derived motion vector (for example, see Non Patent Literature (NPL) 1). Here, in motion detection, a difference value between a current block in a current picture to be coded and a block in a reference picture is calculated, and a block in the reference picture with which the smallest difference value is obtained is determined as a reference block. Then, a motion vector is detected using the current block and the reference block.

CITATION LIST

Non Patent Literature

[0004] [NPL 1] ITU-T Recommendation H.264, “Advanced video coding for generic audiovisual services”, March, 2010

[0005] [NPL 2] JCT-VC, “WD3: Working Draft 3 of High-Efficiency Video Coding”, JCTVC-E603, March 2011

SUMMARY OF INVENTION

Technical Problem

[0006] However, there is a demand for the above conventional technique to achieve an improvement in error resistance in coding and decoding a moving picture using inter prediction.

[0007] In view of this, an object of the present invention is to provide a moving picture coding method and a moving picture decoding method which improves error resistance in coding and decoding a moving picture using inter prediction.

Solution to Problem

[0008] A moving picture coding method according to an aspect of the present invention is a moving picture coding method for calculating a motion vector predictor to be used when coding a motion vector of a current block to be coded, and coding the current block, to generate a bitstream, the method including: determining a maximum number of motion vector predictor candidates each of which is a candidate for the motion vector predictor; deriving one or more first motion vector predictor candidates; determining whether a total number of the one or more first motion vector predictor candidates is smaller than the maximum number; deriving one or more second motion vector predictor candidates when it is determined that the total number of one or more first motion vector predictor candidates is smaller than the maximum number; selecting, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates, the motion vector predictor to be used for coding the motion vector of the current block; and coding, using the determined maximum number, an index for identifying the selected motion vector predictor, and adding the coded index to the bitstream.

[0009] It should be noted that these general and specific aspects may be implemented using a system, a method, an integrated circuit, a computer program, a computer-readable recording medium such as a compact disc read only memory (CD-ROM), or any combination of systems, methods, integrated circuits, computer programs or recording media.

Advantageous Effects of Invention

[0010] According to an aspect of the present invention, it is possible to improve error resistance in coding and decoding a moving picture using inter prediction.

BRIEF DESCRIPTION OF DRAWINGS

[0011] These and other objects, advantages and features of the invention will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present invention.

[0012] In the Drawings:

[0013] FIG. 1A is a diagram for describing an example of a reference picture list for a B-picture;

[0014] FIG. 1B shows an example of a reference picture list for a prediction direction 0 for a B-picture;

[0015] FIG. 1C shows an example of a reference picture list for a prediction direction 1 for a B-picture;

[0016] FIG. 2 is a diagram for describing motion vectors in a temporal motion vector prediction mode;

[0017] FIG. 3 shows examples of motion vectors of adjacent blocks used in a motion vector predictor designating mode;

[0018] FIG. 4 is a diagram for describing an example of a motion vector predictor candidate list for the prediction direction 0;

[0019] FIG. 5 is a diagram for describing an example of a motion vector predictor candidate list for the prediction direction 1;

[0020] FIG. 6 shows examples of assignment of bit strings to motion vector predictor indices;

[0021] FIG. 7 is a flowchart showing an example of coding processing performed when the motion vector predictor designating mode is used;

[0022] FIG. 8A shows an example of calculation of a motion vector predictor;

[0023] FIG. 8B shows an example of calculation of a motion vector predictor;

[0024] FIG. 9 is a block diagram showing an example of a configuration of a moving picture coding apparatus which codes a moving picture using the motion vector predictor designating mode;

[0025] FIG. 10 is a flowchart showing an example of decoding processing performed when the motion vector predictor designating mode is used;

[0026] FIG. 11 is a block diagram showing an example of a configuration of a moving picture decoding apparatus which decodes a moving picture coded using the motion vector predictor designating mode;

[0027] FIG. 12 shows syntax used when a motion vector predictor index is added to a bitstream;

[0028] FIG. 13 is a block diagram showing a configuration of a moving picture coding apparatus according to Embodiment 1;

[0029] FIG. 14 is a flowchart showing processing operation of the moving picture coding apparatus according to Embodiment 1;

[0030] FIG. 15 shows an example of a motion vector predictor candidate list for the prediction direction 0 in Embodiment 1;

[0031] FIG. 16 shows an example of a motion vector predictor candidate list for the prediction direction 1 in Embodiment 1;

[0032] FIG. 17 is a flowchart showing processing for calculating a motion vector predictor candidate and a motion vector predictor candidate list size in Embodiment 1;

[0033] FIG. 18 is a flowchart showing processing for updating the number of available predictor candidates in Embodiment 1;

[0034] FIG. 19 is a flowchart showing processing for adding a new candidate in Embodiment 1;

[0035] FIG. 20 is a flowchart showing processing regarding selection of a motion vector predictor candidate in Embodiment 1;

[0036] FIG. 21 is a block diagram showing a configuration of a moving picture coding apparatus according to Embodiment 2;

[0037] FIG. 22 is a flowchart showing processing operation of the moving picture coding apparatus according to Embodiment 2;

[0038] FIG. 23 is a block diagram showing a configuration of a moving picture decoding apparatus according to Embodiment 3;

[0039] FIG. 24 is a flowchart showing processing operation of the moving picture decoding apparatus according to Embodiment 3;

[0040] FIG. 25 is a flowchart showing processing for calculating the number of available predictor candidates in Embodiment 3;

[0041] FIG. 26 is a flowchart showing processing for calculating a motion vector predictor candidate in Embodiment 3;

[0042] FIG. 27 shows an example of syntax used when a motion vector predictor index is added to a bitstream;

[0043] FIG. 28 shows an example of syntax used when a motion vector predictor candidate list size is fixed to the maximum value of the number of motion vector predictor candidates;

[0044] FIG. 29 is a block diagram showing a configuration of a moving picture decoding apparatus according to Embodiment 4;

[0045] FIG. 30 is a flowchart showing processing operation of the moving picture decoding apparatus according to Embodiment 4;

[0046] FIG. 31 shows an overall configuration of a content providing system for implementing content distribution services;

[0047] FIG. 32 shows an overall configuration of a digital broadcasting system;

[0048] FIG. 33 shows a block diagram illustrating an example of a configuration of a television;

[0049] FIG. 34 shows a block diagram illustrating an example of a configuration of an information reproducing/recording unit that reads and writes information from and on a recording medium that is an optical disk;

[0050] FIG. 35 shows an example of a configuration of a recording medium that is an optical disk;

[0051] FIG. 36A shows an example of a cellular phone;

[0052] FIG. 36B is a block diagram showing an example of a configuration of a cellular phone;

[0053] FIG. 37 illustrates a structure of multiplexed data;

[0054] FIG. 38 schematically shows how each stream is multiplexed in multiplexed data;

[0055] FIG. 39 shows how a video stream is stored in a stream of PES packets in more detail;

[0056] FIG. 40 shows a structure of TS packets and source packets in the multiplexed data;

[0057] FIG. 41 shows a data structure of a PMT;

[0058] FIG. 42 shows an internal structure of multiplexed data information;

[0059] FIG. 43 shows an internal structure of stream attribute information;

[0060] FIG. 44 shows steps for identifying video data;

[0061] FIG. 45 shows an example of a configuration of an integrated circuit for implementing the moving picture coding method and the moving picture decoding method according to each of embodiments;

[0062] FIG. 46 shows a configuration for switching between driving frequencies;

[0063] FIG. 47 shows steps for identifying video data and switching between driving frequencies;

[0064] FIG. 48 shows an example of a look-up table in which video data standards are associated with driving frequencies;

[0065] FIG. 49A is a diagram showing an example of a configuration for sharing a module of a signal processing unit; and

[0066] FIG. 49B is a diagram showing another example of a configuration for sharing a module of the signal processing unit.

DESCRIPTION OF EMBODIMENTS

(Underlying Knowledge Forming Basis of the Present Invention)

[0067] In the moving picture coding scheme referred to as H.264 which has already been standardized, three picture types, namely, I-picture, P-picture, and B-picture are used to compress the amount of information.

[0068] An I-picture is not coded by inter prediction coding processing. Specifically, an I-picture is coded by intra-picture prediction (hereinafter, referred to as intra prediction) coding processing. A P-picture is coded by inter prediction coding by referring to one already coded picture that appears before or after a current picture to be coded in the display time order. A B-picture is coded by inter prediction coding by referring to two already coded pictures that appear before or after the current picture in the display time order.

[0069] In inter prediction coding, a reference picture list for identifying a reference picture is generated. A reference list is a list in which reference picture indices are assigned to coded reference pictures to be referred to in inter prediction. For example, since B-pictures can be coded by referring to two pictures, two reference lists (L0, L1) are generated.

[0070] FIG. 1A is a diagram for describing an example of a reference picture list for a B-picture. FIG. 1B shows an example of a reference picture list 0 (L0) for the prediction direction 0 in bidirectional prediction. Here, in the reference picture list 0, value 0 of the reference picture index 0 is assigned to reference picture 0 at display order 2. Further, value 1 of the reference picture index 0 is assigned to reference picture 1 at display order 1. Further, value 2 of the reference picture index 0 is assigned to reference picture 2 at display order 0. Specifically, reference picture indices having smaller values are assigned to reference pictures in order of temporal proximity to a current picture to be coded in display order.

[0071] FIG. 1C shows an example of the reference picture list 1 (L1) for the prediction direction 1 in bidirectional prediction. Here, in the reference picture list 1, value 0 of the reference picture index 1 is assigned to reference picture 1 at display order 1. Further, value 1 of the reference picture index 1 is assigned to reference picture 0 at display order 2. Further, value 2 of the reference picture index 2 is assigned to reference picture 2 at display order 0.

[0072] In this manner, it is possible to assign reference picture indices having different values for the prediction directions to a reference picture (reference pictures 0 and 1 in FIG. 1A), and reference picture indices having the same value for the prediction directions to a reference picture (reference picture 2 in FIG. 1A).

[0073] Further, in the moving picture coding scheme referred to as H.264 (NPL 1), a motion vector detection mode is used as an inter prediction coding mode for blocks to be coded in a B-picture. In the motion vector detection mode, a difference value between predicted image data and

image data of a current block to be coded, and a motion vector used for generating the predicted image data are coded. Further, in the motion vector detection mode, it is possible to select bidirectional prediction or unidirectional prediction, as the prediction direction. In bidirectional prediction, a predicted image is generated by referring to two already coded pictures which appear before or after a current picture to be coded. In unidirectional prediction, a predicted image is generated by referring to one already coded picture which appears before or after a current picture to be coded.

[0074] Further, in the moving picture coding scheme referred to as H.264, a coding mode referred to as a temporal motion vector prediction mode can be selected when a motion vector is derived in coding a B-picture. An inter prediction coding method in the temporal motion vector prediction mode is described using FIG. 2.

[0075] FIG. 2 is a diagram for describing motion vectors in the temporal motion vector prediction mode. Specifically, FIG. 2 shows the case where block a in picture B2 is to be coded in the temporal motion vector prediction mode.

[0076] Here, motion vector vb is utilized which is used when block b (hereinafter, referred to as “co-located block”) at the same position in picture P3 as that of block a is coded, picture P3 being a reference picture which appears after picture B2. Motion vector vb is a motion vector used when block b is coded by referring to picture P1.

[0077] Two reference blocks for block a are obtained from picture P1 which is a forward reference picture and picture P3 which is a backward reference picture, using motion vectors parallel to motion vector vb. Then, block a is coded by performing bidirectional prediction based on the two obtained reference blocks. Specifically, motion vectors used when block a is coded are motion vector va1 with respect to picture P1 and motion vector va2 with respect to picture P3.

[0078] In addition, a motion vector predictor designating mode is considered to be used (NPL 2) as a method for coding motion vectors of blocks to be coded in a B-picture or a P-picture. In the motion vector predictor designating mode, motion vector predictor candidates are generated based on motion vectors used when coding blocks adjacent to a current block to be coded. Then, a motion vector predictor is selected from among the motion vector predictor candidates, and a motion vector of the current block is coded. At this time, an index of the selected motion vector predictor and the like are added to a bitstream. Consequently, the same motion vector predictor as the motion vector predictor used for coding can be selected also when decoding is performed. A specific example is described with reference to FIG. 3.

[0079] FIG. 3 shows examples of motion vectors of adjacent blocks which are used in the motion vector predictor designating mode. In FIG. 3, adjacent block A is a coded block adjacent to and located at the left of a current block to be coded. Adjacent block B is a coded block adjacent to and located on the current block. Adjacent block C is a coded block adjacent to and located at the upper right of the current block. Adjacent block D is a coded block adjacent to and located at the bottom left of the current block.

[0080] In FIG. 3, the current block is a block which is coded by bidirectional prediction, and has, as a result of motion detection or the like, motion vector MvL0 in the prediction direction 0 as a motion vector relative to a reference picture indicated by reference picture index RefL0 for the prediction direction 0, and motion vector MvL1 in

the prediction direction **1** as a motion vector relative to a reference picture indicated by reference picture index RefL**1** for the prediction direction **1**. Here, MvL**0** is a motion vector for which a reference picture identified using the reference picture list **0** (L**0**) is referred to. Further, MvL**1** is a motion vector for which a reference picture identified using the reference picture list **1** (L**1**) is referred to.

[0081] Adjacent block A is a block coded by unidirectional prediction in the prediction direction **0**. Adjacent block A has motion vector MvL**0**_A in the prediction direction **0** as a motion vector relative to a reference picture indicated by reference picture index RefL**0**_A for the prediction direction **0**. Further, adjacent block B is a block coded by unidirectional prediction in the prediction direction **1**. Adjacent block B has motion vector MvL**1**_B in the prediction direction **1** as a motion vector relative to a reference picture indicated by reference picture index RefL**1**_B for the prediction direction **1**. Adjacent block C is a block coded by intra prediction. Further, adjacent block D is a block coded by unidirectional prediction in the prediction direction **0**. Adjacent block D has motion vector MvL**0**_D in the prediction direction **0** as a motion vector relative to a reference picture indicated by reference picture index RefL**0**_D in the prediction direction **0**.

[0082] In such a case, as a motion vector predictor of a current block to be coded, for example, a motion vector predictor with which a motion vector of the current block can be most efficiently coded is selected from among motion vector predictor candidates generated from motion vectors of adjacent blocks A, B, C and D and a motion vector in the temporal motion vector prediction mode obtained using a co-located block. Then, a motion vector predictor index indicating the selected motion vector predictor is added to a bitstream. For example, if motion vector MvL**0**_A in the prediction direction **0** of adjacent block A is selected as a motion vector predictor when motion vector MvL**0** in the prediction direction **0** of a current block is to be coded, only value “0” of the motion vector predictor index which indicates that the motion vector predictor candidate generated from adjacent block A is used as shown in FIG. 4 is added to a bitstream. Accordingly, the amount of information on motion vector MvL**0** in the prediction direction **0** of the current block can be reduced.

[0083] Here, FIG. 4 shows an example of a motion vector predictor candidate list for the prediction direction **0**. Further, as shown in FIG. 4, in the motion vector predictor designating mode, a candidate with which a motion vector predictor cannot be generated (hereinafter, referred to as “non-available predictor candidate”), and a candidate whose value is the same as the value of another motion vector predictor candidate (hereinafter, “redundant candidate”) are deleted from motion vector predictor candidates. Consequently, the code amount assigned to motion vector predictor indices is reduced by decreasing the number of motion vector predictor candidates. Here, generation of a motion vector predictor being impossible means that an adjacent block is (1) a block coded by intra prediction, (2) a block outside a boundary of a slice or a picture which includes a current block to be coded, or (3) a block which is not coded yet, for instance.

[0084] In the example in FIG. 4, adjacent block C is coded by intra prediction. Accordingly, a predictor candidate indicated by value “3” of the motion vector predictor index is a non-available predictor candidate, and thus is deleted from

the motion vector predictor candidate list. Further, a motion vector predictor in the prediction direction **0** generated from adjacent block D has the same value as the value of a motion vector predictor in the prediction direction **0** generated from adjacent block A, and thus a predictor candidate indicated by value “4” of the motion vector predictor index is deleted from the motion vector predictor candidate list. As a result, the number of motion vector predictor candidates in the prediction direction **0** is eventually reduced to 3, and the motion vector predictor candidate list size for the prediction direction **0** is set to 3.

[0085] FIG. 5 shows an example of a motion vector predictor candidate list for the prediction direction **1**. In the example shown in FIG. 5, the number of motion vector predictor candidates in the prediction direction **1** is eventually reduced to 2 by deleting a non-available predictor candidate and redundant candidates, and the motion vector predictor candidate list size for the prediction direction **1** is set to 2.

[0086] As shown in FIG. 6, bit strings are assigned to motion vector predictor indices according to the motion vector predictor candidate list size, and are variable-length coded. Further, if the motion vector predictor candidate list size is 1, a motion vector predictor index is not added to a bitstream, and a decoding apparatus is caused to estimate that the index is value **0**. In this way, in the motion vector predictor designating mode, bit strings assigned to motion vector predictor indices are changed according to the motion vector predictor candidate list size, thereby reducing the code amount.

[0087] FIG. 7 is a flowchart showing an example of coding processing in the case of using the motion vector predictor designating mode.

[0088] In step S1001, motion vector predictor candidates in a prediction direction X are calculated from adjacent blocks and a co-located block (hereafter, referred to as “prediction block candidates”). Here, X is one of the values “0” and “1”, where 0 represents the prediction direction **0** and 1 represents the prediction direction **1**. Motion vector predictor candidate sMvLX in the prediction direction X is calculated in accordance with the following expression, using motion vector MvLX_N and reference picture index RefLX_N of a prediction block candidate and reference picture index RefLX of a current block to be coded.

$$sMvLX = MvLX_N \times \quad (\text{Expression 1})$$

$$(POC(RefLX) - curPOC) / (POC(RefLX_N) - curPOC)$$

[0089] Here, POC(RefLX) indicates when in the order a reference picture indicated by reference picture index RefLX is displayed, POC(RefLX_N) indicates when in the order a reference picture indicated by reference picture index RefLX_N is displayed, and curPOC indicates when in the order a current picture to be coded is displayed. It should be noted that if a prediction block candidate does not have motion vector MvLX_N in the prediction direction X, motion vector predictor sMvLX is calculated in accordance with Expression 2, using motion vector MvL(1-X)_N in the prediction direction (1-X) and reference picture index RefL(1-X)_N.

$$sMvLX = MvL(1 - X)_N \times (POC(RefLX) - curPOC) / (POC(RefL(1 - X)_N) - curPOC) \quad (\text{Expression 2})$$

[0090] FIGS. 8A and 8B show examples of calculating motion vector predictors using Expressions 1 and 2. It should be noted that as shown by Expressions 1 and 2, if the values of POC(RefLX) and POC(RefLX_N) are the same, namely, the same picture is referred to, scaling can be skipped.

[0091] In step S1002, a redundant candidate and a non-available predictor candidate are deleted from motion vector predictor candidates in the prediction direction X. In step S1003, the motion vector predictor candidate list size is set to the number of motion vector predictor candidates after the deleting processing. In step S1004, a motion vector predictor index to be used for coding a motion vector in the prediction direction X of a current block is determined. In step S1005, the determined motion vector predictor index is variable-length coded using a bit string determined according to the motion vector predictor candidate list size.

[0092] FIG. 9 is a block diagram showing an example of a configuration of a moving picture coding apparatus 1000 which codes a moving picture using the motion vector predictor designating mode.

[0093] As shown in FIG. 9, the moving picture coding apparatus 1000 includes a subtraction unit 1001, an orthogonal transform unit 1002, a quantization unit 1003, an inverse quantization unit 1004, an inverse orthogonal transform unit 1005, an addition unit 1006, a block memory 1007, a frame memory 1008, an intra prediction unit 1009, an inter prediction unit 1010, an inter prediction control unit 1011, a picture type determination unit 1012, a switch 1013, a motion vector predictor candidate calculation unit 1014, a colPic memory 1015, and a variable length coding unit 1016.

[0094] In FIG. 9, the motion vector predictor candidate calculation unit 1014 calculates motion vector predictor candidates. Then, the motion vector predictor candidate calculation unit 1014 transmits the number of calculated motion vector predictor candidates to the variable length coding unit 1016. The variable length coding unit 1016 sets the motion vector predictor candidate list size which is a coding parameter to the number of motion vector predictor candidates. Then, the variable length coding unit 1016 variable-length codes motion vector predictor indices used for coding by assigning thereto bit strings according to the motion vector predictor candidate list size.

[0095] FIG. 10 is a flowchart showing an example of decoding processing in the case of using the motion vector predictor designating mode.

[0096] In step S2001, motion vector predictor candidates in the prediction direction X are calculated from adjacent blocks and a co-located block (prediction block candidates). In step S2002, a redundant candidate and a non-available predictor candidate are deleted from the motion vector predictor candidates. In step S2003, the motion vector predictor candidate list size is set to the number of motion vector predictor candidates after the deleting processing. In step S2004, a motion vector predictor index to be used for decoding a current block is decoded from a bitstream using the motion vector predictor candidate list size. In step S2005, a motion vector is calculated by adding a motion

vector difference to a motion vector predictor candidate indicated by the decoded motion vector predictor index, and a predicted image is generated using the calculated motion vector, thereby performing decoding processing.

[0097] FIG. 11 is a block diagram showing an example of a configuration of a moving picture decoding apparatus which decodes a moving picture coded using the motion vector predictor designating mode.

[0098] As shown in FIG. 11, a moving picture decoding apparatus 2000 includes a variable length decoding unit 2001, an inverse quantization unit 2002, an inverse orthogonal transform unit 2003, an addition unit 2004, a block memory 2005, a frame memory 2006, an intra prediction unit 2007, an inter prediction unit 2008, an inter prediction control unit 2009, a switch 2010, a motion vector predictor candidate calculation unit 2011, and a colPic memory 2012.

[0099] In FIG. 11, the motion vector predictor candidate calculation unit 2011 calculates motion vector predictor candidates. Then, the motion vector predictor candidate calculation unit 2011 transmits the number of calculated motion vector predictor candidates to the variable length decoding unit 2001. The variable length decoding unit 2001 sets the motion vector predictor candidate list size which is a decoding parameter to the number of motion vector predictor candidates. Then, the variable length decoding unit 2001 decodes a motion vector predictor index included in a bitstream using the motion vector predictor candidate list size.

[0100] FIG. 12 shows syntax used when a motion vector predictor index is added to a bitstream. In FIG. 12, inter_pred_flag indicates a prediction direction flag for inter prediction, mvp_idx indicates a motion vector predictor index, and NumMVPCand indicates the motion vector predictor candidate list size. NumMVPCand is set to the number of motion vector predictor candidates after deleting a non-available predictor candidate and a redundant candidate from the motion vector predictor candidates.

[0101] As described above, a moving picture is coded or decoded using the motion vector predictor designating mode. However, in the above motion vector predictor designating mode, the motion vector predictor candidate list size to be used when a motion vector predictor index is coded or decoded is set to the number of motion vector predictor candidates. This number of motion vector predictor candidates is obtained after deleting a non-available predictor candidate or a redundant candidate using reference picture information including information of a co-located block and the like. Thus, if, for instance, there is a difference in the number of motion vector predictor candidates between a moving picture coding apparatus and a moving picture decoding apparatus, different bit strings are assigned to motion vector predictor indices in the moving picture coding apparatus and the moving picture decoding apparatus. As a result, the moving picture decoding apparatus may not be able to decode a bitstream appropriately.

[0102] For example, if information of a reference picture which has been referenced as a co-located block is lost due to a packet loss or the like which has occurred on a transmission channel or the like, a motion vector or a reference picture index of the co-located block will be lost. Thus, information on a motion vector predictor candidate to be generated from the co-located block cannot be obtained. In such a case, a non-available predictor candidate and a redundant candidate cannot be appropriately deleted from

motion vector predictor candidates at the time of decoding. As a result, the moving picture decoding apparatus will not be able to appropriately obtain the motion vector predictor candidate list size, and will not be able to successfully decode a motion vector predictor index.

[0103] In view of this, an object of the present invention is to provide a moving picture coding method which improves error resistance by calculating the motion vector predictor candidate list size to be used when coding or decoding a motion vector predictor index, using a method independent of reference picture information including information of a co-located block and the like.

[0104] In view of this, a moving picture coding method according to an aspect of the present invention is a moving picture coding method for calculating a motion vector predictor to be used when coding a motion vector of a current block to be coded, and coding the current block, to generate a bitstream, the method including: determining a maximum number of motion vector predictor candidates each of which is a candidate for the motion vector predictor; deriving one or more first motion vector predictor candidates; determining whether a total number of the one or more first motion vector predictor candidates is smaller than the maximum number; deriving one or more second motion vector predictor candidates when it is determined that the total number of one or more first motion vector predictor candidates is smaller than the maximum number; selecting, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates, the motion vector predictor to be used for coding the motion vector of the current block; and coding, using the determined maximum number, an index for identifying the selected motion vector predictor, and adding the coded index to the bitstream.

[0105] According to this, an index for identifying a motion vector predictor candidate can be coded using the determined maximum number. Specifically, an index can be coded without depending on the number of motion vector predictor candidates actually derived. Thus, even if information necessary for deriving a motion vector predictor candidate (for example, information of a co-located block and the like) is lost, the decoding apparatus can decode an index, and error resistance can be improved. Further, the decoding apparatus can decode an index, without depending on the number of motion vector predictor candidates actually derived. Specifically, the decoding apparatus can decode an index, without waiting for derivation of a motion vector predictor candidate. In other words, it is possible to generate a bitstream for which deriving a motion vector predictor candidate and decoding an index can be performed in parallel.

[0106] Furthermore, according to this, a second motion vector predictor candidate can be derived if it is determined that the number of first motion vector predictor candidates is smaller than the maximum number. Thus, it is possible to increase the number of motion vector predictor candidates in a range which does not exceed the maximum number, and improve coding efficiency.

[0107] For example, when deriving the first motion vector predictor candidates, the motion vector predictor candidate may be derived as the first motion vector predictor candidate, the motion vector predictor candidate having a motion

vector different from a motion vector of any of the one or more first motion vector predictor candidates which have already been derived.

[0108] According to this, a redundant first motion vector predictor candidate can be deleted. As a result, the number of second motion vector predictor candidates can be increased, and thus the types of selectable motion vectors can be increased. Thus, it is possible to further improve coding efficiency.

[0109] For example, when deriving the one or more first motion vector predictor candidates, the one or more first motion vector predictor candidates may be each derived based on a motion vector used for coding a block spatially or temporally adjacent to the current block.

[0110] According to this, the first motion vector predictor candidate can be derived based on a motion vector used for coding a block spatially or temporally adjacent to the current block.

[0111] For example, when deriving the one or more first motion vector predictor candidates, a motion vector used for coding a block may be derived as the first motion vector predictor candidate, the block being spatially adjacent to the current block, and not being (i) a block coded by intra prediction, (ii) a block located outside a boundary of a slice or a picture which includes the current block, or (iii) a block which is not coded yet.

[0112] According to this, the first motion vector predictor candidate can be derived based on a block suitable for obtaining a motion vector predictor candidate.

[0113] For example, when deriving the one or more second motion vector predictor candidates, the motion vector predictor candidate having a motion vector different from a motion vector of any of the one or more first motion vector predictor candidates may be derived as the second motion vector predictor candidate.

[0114] According to this, a motion vector predictor candidate having a motion vector different from that of any first motion vector predictor candidate can be derived as the second motion vector predictor candidate. Thus, the number of motion vector predictor candidates having different motion vectors can be increased, which allows coding efficiency to be further improved.

[0115] For example, when adding the coded index, information indicating the determined maximum number may be further added to the bitstream.

[0116] According to this, information indicating the determined maximum number can be added to a bitstream. Therefore, the maximum number can be changed in a suitable unit, which allows coding efficiency to be improved.

[0117] For example, the moving picture coding method may further include: switching between first coding processing conforming to a first standard and second coding processing conforming to a second standard; and adding, to the bitstream, identification information indicating the first standard or the second standard to which a corresponding one of the first coding processing and the second coding processing after the switching conforms, wherein when the switch to the first coding processing is made, determining the maximum number, deriving the one or more first motion vector predictor candidates, determining whether the total number of one or more first motion vector predictor candidates is smaller, deriving the one or more second motion vector predictor candidates, selecting the motion vector

predictor, coding the index, and adding the coded index may be performed as the first coding processing.

[0118] According to this, it is possible to switch between the first coding processing conforming to the first standard and the second coding processing conforming to the second standard.

[0119] A moving picture decoding method according to an aspect of the present invention is a moving picture decoding method for calculating a motion vector predictor to be used when decoding a motion vector of a current block to be decoded which is included in a bitstream and decoding the current block, the method including: determining a maximum number of motion vector predictor candidates each of which is a candidate for the motion vector predictor; deriving one or more first motion vector predictor candidates; determining whether a total number of the one or more first motion vector predictor candidates is smaller than the maximum number; deriving one or more second motion vector predictor candidates when it is determined that the total number of one or more first motion vector predictor candidates is smaller than the maximum number; decoding, using the determined maximum number, a coded index added to the bitstream and used for identifying the motion vector predictor; and selecting, based on the decoded index, a motion vector predictor to be used for decoding the current block, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates.

[0120] According to this, an index for identifying a motion vector predictor candidate can be decoded using the determined maximum number. Specifically, an index can be decoded without depending on the number of motion vector predictor candidates actually derived. Therefore, an index can be decoded even if information necessary for deriving a motion vector predictor candidate (for example, information of a co-located block and the like) is lost, which enables error resistance to be improved.

[0121] Furthermore, it is possible to decode an index without waiting for derivation of a motion vector predictor candidate, and also derive a motion vector predictor candidate and decode an index in parallel.

[0122] Furthermore, according to this, if it is determined that the number of first motion vector predictor candidates is smaller than the maximum number, one or more second motion vector predictor candidates can be derived. Therefore, the number of motion vector predictor candidates can be increased in a range which does not exceed the maximum number, and thus a coded image for which coding efficiency has been improved can be decoded.

[0123] For example, when deriving the first motion vector predictor candidates, the motion vector predictor candidate may be derived as the first motion vector predictor candidate, the motion vector predictor candidate having a motion vector different from a motion vector of any of the one or more first motion vector predictor candidates which have already been derived.

[0124] According to this, a redundant first motion vector predictor candidate can be deleted. As a result, the number of second motion vector predictor candidates can be increased, and the types of selectable motion vectors can be increased. Therefore, it is possible to decode a coded image for which coding efficiency has been further improved.

[0125] For example, when deriving the one or more first motion vector predictor candidates, the one or more first

motion vector predictor candidates may be each derived based on a motion vector used for decoding a block spatially or temporally adjacent to the current block.

[0126] According to this, the first motion vector predictor candidate can be derived based on a motion vector used for decoding a block spatially or temporally adjacent to the current block.

[0127] For example, when deriving the one or more first motion vector predictor candidates, a motion vector used for decoding a block may be derived as the first motion vector predictor candidate, the block being a block candidate spatially adjacent to the current block, and not being (i) a block decoded by intra prediction, (ii) a block located outside a boundary of a slice or a picture which includes the current block, or (iii) a block which is not decoded yet.

[0128] According to this, the first motion vector predictor candidate can be derived from a block suitable for obtaining a motion vector predictor candidate.

[0129] For example, when deriving the one or more second motion vector predictor candidates, the motion vector predictor candidate having a motion vector different from a motion vector of any of the one or more first motion vector predictor candidates may be derived as the second motion vector predictor candidate.

[0130] According to this, a motion vector predictor candidate having a motion vector different from that of any first motion vector predictor candidate can be derived as the second motion vector predictor candidate. Thus, the number of motion vector predictor candidates having different motion vectors can be increased, and a coded image for which coding efficiency has been further improved can be decoded.

[0131] For example, when determining the maximum number, the maximum number may be determined based on information indicating a maximum number and added to the bitstream.

[0132] According to this, the maximum number can be determined based on information added to a bitstream. Thus, it is possible to decode an image coded by changing the maximum number in a suitable unit.

[0133] For example, the moving picture decoding method may further include switching between first decoding processing conforming to a first standard and second decoding processing conforming to a second standard, according to identification information indicating the first standard or the second standard and added to the bitstream, wherein when the switch to the first decoding processing is made, determining the maximum number, deriving the one or more first motion vector predictor candidates, determining whether the total number of one or more first motion vector predictor candidates is smaller, deriving the one or more second motion vector predictor candidates, decoding the coded index, and selecting the motion vector predictor may be performed as the first decoding processing.

[0134] According to this, it is possible to switch between the first decoding processing conforming to the first standard and the second decoding processing conforming to the second standard.

[0135] It should be noted that these general and specific aspects may be implemented using a system, a method, an integrated circuit, a computer program, a computer-readable recording medium such as a CD-ROM, or any combination of systems, methods, integrated circuits, computer programs or recording media.

[0136] The following is a specific description of a moving picture coding apparatus and a moving picture decoding apparatus according to an aspect of the present invention, with reference to the drawings.

[0137] Each of the exemplary embodiments described below shows a general or specific example. The numerical values, shapes, materials, constituent elements, the arrangement and connection of the constituent elements, steps, the processing order of the steps and the like described in the following embodiments are mere examples, and thus do not limit the scope of the appended Claims and their equivalents. Therefore, among the constituent elements in the following exemplary embodiments, constituent elements not recited in any one of the independent claims are described as arbitrary constituent elements.

Embodiment 1

[0138] FIG. 13 is a block diagram showing a configuration of a moving picture coding apparatus 100 according to Embodiment 1.

[0139] As shown in FIG. 13, the moving picture coding apparatus 100 includes a subtraction unit 101, an orthogonal transform unit 102, a quantization unit 103, an inverse quantization unit 104, an inverse orthogonal transform unit 105, an addition unit 106, a block memory 107, a frame memory 108, an intra prediction unit 109, an inter prediction unit 110, an inter prediction control unit 111, a picture type determination unit 112, a switch 113, a motion vector predictor candidate calculation unit 114, a colPic memory 115, and a variable length coding unit 116.

[0140] The subtraction unit 101 generates prediction error data by subtracting, for each block, predicted image data from input image data included in an input image sequence. The orthogonal transform unit 102 transforms the generated prediction error data from an image domain into a frequency domain. The quantization unit 103 performs quantization processing on the prediction error data which has been transformed into the frequency domain.

[0141] The inverse quantization unit 104 performs inverse quantization processing on the prediction error data on which quantization processing has been performed by the quantization unit 103. The inverse orthogonal transform unit 105 transforms the prediction error data on which inverse quantization processing has been performed, from the frequency domain into the image domain.

[0142] The addition unit 106 generates reconstructed image data by adding, for each block to be coded, predicted image data and the prediction error data on which inverse quantization processing has been performed by the inverse orthogonal transform unit 105. The block memory 107 stores reconstructed image data on a block-by-block basis. The frame memory 108 stores reconstructed image data on a frame-by-frame basis.

[0143] The picture type determination unit 112 determines which of picture types, namely, I-picture, B-picture, and P-picture, an input image data is to be coded as. Then, the picture type determination unit 112 generates picture type information. The intra prediction unit 109 generates intra-predicted image data of a current block to be coded by performing intra prediction using the reconstructed image data in block units stored in the block memory 107. The inter prediction unit 110 generates inter-predicted image data of a current block to be coded by performing inter prediction using the reconstructed image data in frame units stored in

the frame memory 108, and a motion vector derived by motion detection and the like.

[0144] The switch 113 outputs the intra-predicted image data generated by the intra prediction unit 109 to the subtraction unit 101 and the addition unit 106 as predicted image data of the current block, if intra prediction coding is performed on the current block. On the other hand, the switch 113 outputs the inter-predicted image data generated by the inter prediction unit 110 to the subtraction unit 101 and the addition unit 106 as predicted image data of the current block if the inter prediction coding is performed on the current block.

[0145] The motion vector predictor candidate calculation unit 114 derives motion vector predictor candidates in the motion vector predictor designating mode, using motion vectors of blocks adjacent to the current block and the like and colPic information such as information of a motion vector of a co-located block stored in the colPic memory 115. Then, the motion vector predictor candidate calculation unit 114 calculates the number of available predictor candidates using the method described below. Further, the motion vector predictor candidate calculation unit 114 assigns the values of the motion vector predictor index to the derived motion vector predictor candidates. Then, the motion vector predictor candidate calculation unit 114 sends the motion vector predictor candidates and the motion vector predictor indices to the inter prediction control unit 111. The motion vector predictor candidate calculation unit 114 transmits the number of calculated available predictor candidates to the variable length coding unit 116.

[0146] The inter prediction control unit 111 controls the inter prediction unit 110 so as to cause the inter prediction unit 110 to perform inter prediction coding, using the inter-predicted image generated using a motion vector derived by motion detection. Further, the inter prediction control unit 111 selects, using the method described below, a motion vector predictor candidate most suitable for coding a motion vector used for inter prediction coding. Then, the inter prediction control unit 111 sends a motion vector predictor index corresponding to the selected motion vector predictor candidate, and prediction error information (motion vector difference) to the variable length coding unit 116. Furthermore, the inter prediction control unit 111 transfers colPic information including information of a motion vector of the current block and the like to the colPic memory 115.

[0147] The variable length coding unit 116 performs variable length coding processing on prediction error data on which quantization processing has been performed, a prediction direction flag, picture type information, and a motion vector difference, thereby generating a bitstream. Further, the variable length coding unit 116 sets the motion vector predictor candidate list size to the number of available predictor candidates. Then, the variable length coding unit 116 variable-length codes the motion vector predictor index used for coding a motion vector by assigning, thereto, a bit string according to the motion vector predictor candidate list size.

[0148] FIG. 14 is a flowchart showing processing operation of the moving picture coding apparatus 100 according to Embodiment 1.

[0149] In step S101, the inter prediction control unit 111 determines a prediction direction, a reference picture index, and a motion vector of a current block to be coded by motion detection. Here, in motion detection, a difference value

indicating a difference between a current block to be coded in a picture to be coded and a block in a reference picture is calculated, and a block in the reference picture with which the difference value is smallest is determined as a reference block. Then, a motion vector is obtained based on the position of a current block to be coded and the position of a reference block position using the method for obtaining a motion vector, for instance. Further, the inter prediction control unit **111** performs motion detection on each of reference pictures in the prediction directions **0** and **1**, and determines whether to select the prediction direction **0**, the prediction direction **1** or bidirectional prediction using, for example, the following expression for an R-D optimization model, or the like.

$$\text{Cost} = D + \lambda \times R \quad (\text{Expression 3})$$

[0150] In Expression 3, D denotes coding distortion, and for instance, a sum of absolute differences are used therefor each of which is an absolute difference between a pixel value obtained by coding and decoding a current block using a predicted image generated using a certain motion vector and an original pixel value of the current block. R denotes a generated code amount, and a code amount necessary to code a motion vector used for generating a predicted image is used therefor. Further, λ denotes a Lagrange undetermined multiplier.

[0151] In step S102, the motion vector predictor candidate calculation unit **114** derives motion vector predictor candidates from blocks adjacent to the current block and a co-located block thereof. Further, the motion vector predictor candidate calculation unit **114** calculates the motion vector predictor candidate list size according to the method described below.

[0152] For example, in the case as shown in FIG. 3, the motion vector predictor candidate calculation unit **114** selects motion vectors which adjacent blocks A, B, C, and D have, as motion vector predictor candidates of the current block. Furthermore, the motion vector predictor candidate calculation unit **114** calculates a motion vector, for instance, which is calculated using a temporal prediction mode from a motion vector of the co-located block, as a motion vector predictor candidate.

[0153] The motion vector predictor candidate calculation unit **114** assigns motion vector predictor indices to the motion vector predictor candidates in the prediction directions **0** and **1**, as shown in (a) in FIG. 15 and (a) in FIG. 16. Then, the motion vector predictor candidate calculation unit **114** calculates motion vector predictor candidate lists as shown in (b) in FIG. 15 and (b) in FIG. 16, and the sizes of the motion vector predictor candidate lists by deleting a non-available predictor candidate and a redundant candidate and adding a new candidate, using the method described below.

[0154] The smaller a value of a motion vector predictor index is, the shorter code is assigned to the motion vector predictor index. Specifically, if the value of a motion vector predictor index is small, the amount of information necessary for the motion vector predictor index is small. On the other hand, if the value of a motion vector predictor index is large, the amount of information necessary for the motion vector predictor index is large. Thus, coding efficiency is

increased by assigning a motion vector predictor index having a small value to a motion vector predictor candidate having a high possibility of becoming a motion vector predictor with high precision.

[0155] In view of this, the motion vector predictor candidate calculation unit **114** may measure, for each motion vector predictor candidate, the number of times at which the motion vector predictor candidate has been selected as a motion vector predictor, and assign a motion vector predictor index having a small value to a motion vector predictor candidate whose number of times at which the candidate has been selected is large, for example. Specifically, it is possible to consider identifying a motion vector predictor selected in an adjacent block, and in coding a current block, assigning a motion vector predictor index having a small value to the identified motion vector predictor candidate.

[0156] It should be noted that if an adjacent block does not have information of a motion vector and the like (if the adjacent block is coded by intra prediction, if the adjacent block is located, for instance, outside a boundary of a picture or a slice, if the adjacent block is not coded yet, or the like), the adjacent block cannot be utilized as a motion vector predictor candidate.

[0157] In the present embodiment, a candidate that cannot be utilized as a motion vector predictor candidate is referred to as a non-available predictor candidate. A candidate that can be utilized as a motion vector predictor candidate is referred to as an available predictor candidate. Further, among a plurality of motion vector predictor candidates, a candidate whose value is the same as any one of the other motion vector predictors is referred to as a redundant candidate.

[0158] In the case of FIG. 3, adjacent block C is a block coded by intra prediction, and thus is assumed to be a non-available predictor candidate. Further, motion vector predictor sMvL0_D in the prediction direction **0** generated from adjacent block D has the same value as the value of motion vector predictor MvL0_A in the prediction direction **0** generated from adjacent block A, and thus is assumed to be a redundant candidate.

[0159] In step S103, the inter prediction control unit **111** determines a value of a motion vector predictor index to be used for coding a motion vector in the prediction direction X by using the method described below.

[0160] In step S104, the variable length coding unit **116** variable length-codes motion vector predictor indices of motion vector predictor candidates to be used for coding motion vectors in the prediction direction X by assigning thereto bit strings according to the motion vector predictor candidate list size as shown in FIG. 6.

[0161] In the present embodiment, as shown in (a) in FIG. 15 and (a) in FIG. 16, “0” is assigned as a value of a motion vector predictor index corresponding to adjacent block A. “1” is assigned as a value of a motion vector predictor index corresponding to adjacent block B. “2” is assigned as a value of a motion vector predictor index corresponding to a co-located block. “3” is assigned as a value of a motion vector predictor index corresponding to adjacent block C. “4” is assigned as a value of a motion vector predictor index corresponding to adjacent block D.

[0162] It should be noted that the way to assign motion vector predictor indices is not necessarily limited to this example. For example, if a new candidate is added using the method described below, the variable length coding unit **116**

may assign a small value to a motion vector predictor candidate which is not newly added, and a large value to the new candidate. Specifically, the variable length coding unit **116** may preferentially assign a motion vector predictor index having a small value to a motion vector predictor candidate which is not newly added.

[0163] Further, motion vector predictor candidates are not necessarily limited to be at the positions of adjacent blocks A, B, C, and D. For example, an adjacent block located on bottom-left adjacent block D, for instance, may be used to obtain a motion vector predictor candidate. Further, all the adjacent blocks do not necessarily need to be used to obtain motion vector predictor candidates. For example, only adjacent blocks A and B may be used to obtain motion vector predictor candidates. Alternatively, adjacent blocks may be sequentially scanned by using, for instance, adjacent block A if adjacent block D is a non-available predictor candidate.

[0164] Further, in the present embodiment, although the variable length coding unit **116** adds a motion vector predictor index to a bitstream in step **S104** in FIG. **14**, a motion vector predictor index does not necessarily need to be added to a bitstream. For example, if the motion vector predictor candidate list size is 1, the variable length coding unit **116** may not add a motion vector predictor index to a bitstream. Accordingly, the amount of information can be reduced by that of the motion vector predictor index.

[0165] FIG. **17** is a flowchart showing detailed processing of step **S102** in FIG. **14**. Specifically, FIG. **17** shows a method for calculating motion vector predictor candidates and the motion vector predictor candidate list size. The following is a description of FIG. **17**.

[0166] In step **S111**, the motion vector predictor candidate calculation unit **114** determines, using the method described below, whether a prediction block candidate [N] is an available predictor candidate. Then, the motion vector predictor candidate calculation unit **114** updates the number of available predictor candidates in accordance with the determination result.

[0167] Here, N is an index value for denoting each prediction block candidate. In the present embodiment, N is one of the values from **0** to **4**. Specifically, adjacent block A in FIG. **3** is assigned to a prediction block candidate [0]. Adjacent block B in FIG. **3** is assigned to a prediction block candidate [1]. A co-located block is assigned to a prediction block candidate [2]. Adjacent block C in FIG. **3** is assigned to a prediction block candidate [3]. Adjacent block D in FIG. **3** is assigned to a prediction block candidate [4].

[0168] In step **S112**, the motion vector predictor candidate calculation unit **114** derives a motion vector predictor candidate in the prediction direction X from the prediction block candidate [N] using Expressions 1 and 2 above, and adds the derived candidate to a corresponding one of the motion vector predictor candidate lists.

[0169] In step **S113**, the motion vector predictor candidate calculation unit **114** searches for and deletes a non-available predictor candidate and a redundant candidate from the motion vector predictor candidate lists, as shown in FIGS. **15** and **16**.

[0170] In step **S114**, the motion vector predictor candidate calculation unit **114** adds a new candidate to a corresponding one of the motion vector predictor candidate lists by using the method described below. Here, when a new candidate is added, the motion vector predictor candidate calculation unit **114** may reassign values of motion vector predictor indices

so as to preferentially assign a small motion vector predictor index to a motion vector predictor candidate which is not newly added. Specifically, the motion vector predictor candidate calculation unit **114** may reassign values of motion vector predictor indices so as to assign a motion vector predictor index having a large value to the new candidate. Accordingly, the amount of coding motion vector predictor indices can be reduced.

[0171] In step **S115**, the motion vector predictor candidate calculation unit **114** sets the motion vector predictor candidate list size to the number of available predictor candidates calculated in step **S111**. In the examples of FIGS. **15** and **16**, by using the method described below, “4” is calculated as the number of available predictor candidates in the prediction direction **0**, and the motion vector predictor candidate list size for the prediction direction **0** is set to “4”. Further, “4” is calculated as the number of available predictor candidates in the prediction direction **1**, and the motion vector predictor candidate list size for the prediction direction **1** is set to “4”.

[0172] It should be noted that a new candidate in step **S114** is a candidate newly added to motion vector predictor candidates using the method described below, if the number of motion vector predictor candidates has not reached the number of available predictor candidates. For example, a new candidate may be a motion vector predictor generated from an adjacent block located on bottom-left adjacent block D in FIG. **3**. A new candidate may be a motion vector predictor generated from blocks corresponding to blocks A, B, C, and D adjacent a co-located block, for example. Further, a new candidate may be a motion vector predictor calculated from a total of motion vectors in the entire picture plane or a certain area of a reference picture, for example. In this way, coding efficiency can be improved by the motion vector predictor candidate calculation unit **114** adding a new motion vector predictor as a new candidate if the number of motion vector predictor candidates has not reached the number of available predictor candidates.

[0173] FIG. **18** is a flowchart showing detailed processing of step **S111** in FIG. **17**. Specifically, FIG. **18** shows a method for determining whether the prediction block candidate [N] is an available predictor candidate, and updating the number of available predictor candidates. The following is a description of FIG. **18**.

[0174] In step **S121**, the motion vector predictor candidate calculation unit **114** determines whether a prediction block candidate [N] is (1) intra-predicted, (2) located outside a boundary of a slice or a picture which includes a current block to be coded, or (3) is not coded yet.

[0175] If the determination result in step **S121** is true here (Yes in **S121**), the motion vector predictor candidate calculation unit **114** sets the prediction block candidate [N] as a non-available predictor candidate in step **S122**. On the other hand, if the determination result in step **S121** is false (No in **S121**), the motion vector predictor candidate calculation unit **114** sets the prediction block candidate [N] as an available predictor candidate in step **S123**.

[0176] In step **S124**, the motion vector predictor candidate calculation unit **114** determines whether the prediction block candidate [N] is an available predictor candidate or a co-located block candidate. Here, if the determination result in step **S124** is true (Yes in **S124**), the motion vector predictor candidate calculation unit **114** adds 1 to the number of available predictor candidates, and updates the number of

motion vector predictor candidates in step S5. On the other hand, if the determination result in step S124 is false (No in S124), the motion vector predictor candidate calculation unit 114 does not update the number of available predictor candidates.

[0177] As described above, if a prediction block candidate is a co-located block, the motion vector predictor candidate calculation unit 114 adds 1 to the number of available predictor candidates, irrespective of whether the co-located block is an available predictor candidate or a non-available predictor candidate. Accordingly, even if information of a co-located block is lost due to packet loss or the like, there is no difference in the number of available predictor candidates between the moving picture coding apparatus and the moving picture decoding apparatus.

[0178] The motion vector predictor candidate list size is set to the number of available predictor candidates in step S115 in FIG. 17. Furthermore, in S104 in FIG. 14, the motion vector predictor candidate list size is used for variable-length coding motion vector predictor indices. Accordingly, even if reference picture information including information of a co-located block and the like is lost, the moving picture coding apparatus 100 can generate a bitstream from which a motion vector predictor index can be successfully decoded.

[0179] FIG. 19 is a flowchart showing detailed processing of step S114 in FIG. 17. Specifically, FIG. 19 shows a method for adding a new candidate. The following is a description of FIG. 19.

[0180] In step S131, the motion vector predictor candidate calculation unit 114 determines whether the number of motion vector predictor candidates is smaller than the number of available predictor candidates. Specifically, the motion vector predictor candidate calculation unit 114 determines whether the number of motion vector predictor candidates has not reached the number of available predictor candidates.

[0181] Here, if the determination result in step S131 is true (Yes in S131), the motion vector predictor candidate calculation unit 114 determines in step S132 whether there is a new candidate which can be added to a corresponding one of the motion vector predictor candidate lists as a motion vector predictor candidate. Here, if the determination result in step S132 is true (Yes in S132), the motion vector predictor candidate calculation unit 114 assigns a value of a motion vector predictor index to the new candidate, and adds the new candidate to a corresponding one of the motion vector predictor candidate lists in step S133. Furthermore, in step S134, the motion vector predictor candidate calculation unit 114 adds 1 to the number of motion vector predictor candidates.

[0182] On the other hand, if the determination result in step S131 or step S132 is false (No in S131 or S132), new candidate adding processing ends. Specifically, if the number of motion vector predictor candidates has reached the number of available predictor candidates, or if there is no new candidate, new candidate adding processing ends.

[0183] FIG. 20 is a flowchart showing detailed processing of step S103 in FIG. 14. Specifically, FIG. 20 shows processing regarding selection of a motion vector predictor candidate. The following is a description of FIG. 20.

[0184] In step S141, as initialization, the inter prediction control unit 111 sets motion vector predictor candidate index *mvp_idx* to 0, and sets the smallest motion vector difference to the maximum value.

[0185] In step S142, the inter prediction control unit 111 determines whether the value of motion vector predictor candidate index *mvp_idx* is smaller than the number of motion vector predictor candidates. Specifically, the inter prediction control unit 111 determines whether motion vector differences of all the motion vector predictor candidates have been calculated.

[0186] Here, if there still remains a motion vector predictor candidate for which calculation has not been performed (Yes in S142), the inter prediction control unit 111 calculates a motion vector difference by subtracting a motion vector predictor candidate from a vector obtained as a result of motion detection (motion detection resultant vector) in step S143.

[0187] In step S144, the inter prediction control unit 111 determines whether the motion vector difference obtained in step S143 has a value smaller than the smallest motion vector difference.

[0188] Here, if the determination result in step S144 is true (Yes in S144), the inter prediction control unit 111 updates the smallest motion vector difference and the value of a motion vector predictor index in step S145. On the other hand, if the determination result in step S144 is false (No in S144), the inter prediction control unit 111 does not update the smallest motion vector difference and the value of a motion vector predictor index.

[0189] In step S146, the inter prediction control unit 111 updates a motion vector predictor candidate index by incrementing by +1, and returning back to step S142, the inter prediction control unit 111 determines whether a next motion vector predictor candidate is present.

[0190] On the other hand, if it is determined in step S2 that a motion vector difference has been calculated for all the motion vector predictor candidates (No in S142), the inter prediction control unit 111 fixes, in step S147, the smallest motion vector difference and the motion vector predictor index which are set at last.

[0191] In this way, according to the moving picture coding apparatus 100 according to the present embodiment, the motion vector predictor candidate list size to be used when a motion vector predictor index is coded or decoded can be calculated using a method independent of reference picture information including information of a co-located block and the like. Accordingly, the moving picture coding apparatus 100 can improve error resistance.

[0192] More specifically, the moving picture coding apparatus 100 according to the present embodiment adds 1 to the number of available predictor candidates if a prediction block candidate is a co-located block, irrespective of whether the co-located block is an available predictor candidate. Then, the moving picture coding apparatus 100 determines a bit string to be assigned to a motion vector predictor index using the number of available predictor candidates calculated in this way. Accordingly, the moving picture coding apparatus 100 can generate a bitstream from which a motion vector predictor index can be successfully decoded even if reference picture information including information of a co-located block is lost.

[0193] Further, the moving picture coding apparatus 100 according to the present embodiment can improve coding

efficiency by adding a new candidate having a new motion vector predictor as a motion vector predictor candidate if the number of motion vector predictor candidates has not reached the number of available predictor candidates.

[0194] It should be noted that although in the present embodiment, the moving picture coding apparatus **100** adds a new candidate having a new motion vector predictor as a motion vector predictor candidate if the number of motion vector predictor candidates has not reached the number of available predictor candidates, the present embodiment is not limited to this. For example, the moving picture coding apparatus **100** may set a new candidate having a new motion vector predictor as an initial value of all the motion vector predictor candidates on the motion vector predictor candidate lists when the motion vector predictor candidate lists are created. In this case, the moving picture coding apparatus **100** will calculate a motion vector predictor candidate, and overwrite the new candidate which is an initial value when the calculated motion vector predictor candidate is added to a corresponding one of the motion vector predictor candidate lists. Then, the moving picture coding apparatus **100** determines whether the calculated motion vector predictor candidate is a non-available predictor candidate or a redundant candidate, before the calculated motion vector predictor candidate is added to the corresponding motion vector predictor candidate list. Accordingly, if there is a non-available predictor candidate or a redundant candidate, the new candidate which is an initial value remains in the corresponding motion vector predictor candidate list. It is also possible to add a new candidate as a motion vector predictor candidate by using such a method.

[0195] Although the present embodiment describes an example in which the motion vector predictor designating mode is used in which motion vector predictor candidates are generated from blocks adjacent to a current block to be coded, and a motion vector of the current block is coded, the present embodiment is not necessarily limited to this. For example, a direct mode or a skip mode may be used. In the direct mode or the skip mode, a motion vector difference may not be added to a bitstream by selecting a motion vector predictor from among the motion vector predictor candidates created as shown in (b) in FIG. 15 and (b) in FIG. 16, and directly generating a predicted image using the selected motion vector predictor as a motion vector.

Embodiment 2

[0196] In Embodiment 1 above, the moving picture coding apparatus determines a bit string to be assigned to a motion vector predictor index using the number of available predictor candidates calculated by always adding 1 if a prediction block candidate is a co-located block irrespective of whether the co-located block is an available predictor candidate, the present invention is not limited to this. For example, the moving picture coding apparatus may determine a bit string to be assigned to a motion vector predictor index, using the number of available predictor candidates calculated by always adding 1 also in the case of a prediction block candidate other than the co-located block in step S124 in FIG. 18. Specifically, the moving picture coding apparatus may assign a bit string to a motion vector predictor index using the motion vector predictor candidate list size fixed to the maximum value N of the number of motion vector predictor candidates. In other words, the moving picture coding apparatus may assume that all prediction block

candidates are available predictor candidates, fix the motion vector predictor candidate list size to the maximum value N of the number of motion vector predictor candidates, and code motion vector predictor indices.

[0197] For example, in Embodiment 1 above, the maximum value N of the number of motion vector predictor candidates is 5 (adjacent block A, adjacent block B, co-located block, adjacent block C, adjacent block D), and thus the moving picture coding apparatus may always set the motion vector predictor candidate list size to 5, and code motion vector predictor indices. Further, for example, if the maximum value N of the number of motion vector predictor candidates is 4 (adjacent block A, adjacent block B, adjacent block C, adjacent block D), the moving picture coding apparatus may always set the motion vector predictor candidate list size to 4, and code motion vector predictor indices.

[0198] In this way, the moving picture coding apparatus may determine the motion vector predictor candidate list size according to the maximum value of the number of motion vector predictor candidates. Accordingly, it is possible to generate a bitstream from which the variable length decoding unit of the moving picture decoding apparatus can decode a motion vector predictor index in a bitstream without referring to information of adjacent blocks or a co-located block, which results in a reduction of the amount of processing to be performed by the variable length decoding unit.

[0199] The following is a specific description of a distinguishing configuration of such a moving picture coding apparatus as a moving picture coding apparatus according to Embodiment 2.

[0200] FIG. 21 is a block diagram showing a configuration of a moving picture coding apparatus **200** according to Embodiment 2. The moving picture coding apparatus **200** generates a bitstream by coding an image on a block-by-block basis. The moving picture coding apparatus **200** includes a motion vector predictor candidate derivation unit **210**, a prediction control unit **220**, and a coding unit **230**.

[0201] The motion vector predictor candidate derivation unit **210** corresponds to the motion vector predictor candidate calculation unit **114** in Embodiment 1 above. The motion vector predictor candidate derivation unit **210** derives motion vector predictor candidates. Then, the motion vector predictor candidate derivation unit **210** generates motion vector predictor candidate lists in which, for example, each of the derived motion vector predictor candidates is associated with an index (hereafter, referred to as “motion vector predictor index”) for identifying the motion vector predictor candidate.

[0202] A motion vector predictor candidate is a motion vector which is a candidate for a motion vector predictor to be used for coding a current block to be coded.

[0203] As shown in FIG. 21, the motion vector predictor candidate derivation unit **210** includes a maximum number determination unit **211**, a first derivation unit **212**, an identification unit **213**, a determination unit **214**, and a second derivation unit **215**.

[0204] The maximum number determination unit **211** determines the maximum number of motion vector predictor candidates. Specifically, the maximum number determination unit **211** determines the maximum value N of the number of prediction block candidates.

[0205] For example, the maximum number determination unit **211** determines the maximum number of motion vector predictor candidates, based on features of an input image sequence (sequence, pictures, slices, or blocks). Further, for example, the maximum number determination unit **211** may determine a predetermined number as the maximum number of motion vector predictor candidates.

[0206] The first derivation unit **212** derives one or more first motion vector predictor candidates. Specifically, the first derivation unit **212** derives the one or more first motion vector predictor candidates such that the number of first motion vector predictor candidates does not exceed the maximum number. More specifically, the first derivation unit **212** derives each first motion vector predictor candidate, based on a motion vector used for coding a block spatially or temporally adjacent to a current block to be coded, for example. Then, for example, the first derivation unit **212** registers, into the motion vector predictor candidate lists, the one or more first motion vector predictor candidates derived in this way, each in association with a motion vector predictor index.

[0207] A spatially adjacent block is a block in a picture which includes a current block to be coded, and is a block adjacent to the current block. Specifically, examples of spatially adjacent blocks are adjacent blocks A to D shown in FIG. 3.

[0208] A temporally adjacent block is a block included in a picture different from a picture which includes a current block to be coded, and is a block corresponding to the current block. Specifically, an example of a temporally adjacent block is a co-located block.

[0209] It should be noted that a temporally adjacent block does not necessarily need to be a block at the same position as that of a current block to be coded (co-located block). For example, a temporally adjacent block may be a block adjacent to a co-located block.

[0210] It should be noted that, for example, the first derivation unit **212** may derive, as the first motion vector predictor candidate, a motion vector used for coding a block that is a block spatially adjacent to a current block to be coded, and is not a block which is a non-available predictor candidate. A block which is a non-available predictor candidate is a block coded by intra prediction, a block located outside a boundary of a slice or a picture which includes a current block to be coded, or a block which is not coded yet. Accordingly, the first motion vector predictor candidate can be derived from a block suitable for obtaining a motion vector predictor candidate.

[0211] The identification unit **213** identifies a first motion vector predictor candidate (redundant candidate) having the same motion vector as that of any other first motion vector predictor candidate, if a plurality of the first motion vector predictor candidates are derived. Then, the identification unit **213** deletes the identified redundant candidate from a corresponding one of the motion vector predictor candidate lists.

[0212] The determination unit **214** determines whether the number of first motion vector predictor candidates is smaller than the determined maximum number. Here, the determination unit **214** determines whether the number of first motion vector predictor candidates excluding the identified redundant first motion vector predictor candidate is smaller than the determined maximum number.

[0213] The second derivation unit **215** derives one or more second motion vector predictor candidates if it is determined that the number of first motion vector predictor candidates is smaller than the determined maximum number. Specifically, the second derivation unit **215** derives the one or more second motion vector predictor candidates such that the sum of the number of first motion vector predictor candidates and the number of second motion vector predictor candidates does not exceed the maximum number. Here, the second derivation unit **215** derives the one or more second motion vector predictor candidates such that the sum of the number of first motion vector predictor candidates excluding a redundant candidate and the number of second motion vector predictor candidates does not exceed the maximum number.

[0214] The one or more second motion vector predictor candidates each correspond to the new candidate in Embodiment 1. Therefore, the second derivation unit **215** may derive each second motion vector predictor candidate, based on a motion vector which is used for coding an adjacent block and is different from the first motion vector predictor candidate, for example.

[0215] Furthermore, for example, the second derivation unit **215** may derive, as the second motion vector predictor candidate, a motion vector predictor candidate having a motion vector different from that of any first motion vector predictor candidate. Accordingly, the number of motion vector predictor candidates having different motion vectors can be increased, and thus coding efficiency can be further improved.

[0216] It should be noted that the second derivation unit **215** does not necessarily need to derive a motion vector predictor candidate different from that of any first motion vector predictor candidate, as the second motion vector predictor candidate. Specifically, the second derivation unit **215** may, as a consequence, derive the motion vector predictor candidate which is the same as the first motion vector predictor candidate, as the second motion vector predictor candidate.

[0217] Then, the second derivation unit **215** registers, into the motion vector predictor candidate lists, the one or more second motion vector predictor candidates derived in this way, each in association with a motion vector predictor index, for example. At this time, the second derivation unit **215** may register each second motion vector predictor candidate into a corresponding one of the motion vector predictor candidate lists, such that a motion vector predictor index having a value smaller than that for the second motion vector predictor candidates is assigned to each first motion vector predictor candidate, as in Embodiment 1. Accordingly, if there is a high possibility that the first motion vector predictor candidate will be selected as a motion vector predictor candidate to be used for coding rather than the second motion vector predictor candidate, the moving picture coding apparatus **200** can reduce the code amount, and improve coding efficiency.

[0218] It should be noted that the second derivation unit **215** does not necessarily need to derive the one or more second motion vector predictor candidates such that the sum of the number of first motion vector predictor candidates and the number of second motion vector predictor candidates will be the same as the determined maximum number. If the sum of the number of first motion vector predictor candidates and the number of second motion vector predictor

candidates is smaller than the determined maximum number, there may be a value of a motion vector predictor index which is not associated with a motion vector predictor candidate, for example.

[0219] The prediction control unit **220** selects a motion vector predictor to be used for coding a current block to be coded, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates. Specifically, the prediction control unit **220** selects, from the motion vector predictor candidate lists, a motion vector predictor to be used for coding the current block.

[0220] The coding unit **230** codes an index (motion vector predictor index) for identifying the selected motion vector predictor candidate, using the determined maximum number. Specifically, the coding unit **230** variable-length codes a bit string assigned to the index value of the selected motion vector predictor candidate, as shown in FIG. 6. Furthermore, the coding unit **230** adds the coded index to a bitstream.

[0221] Here, the coding unit **230** may further add information indicating the maximum number determined by the maximum number determination unit **211** to the bitstream. Specifically, the coding unit **230** may also write information indicating the maximum number, for example, into a slice header or the like. Accordingly, the maximum number can be changed in a suitable unit, which can improve coding efficiency.

[0222] It should be noted that the coding unit **230** does not necessarily need to add information indicating the maximum number to a bitstream. For example, if the maximum number is previously determined according to a standard, or if the maximum number is the same as a default value, the coding unit **230** does not need to add information indicating the maximum number to a bitstream.

[0223] Next is a description of various operations of the moving picture coding apparatus **200** constituted as described above.

[0224] FIG. 22 is a flowchart showing processing operation of the moving picture coding apparatus **200** according to Embodiment 2.

[0225] First, the maximum number determination unit **211** determines the maximum number of motion vector predictor candidates (S201). The first derivation unit **212** derives one or more first motion vector predictor candidates (S202). The identification unit **213** identifies a first motion vector predictor candidate having a motion vector which is the same as that of any other first motion vector predictor candidates if a plurality of the first motion vector predictor candidates are derived (S203).

[0226] The determination unit **214** determines whether the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (S204). Here, if it is determined that the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (Yes in S204), the second derivation unit **215** derives one or more second motion vector predictor candidates (S205). On the other hand, if it is not determined that the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (No in S204), the second derivation unit **215** does not derive a second motion vector predictor candidate. These steps S204 and S205 correspond to step S114 in Embodiment 1.

[0227] The prediction control unit **220** selects a motion vector predictor to be used for coding a current block to be coded from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidate (S206). For example, the prediction control unit **220** selects a motion vector predictor with which a motion vector difference is the smallest from the motion vector predictor candidate lists, as in Embodiment 1.

[0228] The coding unit **230** codes an index for identifying the selected motion vector predictor candidate using the determined maximum number (S207). Furthermore, the coding unit **230** adds the coded index to a bitstream.

[0229] As described above, according to the moving picture coding apparatus **200** according to the present embodiment, an index for identifying a motion vector predictor candidate can be coded using the determined maximum number. Specifically, an index can be coded without depending on the number of motion vector predictor candidates actually derived. Therefore, even if information (for example, information of a co-located block and the like) necessary for deriving a motion vector predictor candidate is lost, a decoding apparatus can decode the index, and thus error resistance can be improved. Further, the decoding apparatus can decode an index, without depending on the number of motion vector predictor candidates actually derived. Specifically, the decoding apparatus can decode the index, without waiting for derivation of a motion vector predictor candidate. Specifically, it is possible to generate a bitstream for which deriving a motion vector predictor candidate and decoding an index can be performed in parallel.

[0230] Furthermore, according to the moving picture coding apparatus **200** according to the present embodiment, one or more second motion vector predictor candidates can be derived if it is determined that the number of first motion vector predictor candidates is smaller than the maximum number. Therefore, it is possible to increase the number of motion vector predictor candidates in a range which does not exceed the maximum number, and to improve coding efficiency.

[0231] In addition, according to the moving picture coding apparatus **200** according to the present embodiment, one or more second motion vector predictor candidates can be derived according to the number of first motion vector predictor candidates excluding the redundant first motion vector predictor candidate. As a result, the number of second motion vector predictor candidates can be increased, and the types of selectable motion vectors can be increased. Therefore, it is possible to further improve coding efficiency.

[0232] It should be noted that in the present embodiment, although the moving picture coding apparatus **200** includes the identification unit **213**, the moving picture coding apparatus **200** does not necessarily need to include the identification unit **213**. Specifically, step S203 does not necessarily need to be included in the flowchart shown in FIG. 22. Even in such a case, the moving picture coding apparatus **200** can code an index for identifying a motion vector predictor candidate using the determined maximum number, and thus error resistance can be improved.

[0233] Further, as shown in FIG. 22, in the present embodiment, the first derivation unit **212** derives one or more first motion vector predictor candidates, and thereafter the identification unit **213** identifies a redundant candidate. However, the processing does not necessarily need to be

sequentially performed in this way. For example, in the process of deriving the first motion vector predictor candidates, the first derivation unit 212 may identify a redundant candidate, and derive the first motion vector predictor candidates such that the identified redundant candidate is not included in the first motion vector predictor candidates. Specifically, the first derivation unit 212 may derive a motion vector predictor candidate which is not the same as any first motion vector predictor candidate whose motion vector has already been derived, as the first motion vector predictor candidate. More specifically, for example, when a motion vector predictor candidate based on a left adjacent block is already derived as the first motion vector predictor candidate, if a motion vector predictor candidate based on an upper adjacent block is not the same as the motion vector predictor candidate based on the left adjacent block, the first derivation unit 212 may derive the motion vector predictor candidate based on the upper adjacent block as the first motion vector predictor candidate.

Embodiment 3

[0234] FIG. 23 is a block diagram showing a configuration of a moving picture decoding apparatus 300 according to Embodiment 3.

[0235] As shown in FIG. 23, the moving picture decoding apparatus 300 includes a variable length decoding unit 301, an inverse quantization unit 302, an inverse orthogonal transform unit 303, an addition unit 304, a block memory 305, a frame memory 306, an intra prediction unit 307, an inter prediction unit 308, an inter prediction control unit 309, a switch 310, a motion vector predictor candidate calculation unit 311, and a colPic memory 312.

[0236] The variable length decoding unit 301 performs variable length decoding processing on an input bitstream, and generates picture type information, a prediction direction flag, a quantization coefficient, and a motion vector difference. Further, the variable length decoding unit 301 performs variable length decoding processing on motion vector predictor indices using the number of available predictor candidates described below.

[0237] The inverse quantization unit 302 performs inverse quantization processing on the quantization coefficient obtained by variable length decoding processing. The inverse orthogonal transform unit 303 transforms an orthogonal transform coefficient obtained by inverse quantization processing from a frequency domain into an image domain, to generate prediction error data. The block memory 305 stores decoded image data generated by adding prediction error data and predicted image data, on a block-by-block basis. The frame memory 306 stores decoded image data on a frame-by-frame basis.

[0238] The intra prediction unit 307 generates predicted image data of a current block to be decoded by performing intra prediction using decoded image data in block units stored in the block memory 305. The inter prediction unit 308 generates predicted image data of a current block to be decoded by performing inter prediction using decoded image data in frame units stored in the frame memory 306.

[0239] If intra prediction decoding is performed on the current block, the switch 310 outputs the intra-predicted image data generated by the intra prediction unit 307 to the addition unit 304 as predicted image data of the current block. In contrast, if inter prediction decoding is performed on the current block, the switch 310 outputs the inter-

predicted image data generated by the inter prediction unit 308 to the addition unit 304 as predicted image data of the current block.

[0240] Using, for instance, motion vectors of blocks adjacent to a current block to be decoded and colPic information such as information of a motion vector of a co-located block stored in the colPic memory 312, the motion vector predictor candidate calculation unit 311 derives motion vector predictor candidates in the motion vector predictor designating mode by using the method described below. Further, the motion vector predictor candidate calculation unit 311 assigns a value of a motion vector predictor index to each derived motion vector predictor candidate. Then, the motion vector predictor candidate calculation unit 311 sends the motion vector predictor candidates and the motion vector predictor indices to the inter prediction control unit 309.

[0241] The inter prediction control unit 309 selects, from among the motion vector predictor candidates, a motion vector predictor to be used for inter prediction, based on the decoded motion vector predictor index. Then, the inter prediction control unit 309 calculates a motion vector of the current block, based on the motion vector predictor and a motion vector difference. Then, the inter prediction control unit 309 causes the inter prediction unit 308 to generate an inter-predicted image using the calculated motion vector. Further, the inter prediction control unit 309 transfers colPic information including information of the motion vector of the current block and the like to the colPic memory 312.

[0242] Finally, the addition unit 304 generates decoded image data by adding predicted image data and prediction error data.

[0243] FIG. 24 is a flowchart showing processing operation of the moving picture decoding apparatus 300 according to Embodiment 3.

[0244] In step S301, the variable length decoding unit 301 decodes a prediction direction flag and a reference picture index. Then, the value of the prediction direction X is determined according to the decoded prediction direction flag, and processing of the following steps S302 to S305 is performed.

[0245] In step S302, the motion vector predictor candidate calculation unit 311 calculates the number of available predictor candidates using the method described below. Then, the motion vector predictor candidate calculation unit 311 sets the motion vector predictor candidate list size to the calculated number of available predictor candidates.

[0246] In step S303, the variable length decoding unit 301 variable-length decodes the motion vector predictor index in a bitstream using the calculated motion vector predictor candidate list size. In step S304, the motion vector predictor candidate calculation unit 311 generates motion vector predictor candidates from blocks adjacent to the current block and a co-located block using the method described below. In step S305, the inter prediction control unit 309 adds the decoded motion vector difference to the motion vector predictor candidate indicated by the decoded motion vector predictor index, to calculate a motion vector. Then, the inter prediction control unit 309 causes the inter prediction unit 308 to generate an inter-predicted image using the calculated motion vector.

[0247] It should be noted that if the motion vector predictor candidate list size calculated in step S302 is "1", it may be estimated that a motion vector predictor index is 0, without being decoded.

[0248] FIG. 25 is a flowchart showing detailed processing of step S302 in FIG. 24. Specifically, FIG. 25 shows a method for determining whether a prediction block candidate [N] is an available predictor candidate, and calculating the number of available predictor candidates. The following is a description of FIG. 25.

[0249] In step S311, the motion vector predictor candidate calculation unit 311 determines whether a prediction block candidate [N] is (1) decoded by intra prediction, (2) located outside a boundary of a slice or a picture which includes a current block to be decoded, or (3) not decoded yet.

[0250] Here, if the determination result in step S311 is true (Yes in S311), the motion vector predictor candidate calculation unit 311 sets the prediction block candidate [N] as a non-available predictor candidate in step S312. On the other hand, if the determination result in step S311 is false (No in S311), the motion vector predictor candidate calculation unit 311 sets the prediction block candidate [N] as an available predictor candidate in step S313.

[0251] In step S314, the motion vector predictor candidate calculation unit 311 determines whether the prediction block candidate [N] is an available predictor candidate or a co-located block candidate. Here, if the determination result in step S314 is true (Yes in S314), the motion vector predictor candidate calculation unit 311 adds 1 to the number of available predictor candidates, and updates the value in step S5. On the other hand, if the determination result in step S314 is false (No in S314), the motion vector predictor candidate calculation unit 311 does not update the number of available predictor candidates.

[0252] As described above, if a prediction block candidate is a co-located block, the motion vector predictor candidate calculation unit 311 adds 1 to the number of available predictor candidates, irrespective of whether the co-located block is an available predictor candidate or a non-available predictor candidate. Accordingly, even if information of a co-located block is lost due to packet loss or the like, there is no difference in the number of available predictor candidates between the moving picture coding apparatus and the moving picture decoding apparatus.

[0253] The motion vector predictor candidate list size is set to the number of available predictor candidates in step S302 in FIG. 24. Furthermore, in S303 in FIG. 24, the motion vector predictor candidate list size is used for variable-length decoding motion vector predictor indices. Accordingly, even if reference picture information including information of a co-located block or the like is lost, the moving picture decoding apparatus 300 can successfully decode motion vector predictor indices.

[0254] FIG. 26 is a flowchart showing detailed processing of step S304 in FIG. 24. Specifically, FIG. 26 shows a method for calculating motion vector predictor candidates. The following is a description of FIG. 26.

[0255] In step S321, the motion vector predictor candidate calculation unit 311 calculates, from the prediction block candidate [N], a motion vector predictor candidate in the prediction direction X using Expressions 1 and 2 above, and adds the calculated candidate to a corresponding one of the motion vector predictor candidate lists.

[0256] In step S322, the motion vector predictor candidate calculation unit 311 searches for and deletes a non-available predictor candidate and a redundant candidate from the motion vector predictor candidate lists, as shown in FIGS. 15 and 16.

[0257] In step S323, the motion vector predictor candidate calculation unit 311 adds a new candidate to a corresponding one of the motion vector predictor candidate lists using the same method as in FIG. 19.

[0258] FIG. 27 shows an example of syntax used when a motion vector predictor index is added to a bitstream. In FIG. 27, inter_pred_flag indicates a prediction direction flag, and mvp_idx indicates a motion vector predictor index. NumMVPCand indicates the motion vector predictor candidate list size, and the size is set to the number of available predictor candidates calculated in the processing flow in FIG. 25 in the present embodiment.

[0259] As described above, according to the moving picture decoding apparatus 300 according to the present embodiment, the motion vector predictor candidate list size to be used when a motion vector predictor index is coded or decoded can be calculated by using a method independent of reference picture information including information of a co-located block and the like. Accordingly, the moving picture decoding apparatus 300 can appropriately decode a bitstream having improved error resistance.

[0260] More specifically, the moving picture decoding apparatus 300 according to the present embodiment always adds 1 to the number of available predictor candidates if a prediction block candidate is a co-located block, irrespective of whether the co-located block is an available predictor candidate. Then, the moving picture decoding apparatus 300 determines bit strings to be assigned to motion vector predictor indices using the number of available predictor candidates calculated in this way. Accordingly, even if reference picture information including information of a co-located block is lost, the moving picture decoding apparatus 300 can successfully decode a motion vector predictor index.

[0261] Further, if the number of motion vector predictor candidates has not reached the number of available predictor candidates, the moving picture decoding apparatus 300 according to the present embodiment can appropriately decode a bitstream for which coding efficiency has been improved by adding a new candidate having a new motion vector predictor as a motion vector predictor candidate.

[0262] It should be noted that in the present embodiment, although the moving picture decoding apparatus 300 adds a new candidate having a new motion vector predictor as a motion vector predictor candidate if the number of motion vector predictor candidates has not reached the number of available predictor candidates, the present invention is not limited to this. For example, as in Embodiment 1 described above, when creating the motion vector predictor candidate lists, the moving picture decoding apparatus 300 may set a new candidate having a new motion vector predictor as an initial value of all the motion vector predictor candidates on the motion vector predictor candidate lists.

Embodiment 4

[0263] In Embodiment 3 above, although the moving picture decoding apparatus determines bit strings to be assigned to motion vector predictor indices using the number of available predictor candidates calculated by always adding 1 when a prediction block candidate is a co-located block, irrespective of whether the co-located block is an available predictor candidate, the present invention is not limited to this. For example, the moving picture decoding apparatus may determine bit strings to be assigned to motion

vector predictor indices, using the number of available predictor candidates calculated by also always adding 1 in the case of a prediction block candidate other than a co-located block in step S314 in FIG. 25. Specifically, the moving picture decoding apparatus may assign a bit string to a motion vector predictor index, using the motion vector predictor candidate list size fixed to the maximum value N of the number of motion vector predictor candidates. In other words, assuming that all prediction block candidates are available predictor candidates, the moving picture decoding apparatus may fix the motion vector predictor candidate list size to the maximum value N of the number of motion vector predictor candidates, and decode motion vector predictor indices.

[0264] For example, in Embodiment 3 above, since the maximum value N of the number of motion vector predictor candidates is 5 (adjacent block A, adjacent block B, co-located block, adjacent block C, adjacent block D), the moving picture decoding apparatus may always set the motion vector predictor candidate list size to 5, and decode motion vector predictor indices. Accordingly, the variable length decoding unit of the moving picture decoding apparatus can decode a motion vector predictor index in a bitstream, without referring to information of adjacent blocks or a co-located block. As a result, for example, processing of steps S314 and S315 in FIG. 25, for instance, can be skipped, and thus the amount of processing to be performed by the variable length decoding unit can be reduced.

[0265] FIG. 28 shows an example of syntax used when the motion vector predictor candidate list size is fixed to the maximum value of the number of motion vector predictor candidates. As shown in FIG. 28, NumMVPCand can be deleted from the syntax if the motion vector predictor candidate list size is fixed to the maximum value of the number of motion vector predictor candidates.

[0266] The following is a specific description of a distinguishing configuration of such a moving picture decoding apparatus, as a moving picture decoding apparatus according to Embodiment 4.

[0267] FIG. 29 is a block diagram showing a configuration of a moving picture decoding apparatus 400 according to Embodiment 4. The moving picture decoding apparatus 400 decodes a coded image included in a bitstream on a block-by-block basis. Specifically, the moving picture decoding apparatus 400 decodes, on a block-by-block basis, a coded image included in a bitstream generated by the moving picture coding apparatus 200 according to Embodiment 2, for example. The moving picture decoding apparatus 400 includes a motion vector predictor candidate derivation unit 410, a decoding unit 420, and a prediction control unit 430.

[0268] The motion vector predictor candidate derivation unit 410 corresponds to the motion vector predictor candidate calculation unit 311 in Embodiment 3 above. The motion vector predictor candidate derivation unit 410 derives motion vector predictor candidates. Then, the motion vector predictor candidate derivation unit 410 generates motion vector predictor candidate lists in which each derived motion vector predictor candidate is associated with an index for identifying the motion vector predictor candidate (motion vector predictor index), for example.

[0269] As shown in FIG. 29, the motion vector predictor candidate derivation unit 410 includes a maximum number

determination unit 411, a first derivation unit 412, an identification unit 413, a determination unit 414, and a second derivation unit 415.

[0270] The maximum number determination unit 411 determines the maximum number of motion vector predictor candidates. Specifically, the maximum number determination unit 211 determines the maximum value N of the number of prediction block candidates.

[0271] For example, the maximum number determination unit 411 determines the maximum number of motion vector predictor candidates, using the same method as that used by the maximum number determination unit 211 in Embodiment 2. Further, for example, the maximum number determination unit 411 may determine the maximum number, based on information indicating the maximum number added to a bitstream.

[0272] It should be noted that here, although the maximum number determination unit 411 is included in the motion vector predictor candidate derivation unit 410, the maximum number determination unit 411 may be included in the decoding unit 420.

[0273] The first derivation unit 412 derives one or more first motion vector predictor candidates. Specifically, the first derivation unit 412 derives one or more first motion vector predictor candidates, in the same manner as that for the first derivation unit 212 in Embodiment 2. For example, the first derivation unit 412 derives the first motion vector predictor candidates such that the number of first motion vector predictor candidates does not exceed the maximum number. More specifically, the first derivation unit 412 derives each first motion vector predictor candidate, based on a motion vector used for decoding a block spatially or temporally adjacent to a current block to be decoded, for example. Then, the first derivation unit 412 registers, into the motion vector predictor candidate lists, the one or more first motion vector predictor candidates derived in this way, each in association with a motion vector predictor index, for example.

[0274] It should be noted that the first derivation unit 412 may derive, as the first motion vector predictor candidate, a motion vector used for decoding a block which is spatially adjacent to a current block to be decoded, and is not a non-available predictor candidate, for example. Accordingly, the first motion vector predictor candidate can be derived from a block suitable for obtaining a motion vector predictor candidate.

[0275] If a plurality of the first motion vector predictor candidates are derived, the identification unit 413 identifies a first motion vector predictor candidate (redundant candidate) having the same motion vector as that of any other first motion vector predictor candidates. Then, the identification unit 413 deletes the identified redundant candidate from a corresponding one of the motion vector predictor candidate lists.

[0276] The determination unit 414 determines whether the number of first motion vector predictor candidates is smaller than the determined maximum number. Here, the determination unit 414 determines whether the number of first motion vector predictor candidates excluding the identified redundant first motion vector predictor candidate is smaller than the determined maximum number.

[0277] If it is determined that the number of first motion vector predictor candidates is smaller than the determined maximum number, the second derivation unit 415 derives

one or more second motion vector predictor candidates. Specifically, the second derivation unit **415** derives one or more second motion vector predictor candidates in the same manner as that for the second derivation unit **215** in Embodiment 2.

[0278] For example, the second derivation unit **415** may derive a motion vector predictor candidate having a motion vector different from that of any first motion vector predictor candidate, as the second motion vector predictor candidate. Accordingly, the number of motion vector predictor candidates having different motion vectors can be increased, which allows decoding of a coded image for which coding efficiency has been further improved.

[0279] Then, the second derivation unit **415** registers, into the motion vector predictor candidate lists, the one or more second motion vector predictor candidates derived in this way, each in association with a motion vector predictor index, as in the same manner for the second derivation unit **215** in Embodiment 2, for example.

[0280] The decoding unit **420** decodes, using the determined maximum number, a coded index added to a bitstream and used for identifying a motion vector predictor candidate.

[0281] The prediction control unit **430** selects a motion vector predictor to be used for decoding a current block to be decoded, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates, based on the decoded index. Specifically, the prediction control unit **430** selects, from the motion vector predictor candidate lists, a motion vector predictor to be used for decoding a current block to be decoded.

[0282] Next is a description of various operations of the moving picture decoding apparatus **400** constituted as described above.

[0283] FIG. 30 is a flowchart showing processing operation of the moving picture decoding apparatus **400** according to Embodiment 4.

[0284] First, the maximum number determination unit **411** determines the maximum number of motion vector predictor candidates (**S401**). The first derivation unit **412** derives one or more first motion vector predictor candidates (**S402**). If a plurality of the first motion vector predictor candidates are derived, the identification unit **413** identifies a first motion vector predictor candidate having the same motion vector as that of any other first motion vector predictor candidate (**S403**).

[0285] The determination unit **414** determines whether the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (**S404**). Here, if it is determined that the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (Yes in **S404**), the second derivation unit **415** derives one or more second motion vector predictor candidates (**S405**). On the other hand, if it is not determined that the number of first motion vector predictor candidates excluding a redundant candidate is smaller than the determined maximum number (No in **S404**), the second derivation unit **415** does not derive a second motion vector predictor candidate.

[0286] The decoding unit **420** decodes a coded index added to a bitstream and used for identifying a motion vector predictor candidate, using the determined maximum number (**S406**).

[0287] The prediction control unit **430** selects, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates, a motion vector predictor to be used for decoding a current block to be decoded, based on the decoded index (**S407**).

[0288] It should be noted that here, although an index is decoded (**S406**) after a motion vector predictor candidate is derived, the processing does not necessarily need to be performed in such an order. For example, processing for deriving a motion vector predictor candidate (**S402** to **S405**) may be performed after decoding an index (**S406**). Further, decoding an index (**S406**) and deriving a motion vector predictor candidate (**S402** to **S405**) may be performed in parallel. Accordingly, the decoding processing speed can be increased.

[0289] As described above, according to the moving picture decoding apparatus **400** according to the present embodiment, an index for identifying a motion vector predictor candidate can be decoded using the determined maximum number. Specifically, an index can be decoded without depending on the number of motion vector predictor candidates actually derived. Therefore, even if information necessary for deriving a motion vector predictor candidate (for example, information of a co-located block and the like) is lost, an index can be decoded, and error resistance can be improved. Furthermore, an index can be decoded without waiting for derivation of a motion vector predictor candidate, and thus deriving a motion vector predictor candidate and decoding an index can also be performed in parallel.

[0290] Furthermore, according to the moving picture decoding apparatus **400** according to the present embodiment, if it is determined that the number of first motion vector predictor candidates is smaller than the maximum number, one or more second motion vector predictor candidates can be derived. Therefore, the number of motion vector predictor candidates can be increased in a range which does not exceed the maximum number, and thus a coded image for which coding efficiency has been improved can be decoded.

[0291] Further, according to the moving picture decoding apparatus **400** according to the present embodiment, one or more second motion vector predictor candidates can be derived according to the number of first motion vector predictor candidates excluding a redundant first motion vector predictor candidate. As a result, the number of second motion vector predictor candidates can be increased, and the types of selectable combinations of a prediction direction, a motion vector, and a reference picture index can be increased. Therefore, it is possible to decode a coded image for which coding efficiency has been further improved.

[0292] It should be noted that in the present embodiment, although the moving picture decoding apparatus **400** includes the identification unit **413**, the moving picture decoding apparatus **400** does not necessarily need to include the identification unit **413**, as in Embodiment 2. In other words, step **S403** does not necessarily need to be included in the flowchart shown in FIG. 30. Even in such a case, the moving picture decoding apparatus **400** can decode an index for identifying a motion vector predictor candidate using the determined maximum number, and thus can improve error resistance.

[0293] Further, although in the present embodiment, the first derivation unit **412** derives the first motion vector

predictor candidates, and thereafter the identification unit 413 identifies a redundant candidate as shown in FIG. 30, the processing does not necessarily need to be performed sequentially in this way. For example, the first derivation unit 412 may derive a motion vector predictor candidate having a motion vector that is not the same as that of any first motion vector predictor candidate already derived, as the first motion vector predictor candidate.

[0294] Although the above is a description of the moving picture coding apparatus and the moving picture decoding apparatus according to one or more aspects of the present invention, based on the embodiments, the present invention is not limited to the above embodiments. The herein disclosed subject matter is to be considered descriptive and illustrative only, and the appended Claims are of a scope intended to cover and encompass not only the particular embodiments disclosed, but also equivalent structures, methods, and/or uses.

[0295] Each of the constituent elements in the above-described embodiments may be configured in the form of an exclusive hardware product, or may be realized by executing a software program suitable for the structural element. Each of the constituent elements may be realized by means that a program executing unit such as a CPU and a processor reads and executes the software program recorded on a recording medium such as a hard disc or a semiconductor memory. Here, the software program for realizing the moving picture coding apparatus or the moving picture decoding apparatus according to the above embodiments is a program described below.

[0296] Specifically, the program causes a computer to execute a moving picture coding method for calculating a motion vector predictor to be used when coding a motion vector of a current block to be coded, and coding the current block, to generate a bitstream, the method including: determining a maximum number of motion vector predictor candidates each of which is a candidate for the motion vector predictor; deriving one or more first motion vector predictor candidates; determining whether a total number of the one or more first motion vector predictor candidates is smaller than the maximum number; deriving one or more second motion vector predictor candidates when it is determined that the total number of one or more first motion vector predictor candidates is smaller than the maximum number; selecting, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates, the motion vector predictor to be used for coding the motion vector of the current block; and coding, using the determined maximum number, an index for identifying the selected motion vector predictor, and adding the coded index to the bitstream.

[0297] Alternatively, the program causes a computer to execute a moving picture decoding method for calculating a motion vector predictor to be used when decoding a motion vector of a current block to be decoded which is included in a bitstream and decoding the current block, the method including: determining a maximum number of motion vector predictor candidates each of which is a candidate for the motion vector predictor; deriving one or more first motion vector predictor candidates; determining whether a total number of the one or more first motion vector predictor candidates is smaller than the maximum number; deriving one or more second motion vector predictor candidates when it is determined that the total number of one or more

first motion vector predictor candidates is smaller than the maximum number; decoding, using the determined maximum number, a coded index added to the bitstream and used for identifying the motion vector predictor; and selecting, based on the decoded index, a motion vector predictor to be used for decoding the current block, from among the one or more first motion vector predictor candidates and the one or more second motion vector predictor candidates.

Embodiment 5

[0298] The processing described in each of embodiments can be simply implemented in an independent computer system, by recording, in a recording medium, a program for implementing the configurations of the moving picture coding method (image coding method) and the moving picture decoding method (image decoding method) described in each of embodiments. The recording media may be any recording media as long as the program can be recorded, such as a magnetic disk, an optical disk, a magnetic optical disk, an IC card, and a semiconductor memory.

[0299] Hereinafter, the applications to the moving picture coding method (image coding method) and the moving picture decoding method (image decoding method) described in each of embodiments and systems using thereof will be described. The system has a feature of having an image coding and decoding apparatus that includes an image coding apparatus using the image coding method and an image decoding apparatus using the image decoding method. Other configurations in the system can be changed as appropriate depending on the cases.

[0300] FIG. 31 illustrates an overall configuration of a content providing system ex100 for implementing content distribution services. The area for providing communication services is divided into cells of desired size, and base stations ex106, ex107, ex108, ex109, and ex110 which are fixed wireless stations are placed in each of the cells.

[0301] The content providing system ex100 is connected to devices, such as a computer ex111, a personal digital assistant (PDA) ex112, a camera ex113, a cellular phone ex114 and a game machine ex115, via the Internet ex101, an Internet service provider ex102, a telephone network ex104, as well as the base stations ex106 to ex110, respectively.

[0302] However, the configuration of the content providing system ex100 is not limited to the configuration shown in FIG. 31, and a combination in which any of the elements are connected is acceptable. In addition, each device may be directly connected to the telephone network ex104, rather than via the base stations ex106 to ex110 which are the fixed wireless stations. Furthermore, the devices may be interconnected to each other via a short distance wireless communication and others.

[0303] The camera ex113, such as a digital video camera, is capable of capturing video. A camera ex116, such as a digital camera, is capable of capturing both still images and video. Furthermore, the cellular phone ex114 may be the one that meets any of the standards such as Global System for Mobile Communications (GSM) (registered trademark), Code Division Multiple Access (CDMA), Wideband-Code Division Multiple Access (W-CDMA), Long Term Evolution (LTE), and High Speed Packet Access (HSPA). Alternatively, the cellular phone ex114 may be a Personal Handyphone System (PHS).

[0304] In the content providing system ex100, a streaming server ex103 is connected to the camera ex113 and others via

the telephone network ex104 and the base station ex109, which enables distribution of images of a live show and others. In such a distribution, a content (for example, video of a music live show) captured by the user using the camera ex113 is coded as described above in each of embodiments (i.e., the camera functions as the image coding apparatus according to an aspect of the present invention), and the coded content is transmitted to the streaming server ex103. On the other hand, the streaming server ex103 carries out stream distribution of the transmitted content data to the clients upon their requests. The clients include the computer ex111, the PDA ex112, the camera ex113, the cellular phone ex114, and the game machine ex115 that are capable of decoding the above-mentioned coded data. Each of the devices that have received the distributed data decodes and reproduces the coded data (i.e., functions as the image decoding apparatus according to an aspect of the present invention).

[0305] The captured data may be coded by the camera ex113 or the streaming server ex103 that transmits the data, or the coding processes may be shared between the camera ex113 and the streaming server ex103. Similarly, the distributed data may be decoded by the clients or the streaming server ex103, or the decoding processes may be shared between the clients and the streaming server ex103. Furthermore, the data of the still images and video captured by not only the camera ex113 but also the camera ex116 may be transmitted to the streaming server ex103 through the computer ex111. The coding processes may be performed by the camera ex116, the computer ex111, or the streaming server ex103, or shared among them.

[0306] Furthermore, the coding and decoding processes may be performed by an LSI ex500 generally included in each of the computer ex111 and the devices. The LSI ex500 may be configured of a single chip or a plurality of chips. Software for coding and decoding video may be integrated into some type of a recording medium (such as a CD-ROM, a flexible disk, and a hard disk) that is readable by the computer ex111 and others, and the coding and decoding processes may be performed using the software. Furthermore, when the cellular phone ex114 is equipped with a camera, the video data obtained by the camera may be transmitted. The video data is data coded by the LSI ex500 included in the cellular phone ex114.

[0307] Furthermore, the streaming server ex103 may be composed of servers and computers, and may decentralize data and process the decentralized data, record, or distribute data.

[0308] As described above, the clients may receive and reproduce the coded data in the content providing system ex100. In other words, the clients can receive and decode information transmitted by the user, and reproduce the decoded data in real time in the content providing system ex100, so that the user who does not have any particular right and equipment can implement personal broadcasting.

[0309] Aside from the example of the content providing system ex100, at least one of the moving picture coding apparatus (image coding apparatus) and the moving picture decoding apparatus (image decoding apparatus) described in each of embodiments may be implemented in a digital broadcasting system ex200 illustrated in FIG. 32. More specifically, a broadcast station ex201 communicates or transmits, via radio waves to a broadcast satellite ex202, multiplexed data obtained by multiplexing audio data and

others onto video data. The video data is data coded by the moving picture coding method described in each of embodiments (i.e., data coded by the image coding apparatus according to an aspect of the present invention). Upon receipt of the multiplexed data, the broadcast satellite ex202 transmits radio waves for broadcasting. Then, a home-use antenna ex204 with a satellite broadcast reception function receives the radio waves. Next, a device such as a television (receiver) ex300 and a set top box (STB) ex217 decodes the received multiplexed data, and reproduces the decoded data (i.e., functions as the image decoding apparatus according to an aspect of the present invention).

[0310] Furthermore, a reader/recorder ex218 (i) reads and decodes the multiplexed data recorded on a recording medium ex215, such as a DVD and a BD, or (i) codes video signals in the recording medium ex215, and in some cases, writes data obtained by multiplexing an audio signal on the coded data. The reader/recorder ex218 can include the moving picture decoding apparatus or the moving picture coding apparatus as shown in each of embodiments. In this case, the reproduced video signals are displayed on the monitor ex219, and can be reproduced by another device or system using the recording medium ex215 on which the multiplexed data is recorded. It is also possible to implement the moving picture decoding apparatus in the set top box ex217 connected to the cable ex203 for a cable television or to the antenna ex204 for satellite and/or terrestrial broadcasting, so as to display the video signals on the monitor ex219 of the television ex300. The moving picture decoding apparatus may be implemented not in the set top box but in the television ex300.

[0311] FIG. 33 illustrates the television (receiver) ex300 that uses the moving picture coding method and the moving picture decoding method described in each of embodiments. The television ex300 includes: a tuner ex301 that obtains or provides multiplexed data obtained by multiplexing audio data onto video data, through the antenna ex204 or the cable ex203, etc. that receives a broadcast; a modulation/demodulation unit ex302 that demodulates the received multiplexed data or modulates data into multiplexed data to be supplied outside; and a multiplexing/demultiplexing unit ex303 that demultiplexes the modulated multiplexed data into video data and audio data, or multiplexes video data and audio data coded by a signal processing unit ex306 into data.

[0312] The television ex300 further includes: a signal processing unit ex306 including an audio signal processing unit ex304 and a video signal processing unit ex305 that decode audio data and video data and code audio data and video data, respectively (which function as the image coding apparatus and the image decoding apparatus according to the aspects of the present invention); and an output unit ex309 including a speaker ex307 that provides the decoded audio signal, and a display unit ex308 that displays the decoded video signal, such as a display. Furthermore, the television ex300 includes an interface unit ex317 including an operation input unit ex312 that receives an input of a user operation. Furthermore, the television ex300 includes a control unit ex310 that controls overall each constituent element of the television ex300, and a power supply circuit unit ex311 that supplies power to each of the elements. Other than the operation input unit ex312, the interface unit ex317 may include: a bridge ex313 that is connected to an external device, such as the reader/recorder ex218; a slot unit ex314 for enabling attachment of the recording medium ex216,

such as an SD card; a driver ex315 to be connected to an external recording medium, such as a hard disk; and a modem ex316 to be connected to a telephone network. Here, the recording medium ex216 can electrically record information using a non-volatile/volatile semiconductor memory element for storage. The constituent elements of the television ex300 are connected to each other through a synchronous bus.

[0313] First, the configuration in which the television ex300 decodes multiplexed data obtained from outside through the antenna ex204 and others and reproduces the decoded data will be described. In the television ex300, upon a user operation through a remote controller ex220 and others, the multiplexing/demultiplexing unit ex303 demultiplexes the multiplexed data demodulated by the modulation/demodulation unit ex302, under control of the control unit ex310 including a CPU. Furthermore, the audio signal processing unit ex304 decodes the demultiplexed audio data, and the video signal processing unit ex305 decodes the demultiplexed video data, using the decoding method described in each of embodiments, in the television ex300. The output unit ex309 provides the decoded video signal and audio signal outside, respectively. When the output unit ex309 provides the video signal and the audio signal, the signals may be temporarily stored in buffers ex318 and ex319, and others so that the signals are reproduced in synchronization with each other. Furthermore, the television ex300 may read multiplexed data not through a broadcast and others but from the recording media ex215 and ex216, such as a magnetic disk, an optical disk, and a SD card. Next, a configuration in which the television ex300 codes an audio signal and a video signal, and transmits the data outside or writes the data on a recording medium will be described. In the television ex300, upon a user operation through the remote controller ex220 and others, the audio signal processing unit ex304 codes an audio signal, and the video signal processing unit ex305 codes a video signal, under control of the control unit ex310 using the coding method described in each of embodiments. The multiplexing/demultiplexing unit ex303 multiplexes the coded video signal and audio signal, and provides the resulting signal outside. When the multiplexing/demultiplexing unit ex303 multiplexes the video signal and the audio signal, the signals may be temporarily stored in the buffers ex320 and ex321, and others so that the signals are reproduced in synchronization with each other. Here, the buffers ex318, ex319, ex320, and ex321 may be plural as illustrated, or at least one buffer may be shared in the television ex300. Furthermore, data may be stored in a buffer so that the system overflow and underflow may be avoided between the modulation/demodulation unit ex302 and the multiplexing/demultiplexing unit ex303, for example.

[0314] Furthermore, the television ex300 may include a configuration for receiving an AV input from a microphone or a camera other than the configuration for obtaining audio and video data from a broadcast or a recording medium, and may code the obtained data. Although the television ex300 can code, multiplex, and provide outside data in the description, it may be capable of only receiving, decoding, and providing outside data but not the coding, multiplexing, and providing outside data.

[0315] Furthermore, when the reader/recorder ex218 reads or writes multiplexed data from or on a recording medium, one of the television ex300 and the reader/recorder ex218

may decode or code the multiplexed data, and the television ex300 and the reader/recorder ex218 may share the decoding or coding.

[0316] As an example, FIG. 34 illustrates a configuration of an information reproducing/recording unit ex400 when data is read or written from or on an optical disk. The information reproducing/recording unit ex400 includes constituent elements ex401, ex402, ex403, ex404, ex405, ex406, and ex407 to be described hereinafter. The optical head ex401 irradiates a laser spot in a recording surface of the recording medium ex215 that is an optical disk to write information, and detects reflected light from the recording surface of the recording medium ex215 to read the information. The modulation recording unit ex402 electrically drives a semiconductor laser included in the optical head ex401, and modulates the laser light according to recorded data. The reproduction demodulating unit ex403 amplifies a reproduction signal obtained by electrically detecting the reflected light from the recording surface using a photo detector included in the optical head ex401, and demodulates the reproduction signal by separating a signal component recorded on the recording medium ex215 to reproduce the necessary information. The buffer ex404 temporarily holds the information to be recorded on the recording medium ex215 and the information reproduced from the recording medium ex215. The disk motor ex405 rotates the recording medium ex215. The servo control unit ex406 moves the optical head ex401 to a predetermined information track while controlling the rotation drive of the disk motor ex405 so as to follow the laser spot. The system control unit ex407 controls overall the information reproducing/recording unit ex400. The reading and writing processes can be implemented by the system control unit ex407 using various information stored in the buffer ex404 and generating and adding new information as necessary, and by the modulation recording unit ex402, the reproduction demodulating unit ex403, and the servo control unit ex406 that record and reproduce information through the optical head ex401 while being operated in a coordinated manner. The system control unit ex407 includes, for example, a microprocessor, and executes processing by causing a computer to execute a program for read and write.

[0317] Although the optical head ex401 irradiates a laser spot in the description, it may perform high-density recording using near field light.

[0318] FIG. 35 illustrates the recording medium ex215 that is the optical disk. On the recording surface of the recording medium ex215, guide grooves are spirally formed, and an information track ex230 records, in advance, address information indicating an absolute position on the disk according to change in a shape of the guide grooves. The address information includes information for determining positions of recording blocks ex231 that are a unit for recording data. Reproducing the information track ex230 and reading the address information in an apparatus that records and reproduces data can lead to determination of the positions of the recording blocks. Furthermore, the recording medium ex215 includes a data recording area ex233, an inner circumference area ex232, and an outer circumference area ex234. The data recording area ex233 is an area for use in recording the user data. The inner circumference area ex232 and the outer circumference area ex234 that are inside and outside of the data recording area ex233, respectively are for specific use except for recording the user data. The

information reproducing/recording unit 400 reads and writes coded audio, coded video data, or multiplexed data obtained by multiplexing the coded audio and video data, from and on the data recording area ex233 of the recording medium ex215.

[0319] Although an optical disk having a layer, such as a DVD and a BD is described as an example in the description, the optical disk is not limited to such, and may be an optical disk having a multilayer structure and capable of being recorded on a part other than the surface. Furthermore, the optical disk may have a structure for multidimensional recording/reproduction, such as recording of information using light of colors with different wavelengths in the same portion of the optical disk and for recording information having different layers from various angles.

[0320] Furthermore, a car ex210 having an antenna ex205 can receive data from the satellite ex202 and others, and reproduce video on a display device such as a car navigation system ex211 set in the car ex210, in the digital broadcasting system ex200. Here, a configuration of the car navigation system ex211 will be a configuration, for example, including a GPS receiving unit from the configuration illustrated in FIG. 33. The same will be true for the configuration of the computer ex111, the cellular phone ex114, and others.

[0321] FIG. 36A illustrates the cellular phone ex114 that uses the moving picture coding method and the moving picture decoding method described in embodiments. The cellular phone ex114 includes: an antenna ex350 for transmitting and receiving radio waves through the base station ex110; a camera unit ex365 capable of capturing moving and still images; and a display unit ex358 such as a liquid crystal display for displaying the data such as decoded video captured by the camera unit ex365 or received by the antenna ex350. The cellular phone ex114 further includes: a main body unit including an operation key unit ex366; an audio output unit ex357 such as a speaker for output of audio; an audio input unit ex356 such as a microphone for input of audio; a memory unit ex367 for storing captured video or still pictures, recorded audio, coded or decoded data of the received video, the still pictures, e-mails, or others; and a slot unit ex364 that is an interface unit for a recording medium that stores data in the same manner as the memory unit ex367.

[0322] Next, an example of a configuration of the cellular phone ex114 will be described with reference to FIG. 36B. In the cellular phone ex114, a main control unit ex360 designed to control overall each unit of the main body including the display unit ex358 as well as the operation key unit ex366 is connected mutually, via a synchronous bus ex370, to a power supply circuit unit ex361, an operation input control unit ex362, a video signal processing unit ex355, a camera interface unit ex363, a liquid crystal display (LCD) control unit ex359, a modulation/demodulation unit ex352, a multiplexing/demultiplexing unit ex353, an audio signal processing unit ex354, the slot unit ex364, and the memory unit ex367.

[0323] When a call-end key or a power key is turned ON by a user's operation, the power supply circuit unit ex361 supplies the respective units with power from a battery pack so as to activate the cell phone ex114.

[0324] In the cellular phone ex114, the audio signal processing unit ex354 converts the audio signals collected by the audio input unit ex356 in voice conversation mode into digital audio signals under the control of the main control

unit ex360 including a CPU, ROM, and RAM. Then, the modulation/demodulation unit ex352 performs spread spectrum processing on the digital audio signals, and the transmitting and receiving unit ex351 performs digital-to-analog conversion and frequency conversion on the data, so as to transmit the resulting data via the antenna ex350. Also, in the cellular phone ex114, the transmitting and receiving unit ex351 amplifies the data received by the antenna ex350 in voice conversation mode and performs frequency conversion and the analog-to-digital conversion on the data. Then, the modulation/demodulation unit ex352 performs inverse spread spectrum processing on the data, and the audio signal processing unit ex354 converts it into analog audio signals, so as to output them via the audio output unit ex357.

[0325] Furthermore, when an e-mail in data communication mode is transmitted, text data of the e-mail inputted by operating the operation key unit ex366 and others of the main body is sent out to the main control unit ex360 via the operation input control unit ex362. The main control unit ex360 causes the modulation/demodulation unit ex352 to perform spread spectrum processing on the text data, and the transmitting and receiving unit ex351 performs the digital-to-analog conversion and the frequency conversion on the resulting data to transmit the data to the base station ex110 via the antenna ex350. When an e-mail is received, processing that is approximately inverse to the processing for transmitting an e-mail is performed on the received data, and the resulting data is provided to the display unit ex358.

[0326] When video, still images, or video and audio in data communication mode is or are transmitted, the video signal processing unit ex355 compresses and codes video signals supplied from the camera unit ex365 using the moving picture coding method shown in each of embodiments (i.e., functions as the image coding apparatus according to the aspect of the present invention), and transmits the coded video data to the multiplexing/demultiplexing unit ex353. In contrast, during when the camera unit ex365 captures video, still images, and others, the audio signal processing unit ex354 codes audio signals collected by the audio input unit ex356, and transmits the coded audio data to the multiplexing/demultiplexing unit ex353.

[0327] The multiplexing/demultiplexing unit ex353 multiplexes the coded video data supplied from the video signal processing unit ex355 and the coded audio data supplied from the audio signal processing unit ex354, using a predetermined method. Then, the modulation/demodulation unit (modulation/demodulation circuit unit) ex352 performs spread spectrum processing on the multiplexed data, and the transmitting and receiving unit ex351 performs digital-to-analog conversion and frequency conversion on the data so as to transmit the resulting data via the antenna ex350.

[0328] When receiving data of a video file which is linked to a Web page and others in data communication mode or when receiving an e-mail with video and/or audio attached, in order to decode the multiplexed data received via the antenna ex350, the multiplexing/demultiplexing unit ex353 demultiplexes the multiplexed data into a video data bit stream and an audio data bit stream, and supplies the video signal processing unit ex355 with the coded video data and the audio signal processing unit ex354 with the coded audio data, through the synchronous bus ex370. The video signal processing unit ex355 decodes the video signal using a moving picture decoding method corresponding to the moving picture coding method shown in each of embodiments

(i.e., functions as the image decoding apparatus according to the aspect of the present invention), and then the display unit ex358 displays, for instance, the video and still images included in the video file linked to the Web page via the LCD control unit ex359. Furthermore, the audio signal processing unit ex354 decodes the audio signal, and the audio output unit ex357 provides the audio.

[0329] Furthermore, similarly to the television ex300, a terminal such as the cellular phone ex114 probably have 3 types of implementation configurations including not only (i) a transmitting and receiving terminal including both a coding apparatus and a decoding apparatus, but also (ii) a transmitting terminal including only a coding apparatus and (iii) a receiving terminal including only a decoding apparatus. Although the digital broadcasting system ex200 receives and transmits the multiplexed data obtained by multiplexing audio data onto video data in the description, the multiplexed data may be data obtained by multiplexing not audio data but character data related to video onto video data, and may be not multiplexed data but video data itself.

[0330] As such, the moving picture coding method and the moving picture decoding method in each of embodiments can be used in any of the devices and systems described. Thus, the advantages described in each of embodiments can be obtained.

[0331] Furthermore, the present invention is not limited to embodiments, and various modifications and revisions are possible without departing from the scope of the present invention.

Embodiment 6

[0332] Video data can be generated by switching, as necessary, between (i) the moving picture coding method or the moving picture coding apparatus shown in each of embodiments and (ii) a moving picture coding method or a moving picture coding apparatus in conformity with a different standard, such as MPEG-2, MPEG-4 AVC, and VC-1.

[0333] Here, when a plurality of video data that conforms to the different standards is generated and is then decoded, the decoding methods need to be selected to conform to the different standards. However, since to which standard each of the plurality of the video data to be decoded conform cannot be detected, there is a problem that an appropriate decoding method cannot be selected.

[0334] In order to solve the problem, multiplexed data obtained by multiplexing audio data and others onto video data has a structure including identification information indicating to which standard the video data conforms. The specific structure of the multiplexed data including the video data generated in the moving picture coding method and by the moving picture coding apparatus shown in each of embodiments will be hereinafter described. The multiplexed data is a digital stream in the MPEG-2 Transport Stream format.

[0335] FIG. 37 illustrates a structure of the multiplexed data. As illustrated in FIG. 37, the multiplexed data can be obtained by multiplexing at least one of a video stream, an audio stream, a presentation graphics stream (PG), and an interactive graphics stream. The video stream represents primary video and secondary video of a movie, the audio stream (IG) represents a primary audio part and a secondary audio part to be mixed with the primary audio part, and the presentation graphics stream represents subtitles of the

movie. Here, the primary video is normal video to be displayed on a screen, and the secondary video is video to be displayed on a smaller window in the primary video. Furthermore, the interactive graphics stream represents an interactive screen to be generated by arranging the GUI components on a screen. The video stream is coded in the moving picture coding method or by the moving picture coding apparatus shown in each of embodiments, or in a moving picture coding method or by a moving picture coding apparatus in conformity with a conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1. The audio stream is coded in accordance with a standard, such as Dolby-AC-3, Dolby Digital Plus, MLP, DTS, DTS-HD, and linear PCM.

[0336] Each stream included in the multiplexed data is identified by PID. For example, 0x1011 is allocated to the video stream to be used for video of a movie, 0x1100 to 0x111F are allocated to the audio streams, 0x1200 to 0x121F are allocated to the presentation graphics streams, 0x1400 to 0x141F are allocated to the interactive graphics streams, 0x1B00 to 0x1B1F are allocated to the video streams to be used for secondary video of the movie, and 0x1A00 to 0x1A1F are allocated to the audio streams to be used for the secondary audio to be mixed with the primary audio.

[0337] FIG. 38 schematically illustrates how data is multiplexed. First, a video stream ex235 composed of video frames and an audio stream ex238 composed of audio frames are transformed into a stream of PES packets ex236 and a stream of PES packets ex239, and further into TS packets ex237 and TS packets ex240, respectively. Similarly, data of a presentation graphics stream ex241 and data of an interactive graphics stream ex244 are transformed into a stream of PES packets ex242 and a stream of PES packets ex245, and further into TS packets ex243 and TS packets ex246, respectively. These TS packets are multiplexed into a stream to obtain multiplexed data ex247.

[0338] FIG. 39 illustrates how a video stream is stored in a stream of PES packets in more detail. The first bar in FIG. 39 shows a video frame stream in a video stream. The second bar shows the stream of PES packets. As indicated by arrows denoted as yy1, yy2, yy3, and yy4 in FIG. 39, the video stream is divided into pictures as I pictures, B pictures, and P pictures each of which is a video presentation unit, and the pictures are stored in a payload of each of the PES packets. Each of the PES packets has a PES header, and the PES header stores a Presentation Time-Stamp (PTS) indicating a display time of the picture, and a Decoding Time-Stamp (DTS) indicating a decoding time of the picture.

[0339] FIG. 40 illustrates a format of TS packets to be finally written on the multiplexed data. Each of the TS packets is a 188-byte fixed length packet including a 4-byte TS header having information, such as a PID for identifying a stream and a 184-byte TS payload for storing data. The PES packets are divided, and stored in the TS payloads, respectively. When a BD ROM is used, each of the TS packets is given a 4-byte TP_Extra_Header, thus resulting in 192-byte source packets. The source packets are written on the multiplexed data. The TP_Extra_Header stores information such as an Arrival_Time_Stamp (ATS). The ATS shows a transfer start time at which each of the TS packets is to be transferred to a PID filter. The source packets are arranged in the multiplexed data as shown at the bottom of FIG. 40. The numbers incrementing from the head of the multiplexed data are called source packet numbers (SPNs).

[0340] Each of the TS packets included in the multiplexed data includes not only streams of audio, video, subtitles and others, but also a Program Association Table (PAT), a Program Map Table (PMT), and a Program Clock Reference (PCR). The PAT shows what a PID in a PMT used in the multiplexed data indicates, and a PID of the PAT itself is registered as zero. The PMT stores PIDs of the streams of video, audio, subtitles and others included in the multiplexed data, and attribute information of the streams corresponding to the PIDs. The PMT also has various descriptors relating to the multiplexed data. The descriptors have information such as copy control information showing whether copying of the multiplexed data is permitted or not. The PCR stores STC time information corresponding to an ATS showing when the PCR packet is transferred to a decoder, in order to achieve synchronization between an Arrival Time Clock (ATC) that is a time axis of ATSSs, and an System Time Clock (STC) that is a time axis of PTSs and DTSs.

[0341] FIG. 41 illustrates the data structure of the PMT in detail. A PMT header is disposed at the top of the PMT. The PMT header describes the length of data included in the PMT and others. A plurality of descriptors relating to the multiplexed data is disposed after the PMT header. Information such as the copy control information is described in the descriptors. After the descriptors, a plurality of pieces of stream information relating to the streams included in the multiplexed data is disposed. Each piece of stream information includes stream descriptors each describing information, such as a stream type for identifying a compression codec of a stream, a stream PID, and stream attribute information (such as a frame rate or an aspect ratio). The stream descriptors are equal in number to the number of streams in the multiplexed data.

[0342] When the multiplexed data is recorded on a recording medium and others, it is recorded together with multiplexed data information files.

[0343] Each of the multiplexed data information files is management information of the multiplexed data as shown in FIG. 42. The multiplexed data information files are in one to one correspondence with the multiplexed data, and each of the files includes multiplexed data information, stream attribute information, and an entry map.

[0344] As illustrated in FIG. 42, the multiplexed data information includes a system rate, a reproduction start time, and a reproduction end time. The system rate indicates the maximum transfer rate at which a system target decoder to be described later transfers the multiplexed data to a PID filter. The intervals of the ATSSs included in the multiplexed data are set to not higher than a system rate. The reproduction start time indicates a PTS in a video frame at the head of the multiplexed data. An interval of one frame is added to a PTS in a video frame at the end of the multiplexed data, and the PTS is set to the reproduction end time.

[0345] As shown in FIG. 43, a piece of attribute information is registered in the stream attribute information, for each PID of each stream included in the multiplexed data. Each piece of attribute information has different information depending on whether the corresponding stream is a video stream, an audio stream, a presentation graphics stream, or an interactive graphics stream. Each piece of video stream attribute information carries information including what kind of compression codec is used for compressing the video stream, and the resolution, aspect ratio and frame rate of the pieces of picture data that is included in the video stream.

Each piece of audio stream attribute information carries information including what kind of compression codec is used for compressing the audio stream, how many channels are included in the audio stream, which language the audio stream supports, and how high the sampling frequency is. The video stream attribute information and the audio stream attribute information are used for initialization of a decoder before the player plays back the information.

[0346] In the present embodiment, the multiplexed data to be used is of a stream type included in the PMT. Furthermore, when the multiplexed data is recorded on a recording medium, the video stream attribute information included in the multiplexed data information is used. More specifically, the moving picture coding method or the moving picture coding apparatus described in each of embodiments includes a step or a unit for allocating unique information indicating video data generated by the moving picture coding method or the moving picture coding apparatus in each of embodiments, to the stream type included in the PMT or the video stream attribute information. With the configuration, the video data generated by the moving picture coding method or the moving picture coding apparatus described in each of embodiments can be distinguished from video data that conforms to another standard.

[0347] Furthermore, FIG. 44 illustrates steps of the moving picture decoding method according to the present embodiment. In Step exS100, the stream type included in the PMT or the video stream attribute information included in the multiplexed data information is obtained from the multiplexed data. Next, in Step exS101, it is determined whether or not the stream type or the video stream attribute information indicates that the multiplexed data is generated by the moving picture coding method or the moving picture coding apparatus in each of embodiments. When it is determined that the stream type or the video stream attribute information indicates that the multiplexed data is generated by the moving picture coding method or the moving picture coding apparatus in each of embodiments, in Step exS102, decoding is performed by the moving picture decoding method in each of embodiments. Furthermore, when the stream type or the video stream attribute information indicates conformance to the conventional standards, such as MPEG-2, MPEG-4 AVC, and VC-1, in Step exS103, decoding is performed by a moving picture decoding method in conformity with the conventional standards.

[0348] As such, allocating a new unique value to the stream type or the video stream attribute information enables determination whether or not the moving picture decoding method or the moving picture decoding apparatus that is described in each of embodiments can perform decoding. Even when multiplexed data that conforms to a different standard is input, an appropriate decoding method or apparatus can be selected. Thus, it becomes possible to decode information without any error. Furthermore, the moving picture coding method or apparatus, or the moving picture decoding method or apparatus in the present embodiment can be used in the devices and systems described above.

Embodiment 7

[0349] Each of the moving picture coding method, the moving picture coding apparatus, the moving picture decoding method, and the moving picture decoding apparatus in each of embodiments is typically achieved in the form of an integrated circuit or a Large Scale Integrated (LSI) circuit.

As an example of the LSI, FIG. 45 illustrates a configuration of the LSI ex500 that is made into one chip. The LSI ex500 includes elements ex501, ex502, ex503, ex504, ex505, ex506, ex507, ex508, and ex509 to be described below, and the elements are connected to each other through a bus ex510. The power supply circuit unit ex505 is activated by supplying each of the elements with power when the power supply circuit unit ex505 is turned on.

[0350] For example, when coding is performed, the LSI ex500 receives an AV signal from a microphone ex117, a camera ex113, and others through an AV IO ex509 under control of a control unit ex501 including a CPU ex502, a memory controller ex503, a stream controller ex504, and a driving frequency control unit ex512. The received AV signal is temporarily stored in an external memory ex511, such as an SDRAM. Under control of the control unit ex501, the stored data is segmented into data portions according to the processing amount and speed to be transmitted to a signal processing unit ex507. Then, the signal processing unit ex507 codes an audio signal and/or a video signal. Here, the coding of the video signal is the coding described in each of embodiments. Furthermore, the signal processing unit ex507 sometimes multiplexes the coded audio data and the coded video data, and a stream IO ex506 provides the multiplexed data outside. The provided multiplexed data is transmitted to the base station ex107, or written on the recording medium ex215. When data sets are multiplexed, the data should be temporarily stored in the buffer ex508 so that the data sets are synchronized with each other.

[0351] Although the memory ex511 is an element outside the LSI ex500, it may be included in the LSI ex500. The buffer ex508 is not limited to one buffer, but may be composed of buffers. Furthermore, the LSI ex500 may be made into one chip or a plurality of chips.

[0352] Furthermore, although the control unit ex501 includes the CPU ex502, the memory controller ex503, the stream controller ex504, the driving frequency control unit ex512, the configuration of the control unit ex501 is not limited to such. For example, the signal processing unit ex507 may further include a CPU. Inclusion of another CPU in the signal processing unit ex507 can improve the processing speed. Furthermore, as another example, the CPU ex502 may serve as or be a part of the signal processing unit ex507, and, for example, may include an audio signal processing unit. In such a case, the control unit ex501 includes the signal processing unit ex507 or the CPU ex502 including a part of the signal processing unit ex507.

[0353] The name used here is LSI, but it may also be called IC, system LSI, super LSI, or ultra LSI depending on the degree of integration.

[0354] Moreover, ways to achieve integration are not limited to the LSI, and a special circuit or a general purpose processor and so forth can also achieve the integration. Field Programmable Gate Array (FPGA) that can be programmed after manufacturing LSIs or a reconfigurable processor that allows re-configuration of the connection or configuration of an LSI can be used for the same purpose.

[0355] In the future, with advancement in semiconductor technology, a brand-new technology may replace LSI. The functional blocks can be integrated using such a technology. The possibility is that the present invention is applied to biotechnology.

Embodiment 8

[0356] When video data generated in the moving picture coding method or by the moving picture coding apparatus described in each of embodiments is decoded, compared to when video data that conforms to a conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1 is decoded, the processing amount probably increases. Thus, the LSI ex500 needs to be set to a driving frequency higher than that of the CPU ex502 to be used when video data in conformity with the conventional standard is decoded. However, when the driving frequency is set higher, there is a problem that the power consumption increases.

[0357] In order to solve the problem, the moving picture decoding apparatus, such as the television ex300 and the LSI ex500 is configured to determine to which standard the video data conforms, and switch between the driving frequencies according to the determined standard. FIG. 46 illustrates a configuration ex800 in the present embodiment. A driving frequency switching unit ex803 sets a driving frequency to a higher driving frequency when video data is generated by the moving picture coding method or the moving picture coding apparatus described in each of embodiments. Then, the driving frequency switching unit ex803 instructs a decoding processing unit ex801 that executes the moving picture decoding method described in each of embodiments to decode the video data. When the video data conforms to the conventional standard, the driving frequency switching unit ex803 sets a driving frequency to a lower driving frequency than that of the video data generated by the moving picture coding method or the moving picture coding apparatus described in each of embodiments. Then, the driving frequency switching unit ex803 instructs the decoding processing unit ex802 that conforms to the conventional standard to decode the video data.

[0358] More specifically, the driving frequency switching unit ex803 includes the CPU ex502 and the driving frequency control unit ex512 in FIG. 45. Here, each of the decoding processing unit ex801 that executes the moving picture decoding method described in each of embodiments and the decoding processing unit ex802 that conforms to the conventional standard corresponds to the signal processing unit ex507 in FIG. 45. The CPU ex502 determines to which standard the video data conforms. Then, the driving frequency control unit ex512 determines a driving frequency based on a signal from the CPU ex502. Furthermore, the signal processing unit ex507 decodes the video data based on the signal from the CPU ex502. For example, the identification information described in Embodiment 6 is probably used for identifying the video data. The identification information is not limited to the one described in Embodiment 6 but may be any information as long as the information indicates to which standard the video data conforms. For example, when which standard video data conforms to can be determined based on an external signal for determining that the video data is used for a television or a disk, etc., the determination may be made based on such an external signal. Furthermore, the CPU ex502 selects a driving frequency based on, for example, a look-up table in which the standards of the video data are associated with the driving frequencies as shown in FIG. 48. The driving frequency can be selected by storing the look-up table in the buffer ex508 and in an internal memory of an LSI, and with reference to the look-up table by the CPU ex502.

[0359] FIG. 47 illustrates steps for executing a method in the present embodiment. First, in Step exS200, the signal processing unit ex507 obtains identification information from the multiplexed data. Next, in Step exS201, the CPU ex502 determines whether or not the video data is generated by the coding method and the coding apparatus described in each of embodiments, based on the identification information. When the video data is generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiments, in Step exS202, the CPU ex502 transmits a signal for setting the driving frequency to a higher driving frequency to the driving frequency control unit ex512. Then, the driving frequency control unit ex512 sets the driving frequency to the higher driving frequency. On the other hand, when the identification information indicates that the video data conforms to the conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1, in Step exS203, the CPU ex502 transmits a signal for setting the driving frequency to a lower driving frequency to the driving frequency control unit ex512. Then, the driving frequency control unit ex512 sets the driving frequency to the lower driving frequency than that in the case where the video data is generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiment.

[0360] Furthermore, along with the switching of the driving frequencies, the power conservation effect can be improved by changing the voltage to be applied to the LSI ex500 or an apparatus including the LSI ex500. For example, when the driving frequency is set lower, the voltage to be applied to the LSI ex500 or the apparatus including the LSI ex500 is probably set to a voltage lower than that in the case where the driving frequency is set higher.

[0361] Furthermore, when the processing amount for decoding is larger, the driving frequency may be set higher, and when the processing amount for decoding is smaller, the driving frequency may be set lower as the method for setting the driving frequency. Thus, the setting method is not limited to the ones described above. For example, when the processing amount for decoding video data in conformity with MPEG-4 AVC is larger than the processing amount for decoding video data generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiments, the driving frequency is probably set in reverse order to the setting described above.

[0362] Furthermore, the method for setting the driving frequency is not limited to the method for setting the driving frequency lower. For example, when the identification information indicates that the video data is generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiments, the voltage to be applied to the LSI ex500 or the apparatus including the LSI ex500 is probably set higher. When the identification information indicates that the video data conforms to the conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1, the voltage to be applied to the LSI ex500 or the apparatus including the LSI ex500 is probably set lower. As another example, when the identification information indicates that the video data is generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiments, the driving of the CPU ex502 does not probably have to be suspended. When the identification information indicates

that the video data conforms to the conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1, the driving of the CPU ex502 is probably suspended at a given time because the CPU ex502 has extra processing capacity. Even when the identification information indicates that the video data is generated by the moving picture coding method and the moving picture coding apparatus described in each of embodiments, in the case where the CPU ex502 has extra processing capacity, the driving of the CPU ex502 is probably suspended at a given time. In such a case, the suspending time is probably set shorter than that in the case where when the identification information indicates that the video data conforms to the conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1.

[0363] Accordingly, the power conservation effect can be improved by switching between the driving frequencies in accordance with the standard to which the video data conforms. Furthermore, when the LSI ex500 or the apparatus including the LSI ex500 is driven using a battery, the battery life can be extended with the power conservation effect.

Embodiment 9

[0364] There are cases where a plurality of video data that conforms to different standards, is provided to the devices and systems, such as a television and a cellular phone. In order to enable decoding the plurality of video data that conforms to the different standards, the signal processing unit ex507 of the LSI ex500 needs to conform to the different standards. However, the problems of increase in the scale of the circuit of the LSI ex500 and increase in the cost arise with the individual use of the signal processing units ex507 that conform to the respective standards.

[0365] In order to solve the problem, what is conceived is a configuration in which the decoding processing unit for implementing the moving picture decoding method described in each of embodiments and the decoding processing unit that conforms to the conventional standard, such as MPEG-2, MPEG-4 AVC, and VC-1 are partly shared. Ex900 in FIG. 49A shows an example of the configuration. For example, the moving picture decoding method described in each of embodiments and the moving picture decoding method that conforms to MPEG-4 AVC have, partly in common, the details of processing, such as entropy coding, inverse quantization, deblocking filtering, and motion compensated prediction. The details of processing to be shared probably include use of a decoding processing unit ex902 that conforms to MPEG-4 AVC. In contrast, a dedicated decoding processing unit ex901 is probably used for other processing unique to an aspect of the present invention. Since the aspect of the present invention is characterized by motion compensation in particular, for example, the dedicated decoding processing unit ex901 is used for motion compensation. Otherwise, the decoding processing unit is probably shared for one of the entropy decoding, deblocking filtering, and inverse quantization, or all of the processing. The decoding processing unit for implementing the moving picture decoding method described in each of embodiments may be shared for the processing to be shared, and a dedicated decoding processing unit may be used for processing unique to that of MPEG-4 AVC.

[0366] Furthermore, ex1000 in FIG. 49B shows another example in that processing is partly shared. This example

uses a configuration including a dedicated decoding processing unit ex1001 that supports the processing unique to an aspect of the present invention, a dedicated decoding processing unit ex1002 that supports the processing unique to another conventional standard, and a decoding processing unit ex1003 that supports processing to be shared between the moving picture decoding method according to the aspect of the present invention and the conventional moving picture decoding method. Here, the dedicated decoding processing units ex1001 and ex1002 are not necessarily specialized for the processing according to the aspect of the present invention and the processing of the conventional standard, respectively, and may be the ones capable of implementing general processing. Furthermore, the configuration of the present embodiment can be implemented by the LSI ex500.

[0367] As such, reducing the scale of the circuit of an LSI and reducing the cost are possible by sharing the decoding processing unit for the processing to be shared between the moving picture decoding method according to the aspect of the present invention and the moving picture decoding method in conformity with the conventional standard.

INDUSTRIAL APPLICABILITY

[0368] A moving picture coding method and a moving picture decoding method according to the present invention are applicable to any multimedia data, can improve error resistance of coding and decoding a moving picture, and are useful as a moving picture coding method and a moving picture decoding method in storage, transmission, communication, and the like using cellular phones, DVD apparatuses, and personal computers, for example.

We claim:

1. A moving picture coding apparatus for coding a current block of a picture, the moving picture coding apparatus comprising:

a processor; and

a non-transitory memory,

wherein the processor performs, using the non-transitory memory, processes, the processes including:

generating a motion vector candidate list including motion vector candidates and, for each of the motion vector candidates, a value, the motion vector candidates including a first candidate and a second candidate, the second candidate being arranged before the first candidate in the motion vector candidate list; sorting the first candidate before the second candidate in the motion vector candidate list based on the value of the first candidate and the value of the second candidate;

selecting a motion vector candidate to use to code the current block; and

coding an index to identify the selected motion vector candidate,

wherein the value of the first candidate is higher than the value of the second candidate, and

for each candidate of the motion vector candidates, the value is based on a number of times the candidate is selected as a motion vector predictor for neighboring blocks.

2. The moving picture coding apparatus according to claim 1, wherein the processes further include:

coding information selecting the first candidate or the second candidate.

3. The moving picture coding apparatus according to claim 1, wherein the processes further include:

counting the number of times each candidate of the motion vector candidates is selected as the motion vector predictor for the neighboring blocks.

4. The moving picture coding apparatus according to claim 1, wherein the motion vector candidates are generated from the neighboring blocks.

5. The moving picture coding apparatus according to claim 4,

wherein, in the generating, the motion vector candidate list further includes a new candidate, the new candidate not being generated from the neighboring blocks, the new candidate being arranged before at least one of the first candidate or the second candidate in the motion vector candidate list, and

in the sorting, further sorting the new candidate after the at least one of the first candidate or the second candidate in the motion vector candidate list.

6. The moving picture coding apparatus according to claim 5,

wherein the neighboring blocks include blocks adjacent to the current block, and the new candidate is generated from a block adjacent to one of the neighboring blocks.

7. A moving picture coding method for coding a current block of a picture, the moving picture coding method comprising:

generating a motion vector candidate list including motion vector candidates and, for each of the motion vector candidates, a value, the motion vector candidates including a first candidate and a second candidate, the second candidate being arranged before the first candidate in the motion vector candidate list;

sorting, by at least one of a processor or a circuit, the first candidate before the second candidate in the motion vector candidate list based on the value of the first candidate and the value of the second candidate;

selecting a motion vector candidate to use to code the current block; and

coding an index to identify the selected motion vector candidate,

wherein the value of the first candidate is higher than the value of the second candidate, and

for each candidate of the motion vector candidates, the value is based on a number of times the candidate is selected as a motion vector predictor for neighboring blocks.

8. The moving picture coding method according to claim 7, further comprising:

coding information selecting the first candidate or the second candidate.

9. The moving picture coding method according to claim 7, further comprising:

counting the number of times each candidate is selected as the motion vector predictor for the neighboring blocks.

10. The moving picture coding method according to claim 7, wherein the motion vector candidates are generated from the neighboring blocks.

11. The moving picture coding method according to claim 10,

wherein, in the generating, the motion vector candidate list further includes a new candidate, the new candidate not being generated from the neighboring blocks, the

new candidate being arranged before at least one of the first candidate or the second candidate in the motion vector candidate list, and

in the sorting, further sorting the new candidate after the at least one of the first candidate or the second candidate in the motion vector candidate list.

12. The moving picture coding method according to claim **11**, wherein the neighboring blocks include blocks adjacent to the current block, and the new candidate is generated from a block adjacent to one of the neighboring blocks.

13. A transmitting method for transmitting a bitstream to a terminal, the bit stream being generated by a moving picture coding method for coding a current block of a picture, the moving picture coding method comprising:

generating a motion vector candidate list including motion vector candidates and, for each of the motion vector candidates, a value, the motion vector candidates

including a first candidate and a second candidate, the second candidate being arranged before the first candidate in the motion vector candidate list;

sorting the first candidate before the second candidate in the motion vector candidate list based on the value of the first candidate and the value of the second candidate;

selecting a motion vector candidate to use to code the current block; and

coding an index to identify the selected motion vector candidate,

wherein the value of the first candidate is higher than the value of the second candidate, and

for each candidate of the motion vector candidates, the value is based on a number of times the candidate is selected as a motion vector predictor for neighboring blocks.

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